

Fixed Universal Polynomials

A literal Grill instance, exact degree laws, native witness fibers
and positive-index restoration

Research reports 23 (revision 1), 24, 25, 33 and 34 of one AI-assisted research pipeline,
merged into one report for the ProveIt research-report collection

Sources dated 3 October 2026; merged 3 October 2026

Status. AI-assisted research manuscripts, unrefereed, and not formalized in a proof assistant. The language theorem of Part I is conditional on precisely pinned imports from the Hilbert's-tenth research tree of this repository; the theorems of Parts III and IV are conditional on, or inherit, pinned native-source theorems as each Part states. The exact-degree theorems of Parts I and II are unconditional algebra on the stated sources. Section 0.6 lists what is and is not claimed.

Merge. Text marked [write] was written when the five manuscripts were merged; all other text is the source's own. Section 0.2 gives the provenance.

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0.1 [write] What this report contains

This report prints five research manuscripts that share one object: the fixed universal Diophantine polynomials of the Hilbert’s-tenth research programme in this repository (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`, below *the research tree*) and the native Pell arithmetic block inside them. They share one spine: first the exact resources and exact degree of one literal instance, then the structure of the witness fibers of the native block.

- **Part I** (Report 23, revision 1): one literal Grill program with 397,488 run phases and one complete integer polynomial circuit with an ordinary positive input, five program parameters fixed for each represented language, 797,135 positive witnesses and 3,600,546 binary operations. Its language theorem is conditional on pinned imports; its exact total degree 69,339,973 is proved unconditionally.
- **Part II** (Report 24): for every nonempty run table of the same native template, with $N = 2m + g + 5$, the native unit has exact degree $63N - 4$, the largest residual $12N + 10$, and the complete native polynomial $87N + 16$; residual gluing, recoder and loader maxima, and fresh polynomial inputs. Part I’s degree is the instance $N = 797,011$.
- **Part III** (Report 25): the entire positive fiber of the retained twenty-two-coordinate native component at fixed ports: seventeen coordinates forced, five parametrized by two Pell indices, and the two-term height law with a $\zeta(1/2)$ second term.
- **Part IV** (Reports 33 and 34): the complete⁷⁴ signed¹⁹ index-eliminated child has infinitely many positive zeros with negative restored index at every positive input (Report 33), and the height and counting law of one canonical family of these zeros, with a Kesten-type obstruction to a smooth second term (Report 34).

0.2 [write] Sources, editions and provenance

0.2.1 The five manuscripts

All seven archives of this cluster arrived in commit `db37d18c8` (3 October 2026, “New research reports”) under `docs/incoming/`; they were placed by commit `2f58ab4e9` (batch 82, cluster M3), which retired the archives. The manuscripts are numbered “Research Report NN” by their pipeline; the batch numbers are the manuscript numbers of batch 82. Table 1 lists them. Each manuscript records its own pins, which are the commits of this repository whose research-tree files it imports or cites; they are kept here as provenance.

What each contributed. Source 11 is the base: its `article.tex` and `README.md` were staged as delivered, and its text is Part I. Sources 14, 07, 10 and 05 contributed their manuscripts (Parts II–IV) and their code, data and audit files; their delivered READMEs and PDFs are not shipped. Every result, proof, remark, question and limitation of the five manuscripts is printed.

The edition of Report 23. Two editions of Report 23 arrived. Archive 11 is revision 1: its `11-grill-poly-REVISION.md` calls it “a versioned extension of Report 23” and anchors the SHA-256 digests of the original PDF, LaTeX source, release inventory and of archive 12 itself (`467e2b47...bf9b`, measured equal). Revision 1 changes no gate, coefficient, witness or comparison; it replaces the original’s propagated upper bound 71,731,007 by the exact degree 69,339,973 (new Section 23.13) and keeps the

Report	Batch	Archive (bytes)	Pins	File prefix	Printed as
	82				
23 rev. 1	11	Fixed_Universal_Grill_Polynomial_Package (1).zip (15,508,226)	2d887f0fa	11-grill-poly-	Part I
24	14	Native_Grill_Exact_Degree_Laws_Package.zip (552,226)	2d887f0fa	14-degree-laws-	Part II
25	07	Entire_Native_Witness_Fiber_Package.zip (531,631)	ad634b2d1	07-native-fiber-	Part III
33	10	Failure_of_Positive_Index_Restoration_Package.zip (703,207)	2dde7850f; context 93ad43375	10-index-restore-	Part IV (33.k)
34	05	Counterexample_Height_Expansions_and_Rotation_Discrepancy_Package.zip (591,664)	2dde7850f; transfer 9fa99be93	05-signed19-heights-	Part IV (34.k)
23 orig.	12	Fixed_Universal_Grill_Polynomial_Package.zip (15,390,453)	2d887f0fa	—	superseded

Table 1: The manuscripts. The file prefix is that of the source’s shipped code, data and audit files in this directory; the README lists them.

original’s QA files as `qa/v0-*`. Archive 12 is therefore superseded and nothing of it is staged. Its one open question, “Can the true degree be determined?”, is answered by Section 23.13 and is not printed as open; revision 1 replaces it by “Can a different representation reduce the exact degree?”.

The series citation graph. Report 24 cites Report 23 revision 1 (its optional full regression reads that report’s arithmetic DAG). Report 25 is the sequel of Report 22, “Canonical outer histories and infinite native fibers” (archive `Canonical_Histories_and_Infinite_Fibers_Package.zip` of batch 82, cluster M2, placed by commit `7d2b1b245` as source 23 of the collection report `signal-machine-collision-certificates`, in its new Part VIII); it uses only Report 22’s entire-fiber projection theorem, in the corollary of Section 25.12. Report 34 cites Reports 33 and 25 and carried their LaTeX sources, byte-identical to Parts III and IV, as context (not staged). Reports 23, 24, 25, 33 and 34 all pin research-tree commits. The sibling Report 32 of the same batch, a fixed universal matrix semigroup, is printed as Part V of `group-theoretic-substrates`; it mentions Report 23 only in its delivered audit.

Where the merge had to choose.

1. The edition of Report 23 (above).
2. Placement of Report 25. Report 25 continues Report 22, which goes to `signal-machine-collision-certificates`; Report 25 is printed here because its mathematics is purely the native Pell block that Parts I, II and IV also use, and it uses Report 22 in one corollary only.
3. Part IV prints two manuscripts: Report 34 is a direct sequel of Report 33, studies one family inside Report 33’s construction, and shipped Report 33 as its context.
4. Numbering. Every source keeps its own numbering, prefixed by its Report number: statement $k.m$ of Report NN is printed as `NN.k.m`, its Section k as Section `NN.k`, its equation (k) as `(NN.k)`, and its Appendix X as Section `NN.X`. Tables are numbered consecutively through the whole report. Front-matter sections are numbered $0.k$.
5. Bibliographies. Each source’s bibliography is printed at the end of its own Part, with the source’s numbering; entries naming a sibling Report gained a `[write]` pointer to its place.

6. Macros. Only macro names changed, never renderings: Report 24’s `\sha` and `\fname` are `\shasmall` and `\fnameB`; Report 33’s `\MF` (the native mask M_F^{native}) is `\MFnat` and its `\MFS` (the paid mask M_F^{src}) is `\MFsrc`; Report 34’s `\MF`, which denotes the paid mask, is `\MFsrc`. No mathematical symbol was renamed.
7. Labels. Every label carries the prefix `fup:`; the labels of Reports 23, 24, 25, 33 and 34 carry `fup:gp:`, `fup:dl:`, `fup:nf:`, `fup:ir:` and `fup:hc:` before their delivered names, and the front matter uses `fup:fm:`.

Not shipped. PDFs, checksum ledgers and release seals, the non-base manuscripts and delivery READMEs, byte copies of research-tree files (73 members of the five archives; they match the current files byte for byte and are cited by path), in-archive and cross-archive byte copies (shipped once), Report 34’s context copies of Reports 25 and 33, two heavy regenerable files of Report 23 (the 61,209,290-byte arithmetic DAG and the 1,589,952-byte run table; the README gives the rebuild and the retrieval), and the third-party copies of the Cocke–Minsky (1964) and Neary–Woods (2009) papers that archive 11 contained (PDFs, page scans and text extractions; they are cited in Part I’s bibliography [1, 2] and are not redistributed). All of these survive in the arrival commit.

0.3 [write] Notation across the Parts

Each Part keeps its source’s letters, and inside a Part a letter always has that Part’s meaning. Many letters change meaning between Parts. Table 2 lists those that do, with the reading that would be wrong.

Table 2: Letters with different meanings in different Parts.

Letter	Meanings	Tempting false reading
α	I, II: the lane degree $3N_*$ (I) or $3N$ (II). III: the Pell unit $A + \sqrt{\Delta}$, with $\lambda = \log \alpha$. IV, Report 33: a supplied coordinate in Sections 33.1–33.10, and $\alpha_0 = \log \lambda$ in Lemma 33.6.1; Report 34: the growth logarithm $\log \lambda$, with α_{src} for Report 33’s coordinate.	Report 34 swaps Report 25’s letters: its λ is Report 25’s α and its α is Report 25’s λ (Section 34.9 compares the two Parts).
λ	III: $\log(A + \sqrt{\Delta})$. IV: the unit $A + \sqrt{\Delta}$ (Lemma 33.6.1 and Report 34).	The same letter is a logarithm in Part III and a unit in Part IV.
κ	I: $k_* = \eta + \zeta$ is written k_* ; II: $\kappa = \eta + \zeta$ (native register). III: the counting coefficient of $N(H)$. IV: in Report 33 the input coefficient $u + \delta\Delta$; in Report 34 the counting coefficient of $N(B)$, with κ_{in} for Report 33’s κ .	Part II’s register κ and Part III’s coefficient κ are unrelated; so are Report 33’s and Report 34’s.
δ	II: the degree of a fresh polynomial input. IV: a supplied coordinate in Report 33 (δ_{src} in Report 34); $\log(1 + 1/Y)$ in Report 34.	—
N	I: the varying tape integer $N = N_e(x)$, and $N_* = 797,011$ the fixed lane exponent. II: $N = 2m + g + 5$, which is Part I’s N_* , not its N ; Part II writes N_{tape} for Part I’s N . III: $N(H)$ is a counting function. IV: Report 33’s N is a power-of-five exponent; Report 34’s $N(B)$ is a count and N_0 that exponent.	Part II’s $87N + 16$ is evaluated at Part I’s N_* , never at the tape integer.

Letter	Meanings	Tempting false reading
$K, \mathcal{K}$	I: $K = 2^{2b}$ is the loader radix and $\mathcal{K} = 2^{551890}$ the native radix multiplier. II: K is the native radix multiplier (Part I's $\mathcal{K}$) and K_e the loader radix. IV: $K_0 = D_C + BD_R$ is a compiler constant.	—
$B, \mathcal{B}$	I: $B = 2^{k-1}Q$ in a recoder, $\mathcal{B}$ the native radix. II: $B = KD_0$, Part I's $\mathcal{B}$. III: a generic Pell base, and in Section 25.3 the first Pell base $2XY^2 + 1$. IV: Report 33's $B = 2^d$ is the compiler radix and $P = 2XY^2 + 1$ the first Pell base; Report 34's B is a height bound and B_{comp} the compiler radix.	Part III's first Pell base B is Part IV's P .
Δ, A	II: $\Delta = 4a + 3$ and $A = a^2 + \Delta$, both computed from the register a . III, IV: $A = a + 2$ is a Pell base and $\Delta = A^2 - 1 = a^2 + 4a + 3$; Report 33 writes $H = 4a + 3$ (Report 34: H_0).	Part II's A is Parts III–IV's Δ , and Part II's Δ is Part IV's H . The registers a are different objects in the two presentations.
$p, \mathfrak{p}$	I: $\mathfrak{p} = 2^{551890}(X + Z_0)V_f J_1$, and p_e a program parameter. II: $p = KLVJ_1$, Part I's $\mathfrak{p}$. III, IV: the main Pell index ($p = 2r + 1$ in III).	—
c, a, f, i, j, o, y	I, II: registers and coordinates of the 67-row native kernel of the Grill polynomial. III: coordinates of the twenty-two-coordinate native component pinned at <code>ad634b2d1</code> . IV: coordinates of the signed19 child.	These are different presentations of the programme's native Pell/AND block; the letters play analogous roles but no coordinate of one Part is identified with a coordinate of another.
R	I: registers R_{15}, R_{16} and residuals R_j . III: $R_m = \Delta\psi_A(m)$ and $R = ic^2$. IV: the restored index $R = k - hXY - 1$; in Report 34 also $R, R(B)$ for a number of progression terms.	—
H	I: a joined kernel port. III: a height bound in $N(H)$. IV: $H = 4a + 3$ (Report 33).	—
W	II: witness counts $W_{\text{native}}, W_{\text{total}}$. IV: the computed register $W = C - Z$ (Report 33), $W = 1 - Z$ (Report 34), and the Lambert function W in Section 34.5.	—
ζ	I, II, III: a supplied native coordinate. III also uses the Riemann zeta function $\zeta(1/2)$ and warns about the clash.	—
Q	I: a recoder port and the leading form Q_* . II: $Q = \text{in}(q)$. IV: $Q = q^2 - 1$ (Report 33).	—
$M_F^{\text{native}}, M_F^{\text{src}}$	IV: the native and the paid field mask, $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$.	Report 34's own macro for M_F^{src} had the name of Report 33's macro for M_F^{native} ; the renderings always differed and are unchanged.

0.4 [write] Relation to the Hilbert's-tenth research tree

The five manuscripts take the research tree's own artifacts, at the pinned commits, as their inputs. The tree is maintained separately; this section records what the Parts re-prove or overlap and what is new relative to it at the time of this write.

Byte copies. 73 members of the five archives are byte copies of 51 distinct files of `Computability/HilbertTenthProblem/Papers/` (research tree, 1980/ and `verification/`); every one equals the current file. They are not shipped; the README lists each with its repository path.

The programme’s reviews. The research tree has reviewed these archives, after their arrival:

- `review_incoming_matrix_grill.md` (commit 32da04296, with a helper and receipt) authenticates both editions of Report 23, Report 24 and Report 32 as data, without running their code. It independently parses every row of the 3,600,546-gate DAG, propagates the syntactic degree 71,731,007, checks liveness of every gate and of all 797,141 supplied coordinates and the 260 finalizer rows, and recomputes $87N + 16 = 69,339,973$ at $N = 797,011$. Its verdict: a “useful full numerical Grill milestone”, not an operation improvement on the programme’s 85-operation universal polynomial. It did not rerun the exact-degree verifiers.
- `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) reviews Reports 33, 25 and 22. It finds no gap in Report 33’s construction, independently expands the eight residuals, fourteen definitions and the 23-gate finalizer of the signed19 child, and records the refutation of the signed19 candidate in the programme’s status; it finds Report 25’s classification consistent, notes a domain distinction (Part III, Section 25.4), and does not certify Report 25’s second-term or total-bit-length theorems.
- Report 34 has not been reviewed by the programme.

What the Parts re-prove or overlap.

- The norm cancellation behind the exact degrees, identity (23.10) of Part I and (24.17) of Part II, is identity (4) of `gpcp_shared_selectors774.md` (commit ee507599f), also used in three other notes of the tree; neither Report cites it.
- Part II’s law $87N + 16$ is proved by the same method as the tree’s exact law $58N + 28$ for the PCP affine slope-class history (`pcp_affine_slope_class_history.md`, shipped inside archive 11), for a different template.
- Sections 23.6 and 23.10 of Part I re-state and re-certify the tree’s corrected Grill halt bridge and its composed native history; Part I says so and imports them.
- Report 33’s input-index dichotomy (Section 33.B) is proved independently in the tree’s `complete74_negative_index_refinement.md` (commit 813c1cff4), which goes further: it also proves $4^d > X \geq q^3$ for the defect $d = 2n - p$ on the negative branch, and that for fixed q, X, Y and a fixed bounded representative at most one main index p fits the ratio interval; Report 33’s bounded-window reduction imports the tree’s bootstrap (93ad43375) and says so.

What is new relative to the tree. A literal fixed universal Grill program with a paid ordinary-input composition (conditional on the pinned imports), where the tree recorded that “a numerical universal source and its input-language theorem remain open” for the exact-width Grill loader; the exact degree of a Grill polynomial (the tree recorded only upper bounds, with `exact_degree None`); the parametric laws of Part II; the classification and counting of the entire native fiber in Part III; the signed19 counterexample of Report 33, which answers negatively the open obligation of the tree’s nonlinear-index scout for the signed child (raw29 and positive21 stay open); and the canonical-family counting of Report 34 with its Kesten obstruction.

Records. Part I is not an operation record. The tree’s current universal polynomials by other routes are far smaller: 85 operations with 18 positive witnesses and exact degree 175 (commit 109aaca15), and 86 operations with exact degree 131 (commit 55d248dc5); it also has a 557-operation fixed binary-tag universal polynomial and a 97-tile U15 GPCP construction. Part I does not mention these, and disclaims optimality itself.

0.5 [write] Relation to other reports of the collection

No other report of the collection holds the programme’s universal polynomial, its native block, the Grill route or the complete74 family. Several name the target that Part I approaches, an instantiated audited universal polynomial: `stochastic-and-thermal-exactness`, question `ste:ta:q:instantiate`; `liveness-beyond-halting`, the explicitness boundary of Section `lbh:qd:sec:universal` (“no numerical expanded universal polynomial”); `signal-machine-collision-certificates`, near `smc:cg:sec:barrier`; and `polynomial-witness-histories`, question `pwh:oc:q:numerical`. Part I bears on them without answering them: it gives a literal circuit, conditional on its imports, with exact degree, not an expanded polynomial, and not their particular source machines. Report 22, the predecessor of Part III, is printed in `signal-machine-collision-certificates`; Report 32 in `group-theoretic-substrates`.

Placement of this report beside the Lean development of `Computability/HilbertTenthProblem` confers no formal status: no statement of this report has been formalized, and no source contains or cites Lean or Rocq code.

0.6 [write] Status, claims and non-claims

The five manuscripts are AI-assisted research reports (author lines “Research report 23/24/25” and “Mathematical research report”), unrefereed, and not formalized. Each Part keeps all its source’s disclaimers; the main ones are collected here.

- Part I: the language theorem is conditional on pinned full native-history, binary-recoder and finite-input U15 universality theorems; only the explicit valid five-parameter slices have the machine interpretation; no new general MRDP theorem, no optimality claim, and no materialized full Pell witness or universal accepting execution. The exact degree is unconditional.
- Part II: source-specific algebraic theorems; no universality theorem, no semantic correctness of arbitrary loader attachments, no minimal degree among equivalent representations.
- Part III: native soundness and positive completeness are inherited from pinned source proofs; no change to the native circuit, no minimality, no finite-fold MRDP result, no numerical Pell witness, no literature-wide novelty claim.
- Part IV: Report 33 refutes only the proposed positive-domain restoration for the signed19 child; no improved operation bound, no machine-specific false acceptance, no materialized full child witness; raw29 and positive21 remain open. Report 34 is about one explicit canonical subfamily, not the whole signed19 fiber; its finer second term is equivalent to an unresolved discrepancy estimate.

The suites delivered with the sources were rerun on copies at placement (README); passing them supports the implementations, not the unbounded theorems.

Part I

A fixed universal Grill polynomial: paid ordinary input and exact width through a literal source program

[write] Source 11 of batch 82: *Research report 23, revision 1* (“Revision 1 with the exact degree theorem”), dated 3 October 2026; archive `Fixed_Universal_Grill_Polynomial_Package (1).zip`; pin `2d887f0fa`; the base of this report. Its sections, statements and equations are numbered *23.k* here.

Abstract of the source. We construct one literal Grill program with 397,488 run phases and one complete integer polynomial circuit attached to it. The circuit accepts an ordinary positive integer input, pays for its conversion into the finite input convention of a fixed universal machine, and binds the initial queue to its exact bit width. Its six external coordinates are the ordinary input and five program parameters fixed for each represented language. It has 797,135 positive existential coordinates and 3,600,546 binary arithmetic operations, comprising 803,517 multiplications and 2,797,029 additions or subtractions. Its exact total degree is 69,339,973, established by an all-tuple norm cancellation and an explicit nonzero leading coefficient on a univariate ray.

The language theorem is conditional on precisely pinned full native-history, binary-recorder, and finite-input U15 universality theorems. The exact-degree theorem is unconditional algebra on the unchanged pinned circuit. The new contribution is the explicit composition: corrected right and left tag macros, a unique accepting-head event, phase-preserving normalization, two distinct complete recoders, signed finite-frame arithmetic, exact unary loading, and a unit-safe finalizer. A complete 61,209,290-byte arithmetic DAG and independent whole-source checks accompany the report. The costs describe this actual emitted circuit in a literal-integer arithmetic model; they are not bit complexity or a minimality claim.

[write] The title states universality before its qualifier: the language theorem, Theorem 23.1.1, is conditional on the pinned imports of Section 23.2, as its name, the abstract and the conclusion say. The exact-degree theorem, Theorem 23.13.1, is unconditional. The programme’s review `review_incoming_matrix_grill.md` (commit `32da04296`) authenticated this source and its original edition as data and checked the DAG independently (Section 0.4); it did not rerun the exact-degree verifiers. The source’s files are shipped with the prefix `11-grill-poly-`.

23.1 Result and its mathematical meaning

The purpose of this report is to close the ordinary-input interface for a concrete Grill route to Diophantine representation. A native queue-history relation by itself is insufficient: it may permit different terminal-zero paddings of the same integer, and a substrate-universality slogan does not identify what language a particular numerical input represents. Here both interfaces are explicit and paid.

Theorem 23.1.1 (Conditional fixed universal polynomial). *Assume the pinned theorem imports in Section 23.2. There exists a single polynomial*

$$F \in \mathbb{Z}[x, a, p, s, C, L, y_1, \dots, y_{797135}]$$

represented by the complete arithmetic DAG in the accompanying archive, such that for every computably enumerable set $S \subseteq \mathbb{Z}_{>0}$ there is a fixed tuple

$$\theta_S = (a_e, p_e, s_e, C_e, L_e) \in \mathbb{Z}_{>0}^5$$

with, for every $x \in \mathbb{Z}_{>0}$,

$$x \in S \iff \exists(y_1, \dots, y_{797135}) \in \mathbb{Z}_{>0}^{797135} : F(x, a_e, p_e, s_e, C_e, L_e, \mathbf{y}) = 0.$$

All five coordinates of θ_S have the finite program-dependent recipes in Section 23.7. The polynomial has 3,600,546 emitted binary arithmetic gates and exact total degree 69,339,973 when all six external coordinates and every witness have degree one. The degree assertion is proved independently of these theorem imports in Section 23.13.

The program parameters are not existentially varied with x . An arbitrary positive tuple (a, p, s, C, L) still gives a polynomial slice, but only the explicitly constructed valid tuples have the stated machine interpretation. The theorem covers positive integer inputs and positive integer witnesses. No statement about $x = 0$, negative inputs, or real zeros is intended.

The construction establishes a concrete representation through a selected computational substrate. It is not a new general MRDP theorem, and the operation or witness counts are not asserted to be optimal. The finite audits establish exact source facts and implementation identities; the unbounded existential equivalence additionally uses the imported complete theorems. In particular, no astronomical full Pell witness tuple or complete universal accepting execution has been materialized.

[write] *Not an operation record.* The research tree’s universal polynomials by other routes are far smaller: 85 operations, 18 positive witnesses and exact degree 175 (commit 109aaca15), and 86 operations with exact degree 131 (commit 55d248dc5), besides a 557-operation fixed binary-tag polynomial. This Part does not mention them. It is the Grill-route instance, and its 3,600,546 operations are about 42,000 times the 85-operation record; the programme’s intake review `review_incoming_matrix_grill.md` (commit 32da04296) calls it a “useful full numerical Grill milestone” and not an operation improvement.

23.1.1 An overview of the chain

Fix a language S and its finite semidecider. For the varying $x > 0$, the construction follows

$$\begin{aligned} x &\mapsto u = x + 1 \mapsto \text{canonical LSB bits of } u \mapsto N_e(x), \\ \text{U15 finite tape} &\mapsto \text{corrected deletion-2 tag word} \mapsto \text{normalized Genera word} \\ &\mapsto \text{exact Grill E word and width} \mapsto \text{complete native history relation} \mapsto F = 0. \end{aligned}$$

The two recoders serve different jobs. The first selects the canonical bit length of u . The second extracts the unrestricted exponent $N_e(x) + 1$ used to load a unary tag block. Neither exponent is the native history duration. Keeping these three quantities separate is essential.

Object	Exact size or bound
Refined U15 states	30
Nonhalting U15 instructions	29
Tag symbols and total productions	570
Normalized Genera symbols	1,013
Grill run phases	397,488
Positive runs / zero runs	24,300 / 373,188
Distinct run exponents, including zero	2,030
Largest run exponent	275,944
Positive existential coordinates	797,135
Nonunit comparisons / unit condition	86 / 1
Complete arithmetic operations	3,600,546
Exact total degree	69,339,973

Table 3: The literal source and the actual composed circuit. A run exponent describes a fixed block of commands; a run phase is not an individual command.

23.2 Sources and imported complete theorems

All repository theorem imports use the immutable ProveIt commit

2d887f0fa768fd67f3e545d83f8998b5780e530d.

They are under `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`. The reproducibility packet records Git blob identifiers, SHA-256 digests, frozen texts, and the local adaptations needed to replay the new source. A frozen theorem note is an explicit premise, not a claim that its full historical proof dependency graph has been independently formalized here.

The imports have four distinct roles.

1. *Finite-input universality.* The note `neary_woods_explicit_universal_tm.md` supplies the U15 table and effective finite-frame initialization of a selected one-tape semidecider through the bi-tag construction. The primary Neary–Woods paper is frozen and Table 16 was checked visually. This is the finite-tape machine convention, not an infinite periodic background.
2. *Corrected source compilation.* The corrected `grill_tag_halt_bridge.md` supplies the E/L/R compiler, persistent phase alignment, nonhalting simulation, and exact first-empty cleanup. Its hypotheses are proved for the newly literal normalized source in Sections 23.4–23.6.
3. *Complete native history.* The word-closure, weak-cone, phase-sharing, phase-residual206, composed205, and uniform affine-pair unit notes supply the fixed-arity native relation, positive soundness and converse, restoration, and polynomial rewrites. The scalable emitter is proved identical to the pinned complete native polynomial over *all integer tuples*, not just on valid histories.
4. *Complete binary recoding.* The input-dilation130 source and the generic-width bridge, with their complete parent geometry/AND theorem chain, supply all 34 comparisons and 49 positive auxiliaries at every fixed width $k \geq 4$. The positive converse applies at every permitted exponent $h \geq 2$. The duration-extraction argument uses the unrestricted recoder; it does not import the later extra dyadic-duration predicate.

[write] All imported notes are files of the research tree. Twenty-two of them were shipped inside archive 11 as byte copies (24 members, two of them duplicated under `frozen/arithmetic/`); they are not shipped here and are cited by their repository paths, which the README lists, for example `grill_tag_halt_bridge.md`, `neary_woods_explicit_universal_tm.md` (whose copy in the archive differs from the research-tree file in one path-sanitized command line and is therefore shipped, as `11-grill-poly-input-audit-sources-neary_woods_explicit_universal_tm.md`), `native_binary_input_dilation130.json` (the “complete recoder receipt” below) and `pcp_affine_slope_class_history.md`. The pin `2d887f0fa` descends from the `halt-bridge` commit `8ae0b7c1f` and the `exact-width-loader` commit `1dd95f928`. Every one of the 24 copies equals the current research-tree file, so the pinned bytes are still the tree’s bytes at this write.

Two small literal data objects anchor native transcription. The complete recoder receipt has SHA-256

175c498990a8e63de7a91d1e3d321ef90a19f129f305074f0c6de10967d0a182,

The 67-row native unit kernel has SHA-256

2d88343037c08fe8b073f67dca531ee73309c0735c1d98a23c20b9ddb1eb08ae.

Four complete upstream native fixtures were authenticated and their frozen rows compared independently. Upstream Python is retained only as inert source evidence; none was executed in this work or is required by the offline replay.

The archive retains earlier semantic and local audit reports with their original historical scope. Statements in those reports that a larger circuit remained to be emitted describe that earlier stage. The complete composition proof, current whole-circuit audit, and this report state the final scope.

23.3 A concrete finite U15 input language

Use $c = 0$, $b = 1$, and blank zero outside a finite tape. Refine a U15 control state u and already-read symbol r to

$$i = 2(u - 1) + r \in \{0, \dots, 29\}.$$

The initial refined state is $0 = (u_1, 0)$; the accepting state is $19 = (u_{10}, 1)$. The unique absent instruction in primary Table 16 is (u_{10}, b) . A primary prose sentence instead names c ; the literal construction follows the table, and the discrepancy is recorded rather than silently reconciled.

For a nonhalting instruction $(u, r) \mapsto (v, \epsilon, D)$, the refined machine writes ϵ , moves in direction D , reads r' , and selects $j_{r'} = 2(v - 1) + r'$. There are 15 left-moving and 14 right-moving nonhalting instructions.

Let M, N be the left and right binary tape integers, with the square nearest the head as the least significant bit. The scanned symbol is omitted because the refined control remembers it. A right move changes

$$(i, M, N) \mapsto (j_{N \bmod 2}, 2M + \epsilon, \lfloor N/2 \rfloor),$$

and a left move changes

$$(i, M, N) \mapsto (j_{M \bmod 2}, \lfloor M/2 \rfloor, 2N + \epsilon).$$

These formulas include zero tape integers and indefinitely blank tails.

23.3.1 The chosen semidecider and physical input blocks

For each $S \subseteq \mathbb{Z}_{>0}$ that is computably enumerable, choose a finite semidecider that reads the canonical LSB-first spelling of $u = x + 1$, decodes it, subtracts one, and recognizes S . This is a fixed program choice. It is not an uncharged external algorithm applied anew to x .

Write the physical binary blocks

$$A_i = (01)^{8i-5}11, \quad B_0 = A_2A_1, \quad B_1 = A_1A_2.$$

Both B_0, B_1 have 32 bits. The U15 head scans the final c of the start marker bc . Its left word is the finite encoded program followed by b ; define M_e by reversing that word and taking its LSB value. Its right word is

$$P_e B_{w_0} \cdots B_{w_{n-1}} S_e, \quad w_i = \text{bit}_i(x+1), \quad n = \text{bitlength}(x+1), \quad (23.1)$$

where

$$P_e = (01)^{8q_B}, \quad S_e = A_{\text{right}}A_{\text{left}}, \quad q_B \geq 2.$$

The alphabet size q_B and the two boundary indices depend only on the selected finite program. All squares beyond the displayed words are blank. Put $N_e(x) = \text{val}_{\text{LSB}}(\text{word in (23.1)})$. The suffix ends in 11, hence $N_e(x) > 0$.

23.4 Corrected tag macros and the accepting head

A deletion-2 tag step reads the first symbol, deletes two symbols, and appends that first symbol's production. The canonical word is

$$\mathcal{C}(i, M, N) = A_i \xi_i (a_i \xi_i)^M B_i \xi_i (b_i \xi_i)^N.$$

Internal names are local to the nonhalting refined state. Their numerical identifiers and every production appear in the literal packet.

23.4.1 The right macro

For an instruction writing ϵ and moving right, use the complete rules below. On internal symbols suppress the state index i .

$$\begin{array}{ll} A_i \rightarrow C \xi_i (c \xi_i)^\epsilon, & a_i \rightarrow c \xi_i c \xi_i, \\ B_i \rightarrow S, & b_i \rightarrow s, \\ C \rightarrow D_1 D_0, & c \rightarrow d_1 d_0, \\ S \rightarrow T_1 T_0, & s \rightarrow t_1 t_0, \\ D_1 \rightarrow A_{j_1} \xi_{j_1}, & d_1 \rightarrow a_{j_1} \xi_{j_1}, \\ D_0 \rightarrow \xi_{j_0} A_{j_0} \xi_{j_0}, & d_0 \rightarrow a_{j_0} \xi_{j_0}, \\ T_1 \rightarrow B_{j_1} \xi_{j_1}, & t_1 \rightarrow b_{j_1} \xi_{j_1}, \\ T_0 \rightarrow B_{j_0} \xi_{j_0}, & t_0 \rightarrow b_{j_0} \xi_{j_0}. \end{array}$$

Put $M' = 2M + \epsilon$. Exact intermediate queue cuts are

$$\begin{aligned} & B_i \xi_i (b_i \xi_i)^N C \xi_i (c \xi_i)^{M'}, \\ & C \xi_i (c \xi_i)^{M'} S s^N, \\ & S s^N D_1 D_0 (d_1 d_0)^{M'}. \end{aligned}$$

If $N = 2k + 1$, the shrinking traversal leaves

$$D_1 D_0 (d_1 d_0)^{M'} T_1 T_0 (t_1 t_0)^k,$$

which rotates through the D and T blocks to exactly $\mathcal{C}(j_1, M', k)$. If $N = 2k$, its last deletion consumes D_1 , leaving

$$D_0 (d_1 d_0)^{M'} T_1 T_0 (t_1 t_0)^k.$$

The D_0/d_0 traversal consumes T_1 in its last deletion, including when $M' = 0$, and gives

$$T_0 (t_1 t_0)^k \xi_{j_0} A_{j_0} \xi_{j_0} (a_{j_0} \xi_{j_0})^{M'}.$$

The last T_0/t_0 deletion consumes the extra filler and yields precisely $\mathcal{C}(j_0, M', k)$, including $k = 0$.

The final filler in $t_0 \rightarrow b_{j_0} \xi_{j_0}$ is deliberate. The inspected printed Cocke–Minsky production omits it although the next displayed target retains paired repetitions. With $M = 0, N = 2, \epsilon = 0$, the printed rule produces a five-symbol target after ten steps, missing its sixth filler. The construction here proves its own corrected rule; it does not present an asserted published erratum.

23.4.2 The left macro and explicit rotation

For a left move replace the initial four right rules by

$$\begin{array}{ll} A_i \rightarrow L_i \xi_i, & a_i \rightarrow l_i \xi_i, \\ B_i \rightarrow C \xi_i (c \xi_i)^\epsilon, & b_i \rightarrow c \xi_i c \xi_i, \\ L_i \rightarrow S, & l_i \rightarrow s. \end{array}$$

One traversal gives

$$L_i \xi_i (l_i \xi_i)^M C \xi_i (c \xi_i)^{2N+\epsilon}.$$

Keep the $C/c/S/s$ preparation. Replace target A_j, a_j in the D/d rules by temporary B'_j, b'_j , and replace target B_j, b_j in the T/t rules by actual A_j, a_j . Add

$$B'_j \rightarrow B_j \xi_j, \quad b'_j \rightarrow b_j \xi_j.$$

The right parity calculation with the tape roles exchanged yields

$$B'_j \xi_j (b'_j \xi_j)^{2N+\epsilon} A_j \xi_j (a_j \xi_j)^{\lfloor M/2 \rfloor}, \quad j = j_M \bmod 2.$$

Moving the leading prime block to the tail gives the exact canonical left-move target. This finite rotation is the missing content behind an informal appeal to left/right symmetry.

23.4.3 Safety before acceptance and uniqueness at acceptance

Every canonical word has at least four symbols. All preacceptance productions except the shrinking B_i/b_i or L_i/l_i rules have length at least two. During the shrinking phase an untouched $C \xi_i (c \xi_i)^{M'}$ block of length at least two remains, and an S remains at the finish. Therefore every preacceptance intermediate queue has length at least three. Prime rotation preserves length. No valid rejecting run can halt by an accidental short queue.

The exact cuts create one next canonical head. It is read only after all old-macro symbols preceding it are consumed; at that read the whole word is canonical. In particular the first active A_{19} occurs at a genuine accepting boundary, with no delayed old head lineage elsewhere in the queue.

Set $A_{19} \rightarrow H$, with a fresh halt symbol H . Give the accepting suffix symbols inert productions ZZ , and $Z \rightarrow ZZ$. No other nonhalting rule emits H . Once A_{19} is read, the remaining accepting-state symbols cannot recreate either A_{19} or H . Thus exactly one fresh H is emitted. An occurrence in the second deleted position is never mistaken for an active head event.

23.5 Normalization with a persistent phase

Give each original tag symbol width one. Its unnormalized Genera production is its tag production at phase zero and empty at phase one. Two FIFO reads implement one deletion-2 step. The phase is persistent across generation boundaries, so a generation ending after an odd number of original reads is not reset.

Introduce a width-zero dummy d , with $d \rightarrow dd$. Pad each production to four symbols using trailing d , and replace its two consecutive pairs by fresh width-zero pair symbols. A pair $[u, v]$ expands to uv at either phase. Every normalized production now has exactly two outputs.

Lemma 23.5.1 (Two-layer projection). *Starting in an original-symbol layer, erasing dummies after two normalized generations gives exactly one unnormalized generation, with exactly its final persistent phase.*

Proof. Only original symbols have width one, so only the first pass changes the phase. Its outputs are pairs and width-zero dummies. The second pass expands each pair, preserves order, and leaves the phase unchanged. Dummy expansion erases to the empty word. Applying these facts successively proves both the word and phase identities, also for odd erased word length. $\square$

A first literal H occurs only when its containing pair expands. The preceding prefix of that output contains original symbols and dummies. Their hypothetical next outputs contain only pairs and dummies, never H . This verifies the extra prefix condition in the Genera halt convention, not merely the existence of a halt symbol.

The literal normalized alphabet has 569 nonhalting original symbols, 442 ordered pairs, one dummy, and one halt symbol, for $N_G = 1013$. There are 2,024 nonhalting phase rows, each with two outputs; the two ignored halt rows are also total. Its unique halt-containing pair is (H, d) , numeric ID 694; the halt has ID 1012. Only phase-zero A_{19} produces that pair. The complete literal table has SHA-256

3c8924dbb1b5d6e6b8897e59550b0e38a703654b87442c73487f411e94ae355b.

23.6 The literal corrected Grill program

Set

$$a = 28(N_G + 1) = 28392, \quad b = 7a = 198744, \quad m = 14a = 397488.$$

At one run step a zero is removed without output; a one is removed and appends $g(n_p) = 0(10)^{n_p}$; the phase advances modulo m . An exponent n_p denotes n_p commands 11 followed by command 10. All source times below count run steps.

For a nonhalting source symbol y of width $w_y \in \{0, 1\}$ put $e_y = 7a(1 - w_y)$, and put $e_H = 3a/2$. The corrected blocks are

$$\begin{aligned} E_y &= 0^{14y+7}g(a-3)g(7)g(a-4)0^{3a-14y-10+e_y}, \\ L_y &= 0^{a-3}g(14y+7)g(3)g(3a-14y-10)(10)^{e_y}, \\ R_y &= 0g(14y+7)g(3)g(3a-14y-10+e_y)0^{a-4}. \end{aligned}$$

For $y = H$, the displayed E_H is only a cleanup intermediate E_H^* , not an initial source block. The second long E grill uses $a - 4$, correcting the earlier informal $a - 2$ expression.

23.6.1 Every run exponent is specified

Use zero-based positions relative to the start $7ap$ of each half $p \in \{0, 1\}$ of length $7a$. Place the values $a - 3, 7, a - 4$ in the listed order at

$$28y + \{17, 19, 21\} \quad \text{and} \quad 28y + a + \{13, 15, 17\},$$

respectively, for every symbol y , including H . If nonhalt y outputs (u, v) at source phase p , place

$$14u + 7, 3, 3a - 14u - 10 + e_u, 0, 14v + 7, 3, 3a - 14v - 10 + e_v$$

at the seven positions $14y + 2a + \{3, 5, 7, 9, 11, 13, 15\}$, in that order. All other exponents are zero. There are no E-production positions for H . The source rows therefore determine the entire little-endian uint32 array; its hash is

fa658fcdfe1dae2be3a8e2bf97bd613db59558949ea23ca03f549b77201dad01.

Independent reconstruction agrees at every entry and all 1,589,952 bytes.

23.6.2 Phase alignment and exact cleanup

At a normal start phase $7ap$, a nonhalt E block produces the appropriate $L_u R_v$. Its phase advance is $7aw_y \pmod{14a}$. A nonhalt L/R block has length $7a$ or $21a$, both congruent to $7a$ modulo $14a$. A complete two-output LR generation is phase-neutral. These facts preserve the source's persistent phase.

Every active run position modulo $7a$ is odd and below $5a/2$. Translating it by $3a$ lands in the zero-run region. At a start phase shifted by $3a$, every one in an E/L/R block therefore selects exponent zero, producing one zero, while every zero produces nothing.

Suppose the first halting source output is an even word v with its unique H at zero-based index j . The H block has length $10a$ and shifts all later blocks by $3a$. The three remaining generation lengths from that LR boundary are

$$\begin{aligned} L_1 &= 10a + \sum_{i \neq j} [7a + 14a(1 - w_{v_i})], \\ Z &= \sum_{i > j} [3a + 7a(1 - w_{v_i})], \\ L_2 &= 17a/2 + \sum_{i < j} [7a + 7a(1 - w_{v_i})] + Z, \\ L_3 &= 2a(j + 1). \end{aligned}$$

The last produced word is all zeros. All three lengths are positive, and a FIFO queue cannot empty while an initial-generation suffix remains. Thus $L_1 + L_2 + L_3$ is the exact first-empty time from that boundary. For the actual H-producing pair, the lengths are 880,152, 525,252, and 56,784, totaling 1,462,188 further runs.

Nonhalting E boundaries stay nonempty. Consequently the correctly normalized source halts if and only if the Grill queue from its exact E word empties. This equivalence is at the literal width. Allowing arbitrary high-zero padding of the same numeral would be a different and unjustified interface.

[write] *Re-statement.* The corrected E/L/R compiler, the persistent phase alignment, the $3a$ cleanup shift and the exact first-empty lengths L_1, L_2, L_3 are those of the research tree's `grill_tag_halt_bridge.md`

(commit 8ae0b7c1f), imported in Section 23.2; its $a - 2 \rightarrow a - 4$ correction of the E encoding is the tree's `review_grill_encoding_e.md`. New here are the literal 397,488-entry table and its instantiated totals 880,152, 525,252 and 56,784. The tree's halt bridge itself does not give an ordinary-input universal Grill recognizer; this Part supplies that composition.

23.7 All five valid program parameters

The first three parameters record the finite U15 frame:

$$a_e = \text{val}_{\text{LSB}}(\mathbf{P}_e) = \frac{2(4^{8q_B} - 1)}{3}, \quad p_e = 2^{16q_B}, \quad s_e = \text{val}_{\text{LSB}}(\mathbf{S}_e). \quad (23.2)$$

Each is positive and fixed once the finite program is chosen.

For original symbols define $E_y^\# = \text{val}_{\text{LSB}}(E_y)$. Let

$$\begin{aligned} \gamma(t) &= 2 \sum_{j=0}^{t-1} 4^j = \frac{2(4^t - 1)}{3}, \\ A_* &= 2^7[\gamma(a - 3) + 2^{2a-5}\gamma(7) + 2^{2a+10}\gamma(a - 4)]. \end{aligned}$$

Then $E_y^\# = A_* 2^{14y}$ and every original block has width b . Put

$$K = 2^{2b}, \quad W_y = E_y^\# + 2^b E_{539}^\#.$$

The literal start IDs are $A_0 = 0$, $a_0 = 270$, $B_0 = 30$, $b_0 = 300$, and $\xi_0 = 539$. The remaining parameters are

$$\begin{aligned} C_e &= W_0 + KW_{270} \sum_{j=0}^{M_e-1} K^j + K^{M_e+1} W_{30}, \\ L_e &= K^{M_e+2}. \end{aligned} \quad (23.3)$$

The empty sum when $M_e = 0$ is zero. These are the value and dyadic width scale of

$$E(A_0 \xi_0 (a_0 \xi_0)^{M_e} B_0 \xi_0).$$

Equations (23.2)–(23.3) are finite recipes for fixed program integers. Their powers and finite sums are not runtime operations of the universal polynomial. By contrast, every multiplication involving any of these five *external coordinates* in the universal DAG is charged.

The full initial tag word is the displayed fixed prefix followed by $(b_0 \xi_0)^{N_e(x)}$. That repetition count varies with the ordinary input and must be paid inside the polynomial; it is the task of the second recoder.

23.8 Two complete recoders and the signed frame

The complete generic recoder at fixed width $k \geq 4$, on positive input $z < 2^h$ and exponent $h \geq 2$, retains 49 positive auxiliaries and 34 comparison pairs. At a complete zero its ports satisfy

$$\begin{aligned} q &= 2^h, \quad Q = q^k, \quad B = 2^{k-1}Q, \quad P = B^h, \\ J &= \sum_{j < h} B^j, \quad K_{\text{mask}} = \sum_{j < h} (2B)^j, \quad z_{\text{out}} = \sum_{j < h} \text{bit}_j(z) 2^{kj}. \end{aligned} \quad (23.4)$$

The full converse supplies every retained positive auxiliary at every such exponent. Selected outer equations alone would not justify this interface.

Typing does not depend circularly on the conclusions. Before the recoder equalities, $B = 2^{k-1}q^k \geq 8q^2$ for positive q and $k \geq 4$, and the conceptual prescribed-scale AND ports $zJ + 1$, $K_{\text{mask}} + 1$, its positive supplied output hat, and qP are positive. Generic width changes only the fixed power chain and coefficient in the complete 130-row source; all remaining comparisons and witnesses remain.

23.8.1 Canonical input with an inclusive upper endpoint

Pay $u = x + 1$ and apply the width-32 recoder with positive spread output R . Its existing positive slack s_0 satisfies $u + s_0 = q_0$. Supply positive β and impose

$$s_0 + \beta = u + 1.$$

Then $u < q_0 \leq 2u$. Since q_0 is dyadic, it is exactly 2^n for $n = \text{bitlength}(u)$. For dyadic u the upper endpoint is necessary and $\beta = 1$. Conversely, $s_0 = 2^n - u > 0$ and $\beta = 2u + 1 - 2^n > 0$ give the required extension. As $x > 0$, $u \geq 2$, so $n \geq 2$ is in the imported domain.

With $K_0 = 2^{32}$ and $m_0 = K_0 - 1$, the literal block values are

$$c_0 = 3941247658, \quad c_1 = 3937053418, \quad c_1 - c_0 = -4194240.$$

Supply positive N and impose the denominator-cleared equation

$$\begin{aligned} m_0 N &= m_0 a_e + p_e c_0 (Q_0 - 1) \\ &+ p_e m_0 (c_1 - c_0) R + p_e m_0 s_e Q_0. \end{aligned} \tag{23.5}$$

The recoder gives $Q_0 = K_0^n$ and $R = \sum_{i < n} w_i K_0^i$. Hence (23.5) is exactly the LSB concatenation formula for (23.1). The signed coefficient is retained; it must not be changed to a presumed positive block difference. N is a positive supplied witness, so the off-zero sign of the computed right side does not become an illicit native positivity assumption.

23.8.2 An unrestricted exponent equal to the tape integer plus one

Use a disjoint second recoder at fixed width $k = 2b = 397488$, with input and spread output both equal to one. Its theorem gives $h \geq 2$ and (23.4). It receives no canonical-length guard and no dyadic-duration AND predicate. The two namespaces share no private witness.

Supply positive ℓ, v, g and impose

$$(B - 1)v + \ell = J, \quad \ell + g = B - 1, \quad \ell = N + 1. \tag{23.6}$$

Since $J \equiv h \pmod{B - 1}$, the first equation gives $\ell \equiv h$. The second bounds $1 \leq \ell \leq B - 2$. Also

$$B = 2^{kh+k-1} \geq 2^{4h+3} > h + 1,$$

so $2 \leq h \leq B - 2$. The representatives lie in one modulus interval and are equal: $h = \ell = N + 1$. No modular alias survives.

For the converse choose the full recoder extension at $h = N + 1$ and take

$$v = \sum_{j=1}^{h-1} \sum_{i=0}^{j-1} B^i > 0, \quad g = B - 1 - h > 0, \quad \ell = h.$$

The case $N = 1, h = 2$ is already valid; the inner quotient is positive. $N = 0$ would require the excluded $h = 1$, but the physical suffix has proved $N > 0$. Nondyadic exponents are allowed and are essential.

23.9 Exact unary loading and positive width

The fixed repeated pair has value

$$V = E_{300}^{\#} + 2^b E_{539}^{\#} = A_* 2^{14 \cdot 300} + A_* 2^{b+14 \cdot 539}$$

and exact width $k = 397488$. Its value has 319,867 bits. Those two numbers are different because its terminal zeros are part of the word.

The second recoder has $Q_1 = K^{N+1}$. Supply positive T, X and impose

$$KT = Q_1, \tag{23.7}$$

$$(K - 1)X = (K - 1)C_e + L_e V(T - 1),$$

$$P_0 = L_e T. \tag{23.8}$$

The first equation gives $T = K^N$. The second uniquely gives

$$X = C_e + L_e V \sum_{j=0}^{N-1} K^j,$$

precisely the numeral of the full initial E word. Equation (23.8) fixes its exact dyadic width scale.

The native source already has a positive width slack Z_0 and computes $P_0 = X + Z_0$. This coordinate is retained, and (23.8) is an additional comparison. Substituting an arbitrary signed computed difference for a positive input would not be a valid replacement.

Every original E block is nonzero and ends with at least two zero bits. In base 2^b , each block value is below one quarter of that base. A finite geometric sum therefore gives $0 < 3X < P_0$ for the exact word. On a genuine halt the strong native converse applies at that same width, with positive slack $P_0 - 3X$. The established strong-to-weak map yields $Z_0 = P_0 - X > 0$ while preserving the word, width, and history. No optional padding is used.

23.10 A complete scalable native history

The complete native component is much larger than the loader. Its correctness transfer must therefore cover its entire source, not just a few ports or small instances. The accompanying `native_history_proof.md` proves the scalable emitter identically equal to the pinned complete weak native polynomial after a bijective coordinate renaming. The following formulas expose the main identities.

[write] *Re-certification*. The native history polynomial of this section is the research tree's word-closure, weak-cone, phase-sharing, phase-residual206 and composed205 polynomial for a fixed program; the new content is a scalable emitter proved identical to it over all integer tuples. The tree states for its own small programs that “no ordinary-input decoder or fixed universal Grill recognizer has been supplied” (`grill_tag_native_weak_cone.md`) and that its word closure “supplies no fixed universal Grill program” (`review_grill_native_word_closure.md`).

23.10.1 Coordinates and grouped affine transports

For a fixed program $n_0, \dots, n_{m-1}$, let $s = 2m$ and let g be the number of distinct exponents. The two tiles at phase i are

$$(2, 0, 1, 0), \quad (2, 1, b(n_i), d(n_i)),$$

where $b(n) = 2^{2n+1}$ and $d(n) = 2(4^n - 1)/3$. These are fixed integer coefficients. Distinct exponent classes retain separate selected-history coordinates, while every original chronological selector remains present.

There are s positive selector hats $\hat{S}_i$, g positive history hats $\hat{Z}_j$, seven positive outer coordinates, and sixteen positive native-kernel coordinates. Thus the native count is $s + g + 23$. For the literal program, $s = 794976$ and $g = 2030$, giving 797,029.

Let $I_j = \{2i + 1 : n_i = e_j\}$, with groups ordered by first occurrence, and put

$$\sigma_i = \hat{S}_i - 1, \quad G_j = \sum_{i \in I_j} \sigma_i, \quad z_j = \hat{Z}_j - 1, \quad J = \sum_i \sigma_i.$$

To avoid collision with the loader's K, B , write the native radix multiplier as $\mathcal{K}$ and radix as $\mathcal{B}$. Its exact fixed value is

$$\mathcal{K} = 2^{\max(3, \lceil \log_2(s+4) \rceil, 2 \max n_i + 2)} = 2^{551890}.$$

The implementation computes the integer ceiling by bit length, without floating point. Define

$$\begin{aligned} P_0 &= X + Z_0, & U_f &= P_0 V_f + X, \\ D &= U_f + q_{\text{init}} + h_{\text{slack}}, & \mathcal{B} &= \mathcal{K} D, & P &= (\mathcal{B} - 1)J + 1. \end{aligned}$$

Grouped transports are

$$\begin{aligned} \text{next } U &= 2H_U + \sum_j G_j, \\ \text{next } V &= H_V + \sum_j (b(e_j) - 1)z_j + \sum_j d(e_j)G_j. \end{aligned}$$

Distributivity gives exactly the ungrouped template. The first three outer comparisons are

$$\begin{aligned} H_U + H_V + \sum_j \hat{Z}_j + g_{\text{bound}} &= P, \\ \mathcal{B} \text{ next } U + 1 &= H_U + P U_f, \\ \mathcal{B} \text{ next } V + 1 &= H_V + P V_f. \end{aligned}$$

Positive hats in the first equality are intentional; they implement the precise global bound, including its g offset.

23.10.2 The exact phase residual

Let $\pi_i = \hat{S}_{2i} + \hat{S}_{2i+1}$, $S_i = \pi_i - 2$, and

$$T_{\text{ph}} = \sum_{i=1}^{m-1} (m-i)S_i, \quad J = \sum_i S_i.$$

The streamed nested prefix sum gives coefficient $m-i$ on π_i ; subtracting $m(m-1)$ removes its hat offsets. Every prefix and accumulator addition is an emitted gate.

For proof purposes write $\text{Next} = mJ - T_{\text{ph}}$ and $Q = (m + 1)J - mS_0 - T_{\text{ph}}$. Substitution yields the polynomial identity

$$\begin{aligned} \mathcal{B}\text{Next} + q_{\text{init}} - Q - mP \\ = m\pi_0 - (\mathcal{B} - 1)T_{\text{ph}} + q_{\text{init}} - (J + 3m). \end{aligned}$$

This is an all-tuple identity over a commutative ring. It assumes neither Boolean selectors nor a valid phase path nor the vanishing of another residual. For $m = 1$ it reduces to $q_{\text{init}} - 1$.

23.10.3 Exact packed ports and the fixed kernel

Use $\text{pack}(v) = \sum_i v_i P^i$ and $\text{rep}(r) = \sum_{i < r} P^i$. Horner packing and the paid identities

$$\text{rep}(2r) = \text{rep}(r)(1 + P^r), \quad \text{rep}(2r + 1) = \text{rep}(2r) + P^{2r}$$

are polynomial identities also at $P = 0$ and $P = 1$. All exponents are fixed by the literal program; square-and-multiply emits ordinary paid products.

Define

$$\begin{aligned} S &= \text{pack}(\sigma), \quad M_c = J \text{rep}(s), \quad T_c = H_U + PH_V, \\ G_* &= \text{pack}(G), \quad H_b = H_V \text{rep}(g), \quad M_b = (\mathcal{B} - 1)G_*, \quad Z_b = \text{pack}(z), \\ R_M &= (D - 1)J(1 + P). \end{aligned}$$

With $e_c = g$, $e_r = g + s$, $e_t = g + s + 2$, $e_i = g + s + 4$, put

$$\begin{aligned} W &= P^{e_c} S + P^{e_r} T_c, \\ H &= H_b + W + P^{e_t} \mathcal{B} + P^{e_i} P_0, \\ M &= M_b + P^{e_c} M_c + P^{e_r} R_M + P^{e_t} (\mathcal{B} - 1) + P^{e_i} (P_0 - 1), \\ Z &= Z_b + W, \quad \text{scale} = P^{g+s+5}. \end{aligned}$$

These are the exact controller, range, top/radix, and initial-width lanes of the pinned native adapter. Here the scale exponent is 797,011.

The frozen 67-row kernel receives exactly (H, M, Z, scale) . It retains sixteen positive coordinates, all six native comparison pairs, and a four-factor integer unit

$$U = R_{15} P_{17} \text{first_unit bs}_q.$$

Together with the four outer comparisons this gives ten nonunit native comparisons. In particular the strong norm-coefficient equality is retained. Induction over the frozen rows, combined with the displayed port identities, proves equality of every kernel register, residual, unit factor, and complete native polynomial on all integer tuples.

23.10.4 The semantic import at its positive domain

On supplied positive coordinates and $X > 0$, $P_0 \geq 2$, $D \geq 5$, $\mathcal{B} > D$, and the decoded hats are nonnegative. At a complete zero the global equality forces $P > 1$, $J > 0$ and the history bounds. Native classification gives dyadic scale and width and separates the lanes. The retained transports and phase relation recover a common reverse affine history. Its endpoint yields the word closure, hence an actual halt at or before the proposed closure length.

Conversely a genuine finite halt at the specified admissible P_0 supplies reversed sentinel histories and selector hats. Choose an adequate dyadic height, pack at $\mathcal{B} = \mathcal{K}D$ and $P = \mathcal{B}^t$, and use the positive global-bound estimate from the complete theorem. The prescribed-scale native AND converse supplies its full positive witnesses; the unit projection supplies all sixteen retained coordinates.

A native closure may include a post-halt suffix. Soundness gives a first actual halt no later than the closure length; it does not claim legal physical execution after an empty queue. The separate Grill cleanup theorem identifies the actual first-empty boundary. There is no externally imposed time cutoff.

23.11 One unit-safe finalizer and the full proof

The complete residual list is shown in Table 4. It includes both full recoders, all loader constraints, all native comparisons, and the exact-width equality once.

Source of nonunit residuals	Number
Complete native history	10
Canonical width-32 recoder	34
Canonical length guard and signed frame	2
Unrestricted width-397488 recoder	34
Exponent extraction and binding	3
Unary scale, unary value, exact native width	3
Total	86

Table 4: All 86 residuals occur inside the same unit product.

The emitted polynomial is exactly

$$F = U \left(1 + \sum_{j=0}^{85} R_j^2 \right) - 1. \quad (23.9)$$

For any integer tuple, the second factor is a positive integer. If $F = 0$, it divides one, so it equals one, every $R_j = 0$, and $U = 1$. Conversely those conditions give $F = 0$. This argument does not assume that U or a previously finalized native polynomial is nonnegative away from its zero set. Adding an unweighted external sum of squares to a signed native polynomial would not provide the same implication.

Proof of Theorem 23.1.1. Fix a valid θ_S from Section 23.7. Suppose F has a positive zero at $x > 0$. The finalizer restores every comparison and the native unit. The first recoder fixes the canonical bits of $x + 1$ and the frame equation fixes $N = N_e(x)$. The second recoder and the bounded residue extraction fix $h = N + 1$, hence $T = K^N$. The unary equation and width comparison identify the exact literal E word, with no independent padding choice. Native soundness implies a finite Grill halt. The corrected halt bridge, phase-preserving normalization, and accepting-head invariant imply U15 acceptance. The selected finite program recognizes S on the decoded input x , so $x \in S$.

Conversely let $x \in S$. Its selected U15 computation accepts. The corrected tag macros avoid premature short queues and emit one halt from the accepting head. Normalization satisfies the source halt convention, and the corrected Grill bridge gives a genuine finite first empty time from the exact E word. Choose the full first recoder extension at $n = \text{bitlength}(x + 1)$ and its positive guard. Set the positive genuine N , then choose the full second extension at $h = N + 1$ and the positive ℓ, v, g in

Section 23.8. Set $T = K^N$ and the literal positive X . The E-word margin permits the strong native converse at the same width and its positive transfer to the retained weak slack. All private witness sets are disjoint except for the deliberately shared ports. Thus every added coordinate can be positive simultaneously, every residual vanishes, and $U = 1$. Equation (23.9) vanishes. $\square$

23.12 What the arithmetic ledger counts

The source is a straight-line DAG with binary addition, subtraction, and multiplication over integers. Reading an exact literal coefficient is free in this model, but multiplying a runtime register by any nontrivial coefficient is an emitted multiplication. A fixed-exponent runtime power is emitted as binary multiplication gates. There are no runtime division or variable-exponentiation gates.

Stage	Multiplications	Additive gates	Total
Two full recoders and loader	173	137	310
Native history before finalizer	803,257	2,796,719	3,599,976
One finalizer	87	173	260
Complete polynomial	803,517	2,797,029	3,600,546

Table 5: Measured operations in the actual emitted source, not an inherited example.

The two recoders contribute 98 private positive auxiliaries. The first spread output and the seven added loader coordinates $\beta, N, \ell, v, g, T, X$ contribute eight, for 106 new positive witnesses. Added to the 797,029 native witnesses, this gives 797,135. The six external positive coordinates are separate.

The width-32 recoder uses 133 operations. The width-397488 recoder on input/output one uses 150; its one multiplication $1 \cdot J$ is an alias, with all consumers and comparisons retained. The other loader operations total 27. The finalizer consists of 86 differences, 86 squares, 85 sum additions, addition of one, multiplication by U , and subtraction of one. All 3,600,546 emitted gates and every supplied coordinate reach the output.

23.12.1 Typed constant recipes

The manifest has 8,160 interned constant recipes. The allowed constructors are a decimal integer, a fixed 2^t , a fixed finite geometric sum $(4^t - 1)/3$, and addition, subtraction, or multiplication of earlier constant handles. The exponent t is a fixed nonnegative source integer. A recipe cannot refer to an input or gate. These rules define ordinary finite integer coefficients; they do not smuggle in a witness-dependent power or unproved division.

The largest exact fixed coefficient in the audited pool has 551,891 bits. Constant interning saves repeated storage. It neither removes runtime scalar multiplications nor reduces quantified coordinates. Some interned recipe rows are unused coefficient-construction remnants; they are not arithmetic gates, and do not contradict full gate and input liveness.

23.12.2 Degree and resource scope

Ordinary degree propagation gives $d(A \pm B) \leq \max(d(A), d(B))$ and $d(AB) \leq d(A) + d(B)$, starting every external coordinate and witness at degree one. It produced the valid historical bound 71,731,007. Two initial highest-form probes returned zero and did not certify attainment. The cancellation-aware

analysis in Section 23.13 now proves exact total degree 69,339,973 for the very same source. No gate, coefficient, coordinate, or comparison changes.

The DAG occupies 61,209,290 bytes, including its eight-byte header. Its SHA-256 is

a07c1ba41e3a18fafd05c19b8ac39211b0475e30ed8a08ddf43c45a9f5f440f2.

The recorded full emission and backwards liveness audit took 5.47 seconds and 107,515,904 bytes peak RSS. Limits were six million gates, 200,000 constants, 600 seconds, and 1,800 MiB RSS; none was reached. These measurements describe one execution of a streamed construction, not evaluation of a huge positive solution or a performance guarantee.

Earlier 205-operation examples, and the 16,291-operation exact-width illustration with its distinct 16,289-operation factored successor, describe different sources. None supplies the present ledger. The current all-tuple identity proof transfers mathematics between schedules; it does not identify their operation counts.

[write] *Stale bytes in two shipped files.* The frozen reproducibility notes of the delivery keep the original upper-bound wording byte for byte, as 11-grill-poly-REVISION.md explains: 11-grill-poly-README.md (delivered reproducibility/README.md) says that 71,731,007 “is an upper bound, not a proven exact degree”, 11-grill-poly-arithmetic-README.md says “no exact-degree claim”, and the manifest field `degree_upper` keeps 71,731,007. Section 23.13 (revision 1) proves the exact degree 69,339,973; the upper bound remains valid.

23.13 The exact total degree of the unchanged polynomial

The initial edition reported the sound propagated upper bound 71,731,007. A cancellation in the first native norm factor explains why that number was not attained. Resolving this cancellation in the *analysis* of the source gives exact degree 69,339,973. The arithmetic DAG, its coefficients, all comparisons, and all supplied coordinates are unchanged. This is not a new circuit or a replacement polynomial with the same zero set.

Theorem 23.13.1 (Exact degree of the pinned source). *Assign degree one to all 797,141 supplied coordinates of the frozen DAG, including its six external coordinates. The integer polynomial computed by that source has total degree exactly*

$$\deg F = 69,339,973.$$

This assertion is unconditional algebra on the pinned source. It uses neither witness positivity nor a residual equality, and it does not assume any native-history or universality theorem.

The main DAG digest is the one in Section 23.12. The unchanged arithmetic manifest has SHA-256

41e6f754df8f60448e1207ef36e6161729b52004fac719a580d6ec643d039d49.

The kernel digest is the one in Section 23.2. Both new checkers authenticate these three inputs before reading the complete source. Gate and input indices in this section are zero based.

23.13.1 The first norm cancels identically

For local notation in this section, let u, a, c, γ, d denote these actual registers:

Symbol	Native register	Source location
u	<code>and__wn2</code>	gate 3,600,220
a	<code>and__R12</code>	gate 3,600,228
c	<code>and__R10a</code>	gate 3,600,226
γ	<code>and__ga</code>	input 797,137
$d = 4a + 3$	<code>and__a4m5</code>	gate 3,600,232

These local u, a are not the earlier shifted ordinary input or the source alphabet parameter. The first unit factor R_{15} , gate 3,600,240, is

$$R_{15} = (u + ac + \gamma d)^2 - (a^2 + d)c^2.$$

Expanding in four independent indeterminates u, a, c, γ and substituting $d = 4a + 3$ gives the identity

$$R_{15} = u^2 + 2uac + 2u\gamma d + 2ac\gamma d + \gamma^2 d^2 - dc^2. \quad (23.10)$$

The two a^2c^2 terms cancel. There is no appeal to a norm equation, a valid history, or any zero-set restriction. Independent sparse expansion of both sides gives identical coefficient dictionaries with eleven nonzero monomials. The check is in four abstract indeterminates, not an expansion in hundreds of thousands of coordinates.

[write] *Attribution.* Identity (23.10) is identity (4) of the research tree’s `gpcp_shared_selectors774.md` (commit `ee507599f`, 1 October 2026), written there as $D^2 - \Delta c^2 = X^2 + 2Xac + 2XG + 2acG + G^2 - (4a + 3)c^2$ with $D = X + ac + G$, $G = \gamma(4a + 3)$ and $\Delta = a^2 + 4a + 3$; replace X by u and $4a + 3$ by d . The tree also uses it in `gpcp_history_computed_fields744.md`, `gpcp_index_linear_units757.md` and `gpcp_coupled_index_units754.md`. This Part does not cite it; the identity is the tree’s, and its application to the Grill source with the exact degree that follows is new. Part II proves it again as (24.17).

Before this identity is used, both checkers match all 67 native-kernel rows to the raw DAG, including operation codes, operand handles, supplied coordinates, constants, caller ports and unit factors. Thus (23.10) analyzes the factor actually present in the frozen source.

23.13.2 The outer leading forms

For a nonzero polynomial R , denote its highest homogeneous part by R_* . Put

$$s = 794976, \quad g = 2030, \quad N_* = g + s + 5 = 797011, \quad \alpha = 3N_* = 2391033.$$

Here N_* is a fixed lane exponent, distinct from the varying tape integer N . Let

$$J_1 = \sum_{j=0}^{s-1} \widehat{S}_j, \quad \mathfrak{p} = 2^{551890}(X + Z_0)V_f J_1, \quad Q_* = 16\mathfrak{p}^{N_*}.$$

X is the supplied loader witness, not ordinary input x . From the actual definitions

$$D = (X + Z_0)V_f + X + q_{\text{init}} + h_{\text{slack}}, \quad P = (2^{551890}D - 1)(J_1 - s) + 1,$$

we have $P_* = \mathfrak{p}$ and $\deg P = 3$. Its visibly nonzero leading form has positive coefficients. The independent checker also traverses the raw selector-sum graph and checks that each of the s supplied selector hats occurs exactly once.

The scale port is P^{N*} , so the kernel's radix $q = 16$ scale has leading form Q_* and exact degree α . In the joined Z port, the uniquely highest summand is $P^{g+s+1}H_V$. Indeed the selector lane has degree $3(g+s) - 2$ and the history lane has degree $3(g+s) + 4$; the low group pack is smaller. Hence

$$Z_* = \mathfrak{p}^{N*-4}H_V, \quad \deg Z = \alpha - 11.$$

The joined H and M ports have degree at most $\alpha - 2$. These are source-polynomial statements before imposing any comparison.

Write β_* for the native input `and__odd_half`, put $b_* = 2\beta_*$, and put $k_* = \eta + \zeta$. The symbols $w, f, i, \gamma, \tau, \eta, \zeta$ denote their corresponding supplied native auxiliaries; τ is `and__tau_gap`. They are independent coordinates. The relevant kernel leading forms are

$$u_* = wQ_*, \quad (\text{sn2})_* = b_*Q_*, \quad c_* = k_*b_*Q_*, \quad a_* = wb_*Q_*^2.$$

Their exact degrees are $\alpha + 1$, $\alpha + 1$, $\alpha + 2$, and $2\alpha + 2$, respectively.

23.13.3 The degree of every unit factor

The six terms in (23.10), in the displayed order, have degree bounds

$$2\alpha + 2, \quad 4\alpha + 5, \quad 3\alpha + 4, \quad 5\alpha + 7, \quad 4\alpha + 6, \quad 4\alpha + 6.$$

The fourth is uniquely highest. Since $d_* = 4a_*$, it gives

$$(R_{15})_* = 8\gamma a_*^2 c_* = 8\gamma w^2 b_*^3 k_* Q_*^5 \neq 0.$$

Therefore $\deg R_{15} = 5\alpha + 7 = 11,955,172$. Naive max/sum propagation at the original subtraction gave $6\alpha + 8 = 14,346,206$. The precise reduction is $\alpha + 1 = 2,391,034$.

The other unit-factor calculations use the source's highest homogeneous pieces

$$(R_{16})_* = a_*^2 f^2 = w^2 b_*^2 f^2 Q_*^4, \\ (\text{bs_packed})_* = 16Q_*^3 Z_*, \quad (H_{17})_* = -32Q_*^3 Z_*.$$

Collecting their terms gives Table 6. The linear factor $\tau - k_*$ is nonzero because τ, η, ζ are independent supplied coordinates. Each displayed leading form is consequently nonzero in the unrestricted integer polynomial ring.

Factor	Highest homogeneous part	Exact degree
R_{15}	$8\gamma w^2 b_*^3 k_* Q_*^5$	$5\alpha + 7 = 11,955,172$
P_{17}	$1024w^2 b_*^2 f^2 Q_*^{10} Z_*^2$	$12\alpha - 16 = 28,692,380$
<code>first_unit</code>	$4wb_*^2 k_*(\tau - k_*)Q_*^3$	$3\alpha + 5 = 7,173,104$
<code>bs_q</code>	Q_*	$\alpha = 2,391,033$
U	Product of these four forms	$21\alpha - 4 = 50,211,689$

Table 6: Exact degrees of all four native unit factors and their product.

23.13.4 All residuals and the highest square

The ten native residual bounds, in their actual zero-based order, are

index	0	1	2	3	4	5	6	7	8	9
bound	3	5	4	3	$4\alpha - 11$	$4\alpha - 11$	$4\alpha + 10$	$4\alpha - 11$	$\alpha - 2$	$\alpha - 2$.

The next 75 residuals are the complete loader's comparisons, each of degree at most 397,489. The final exact-width residual has degree at most two. These bounds account for all 86 residuals; they are propagated from the full source, not inferred from satisfying assignments.

Residual 6 is uniquely largest. Its comparison is $\text{ic22} = R_{16}$, where $\text{ic22} = (ic^2)^2$. Therefore

$$(\text{ic22} - R_{16})_* = i^2 c_*^4, \quad \deg(\text{ic22} - R_{16}) = 4\alpha + 10 = 9,564,142.$$

Its square has leading form $i^4 c_*^8$ and degree $8\alpha + 20 = 19,128,284$. Every other residual square has a strictly smaller bound. Thus the leading form of $1 + \sum_{j=0}^{85} R_j^2$ is exactly $i^4 c_*^8$.

Since the polynomial ring over $\mathbb{Z}$ is an integral domain, degrees of products of nonzero polynomials add. Subtracting one changes no positive-degree leading term. Applying this to the unchanged finalizer yields

$$\deg F = (21\alpha - 4) + (8\alpha + 20) = 29\alpha + 16 = 69,339,973,$$

proving Theorem 23.13.1.

[write] Theorem 23.13.1 is the instance $N = N_* = 797,011$ of Part II's law (Theorem 24.1.1): $87 \cdot 797,011 + 16 = 69,339,973$, and also $63N_* - 4 = 21\alpha - 4$ for $\deg U$, $12N_* + 10 = 4\alpha + 10$ for residual 6, and $29N_* - 8$ for the exponent of $\mathbf{p}$. Part II's ray coefficient $-2^{148}(2Ks)^{29N-8}$ with $K = 2^{551890}$ and $s = 794,976$ is (23.12). Both statements are printed: this one is proved on the literal source, Part II's for every table of the template.

23.13.5 The complete leading form and a nonzero ray coefficient

Multiplying the four unit-factor leading forms and the highest residual square, then substituting $b_* = 2\beta_*$ and $Q_* = 16\mathbf{p}^{N_*}$, gives the entire highest homogeneous component:

$$F_* = 2^{138} \mathbf{p}^{29N_*-8} \gamma w^5 \beta_*^{15} k_*^{10} f^2 i^4 (\tau - k_*) H_V^2. \quad (23.11)$$

[write] *An observation not made by the source* (checked in this write from (23.11)). The leading form F_* involves only native witnesses and the loader witness X (through $\mathbf{p}$); none of the six external coordinates x, a, p, s, C, L occurs in it. Every monomial of F of top degree therefore survives any specialization of the external coordinates, so the exact degree 69,339,973 in the remaining coordinates persists on every fixed-program slice, and also with x fixed. Part II's caution that specialization can lower degree (Section 24.7) concerns the loader residuals, which are not the dominant ones here.

The exponent $29N_* - 8$ is 23,113,311. The remaining factors have total degree 40, so (23.11) has degree

$$3(23,113,311) + 40 = 69,339,973.$$

The independent symbolic checker expands this expression into 23 monomials in the opaque degree-three base $\mathbf{p}$ and the native input atoms, and checks exact coefficient-dictionary equality with the form propagated independently from raw DAG rows.

There is also an explicit univariate certificate. Set *every supplied coordinate*, including all six externals, equal to a common indeterminate t . Then $\mathfrak{p}$ has leading coefficient $2^{551891}s$, k_* has coefficient two, and $\tau - k_*$ has coefficient minus one. The coefficient of $t^{69339973}$ in this ray specialization is exactly

$$-2^{148}(2^{551891} \cdot 794976)^{23113311} \neq 0. \quad (23.12)$$

This finite negative integer is specified by a compact exact expression; it is not expanded into a giant decimal numeral. Its two independently checked residues are

modulus	coefficient residue
17	3
1, 000, 000, 007	53, 942, 795.

Either nonzero residue alone certifies a nonzero integer coefficient. The certificates are deterministic; they do not invoke random identity testing or require an assumption that a modulus is prime.

23.13.6 Two independent verifiers and their scope

The cancellation-aware verifier reads all 3,600,546 original rows. For each gate it carries a valid degree bound and the coefficient at that bound on the all-coordinate ray, modulo each stated modulus. A zero coefficient at an intermediate bound never promotes that bound to an exact degree. The sole improved rule is at R_{15} , justified by the structurally authenticated identity (23.10). At the output the improved bound and a nonzero coefficient agree, and the coefficient matches (23.12).

The second verifier independently propagates exact sparse highest forms using an opaque degree-three base, authenticates the selector sum and all kernel rows, and proves equality with (23.11). Both execute newly written standard-library code, read the frozen source as data, and retain explicit checks under optimized Python. Their receipts record source pins, kernel and port degrees, all residual bounds, factor degrees, and the two coefficient residues.

The modular verifier uses at most four million source nodes, 800,000 coordinates, 256 monomials for its four-indeterminate identity, 512 MiB address space, and a 180-second ceiling. It keeps compact degree/coefficient arrays, not a degree-sized univariate polynomial or a full multivariate expansion. The independent symbolic verifier has its own explicit limits and receipt. These operational bounds constrain the checks; they are not bounds on witness size or computational complexity of the represented languages.

The historical manifest and original reproducibility subtree remain byte-for-byte unchanged, including their valid syntactic degree bound and earlier inconclusive probes. The separate exact-degree extension supplies the stronger invariant for the *same* source. This preserves an auditable distinction between an improved theorem about a circuit and an alteration of that circuit.

23.14 Independent evidence and reproducibility

The evidence has several layers, with deliberately different purposes.

1. *Semantic audit.* Independent checks cover 29,696 sparse-tape instruction contexts, 1,056 signed input-frame cases, 4,132 canonical-length cases, 192 exponent cases, and 192 exact-width E cases with rejected padding variants. A separate normalization audit checks 819,907 two-generation identities and 237,750 unique-H prefix cases. The accompanying proofs, rather than these finite counts, establish the parametric statements.

2. *Literal-table audit.* All 29 instructions and all numeric tag/Genera productions are reconstructed independently. Dynamics include 18,125 macro comparisons and 2,005,675 tag steps, with all 534 operative nonhalt tag symbols exercised, minimum preacceptance length three, and 300 accepting entries. Original-source code is read as data, never executed.
3. *Grill table audit.* Every one of the 397,488 exponents is derived by an independent position formula and compared byte for byte. Exact local E/L/R, shifted-erasure, and complete cleanup tests check the actual source rows, including the unique halt-producing pair.
4. *Complete arithmetic audits.* Independent readers check the binary DAG's typed references, constant-only recipes, six-external signature, witness count, all comparisons, one-unit finalizer, gate/input liveness, ledger, and degree bound. An exact motif audit consumes every row and compares every runtime coefficient by integer equality; all 67 native rows match authenticated complete fixtures. A separate formula evaluator checks the full source modulo independent primes without importing the producer.

The mathematical text, source emitters, complete DAG, literal table, frozen proof imports, audit code, and results are included in the archive. Revision 1 adds a separate exact-degree proof and two independent verifiers using the existing DAG without duplicating it. The original scientific subtree and its historical degree metadata remain byte-for-byte unchanged. The portable replay uses Python 3.10 or newer on Linux and only its standard library; the resource-limited checkers use the Unix resource module and Linux RSS units. It needs no network or upstream Python execution. It works from a freshly extracted archive and is checked both normally and with `python3 -O`. Explicit exceptions, rather than removable assertions, enforce validation. PDF rebuilding additionally requires a conventional LaTeX installation and the packages named in the preamble.

[write] In this collection the archive's files are shipped under prefixed names (README), except for the arithmetic DAG and the run table, which are rebuilt by the shipped code or extracted from the arrival archive (README, "Reconstructing the excluded data"), the byte copies of research-tree files (cited by path), the third-party paper copies, and the release seals. The delivered verifiers check sealed inventories, so they run in an extraction of the arrival archive, not in this directory.

The source-provenance record distinguishes original frozen hashes from documented portability transformations, including path adaptation and replacing local validation assertions by explicit exceptions. The inventory authenticates the delivered file set, including the actual arithmetic DAG, and rejects missing, extra, and altered files. Regeneration is performed in a separate work directory; it does not rewrite the supplied scientific evidence. Timing fields may vary, while mathematical metadata and DAG bytes are deterministic.

The binary format is deliberately simple. Its header is CDAGv1 followed by two zero bytes. Each subsequent 17-byte record is little-endian <Bqq: opcode 0 for addition, 1 for subtraction, or 2 for multiplication, followed by two signed 64-bit handles. A negative handle selects a constant recipe; a handle at least 2^{50} selects a supplied coordinate; other nonnegative handles refer to earlier gates. The JSON manifest gives compact coordinate families, exact coefficient recipes, comparisons, and the final output handle. This fully specifies the polynomial without expanding a prohibitively large monomial list.

23.15 Further mathematical questions

Can a different representation reduce the exact degree? The exact degree of this frozen polynomial is now settled, with all program parameters still variable. Lowering degree therefore requires

a genuinely different polynomial, rather than a sharper analysis of these same gates. One concrete direction is to isolate the largest native comparison and ask whether a paid auxiliary-variable interface can reduce total degree while controlling witness count, coefficient height and operation count. Such a change needs its own complete positive soundness and converse.

Can ordinary input be loaded without unary expansion? The second recoder pays for an exponent equal to the finite tape integer plus one. This is finite and arithmetically explicit but encodes an enormous unary tag word. A direct binary-input source compiler could remove that conceptual expansion. Such a proposal needs an actual fixed source-language theorem and an exact-width loader, rather than substituting a familiar universality claim.

Which selectors can genuinely be shared? The current emitter preserves all $2m$ chronological selectors. Equal run exponents share justified history classes, not phase identities. Eliminating selectors requires a sound and complete reconstruction of their phase chronology, especially across repeated zero exponents. A smaller numerical table could reduce both selectors and gates, but must retain the complete halt protocol.

Can the five program parameters be tied together? The valid tuple recipes have substantial algebraic dependence on a selected finite program. Encoding those dependences with fewer external parameters is a distinct task: it requires paying for a program decoder and proving that its range is the valid slice. Existentially freeing the present five parameters would change the recognized language and is not an acceptable shortcut.

Can the imported theorem boundary be reduced? A proof-assistant development could first formalize the all-tuple native identities, typed coefficient language, and exact source compiler, leaving the complete Pell/AND theorem as one explicit interface. A fuller development would certify that interface and the finite U15 initialization. These are separable proof obligations, and neither is supplied by increasing the number of bounded tests.

What should an explicit accepting witness demonstrate? Constructing a full positive tuple for a small genuine accepting source would test the most remote native interfaces, especially prescribed scale and restoration. It would still not prove universality. A feasible experiment should first estimate witness bit lengths and distinguish a full native solution from outer port assignments or modular residues.

What is the right cost comparison? The emitted source offers a reproducible arithmetic-operation baseline. Comparing substrates also requires coefficient height, source-program size, witness arity, maximum witness growth, exact-degree information, and bit cost. A source with fewer arithmetic gates can be worse on the other measures. Reporting those tradeoffs is more informative than collapsing them into a single unqualified complexity figure.

Conclusion

The completed construction has one fixed numerical Grill program, one fully emitted polynomial circuit, explicit ordinary-input and program-parameter semantics, and an exact initial-width interface. The paid composition uses 797,135 positive witnesses and 3,600,546 operations. Its exact total degree is 69,339,973, certified on the unchanged source by the explicit full leading form and nonzero ray coefficient. The

central logical safeguards are the accepting-head invariant, phase-preserving normalization, unrestricted second recoder, same-width positive native converse, and one integer-unit finalizer. Under the pinned complete theorem imports, these give the fixed-language equivalence of Theorem 23.1.1 for every positive ordinary input.

References of Report 23 revision 1

- [1] T. Neary and D. Woods, *Four small universal Turing machines*, Fundamenta Informaticae 91 (2009), 123–144. The author’s publication list notes incorrect page numbering in the distributed PDF; Table 16 appears on its printed page 121.
https://mural.maynoothuniversity.ie/id/eprint/12416/1/Woods_FourSmall_2009.pdf.
 Publication metadata: <https://dna.hamilton.ie/woods/>. The archive preserves the inspected primary PDF and page image; theorem use follows its concrete finite-frame table and the pinned initialization note. [write] The archive’s copies of this paper (two PDFs, page images including Table 16, and text extractions) are not redistributed in this collection; they survive in the arrival archive.
- [2] J. Cocke and M. Minsky, *Universality of Tag Systems with $P=2$* , Journal of the ACM 11 (1964), 15–20. <https://cs.famaf.unc.edu.ar/~hoffmann/cc18/p15-cocke.pdf>. The corrected construction in this report is distinguished from the printed production. [write] The archive’s copies of this paper (two PDFs, page images and a text extraction) are not redistributed in this collection; they survive in the arrival archive.
- [3] *Grill Tag*, creator construction, revision 181950.
https://esolangs.org/w/index.php?title=Grill_Tag&oldid=181950.
- [4] *Genera Tag*, creator semantics, revision 182460.
https://esolangs.org/w/index.php?title=Genera_Tag&oldid=182460.
- [5] V. Reshetnikov, *ProveIt native stream queue research notes*, commit 2d887f0fa768fd67f3e545d83f8998b5780e530d. <https://github.com/VladimirReshetnikov/ProveIt/tree/2d887f0fa768fd67f3e545d83f8998b5780e530d/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue>. The exact note names, individual blob identifiers, and content hashes used here are listed in the included provenance inventory.
- [6] *Report 23 reproducibility packet, revision 1*, 3 October 2026. Complete composition and component proofs, literal tables, emitted arithmetic DAG, source provenance, independent semantic and circuit audits, the exact-degree proof and certificates, and offline replay. The archive is the accompanying research artifact, not an external theorem citation. [write] Shipped in this directory with the prefix 11-grill-poly-, except the files the README lists as not shipped.

Part II

Parametric exact degree laws for the native Grill polynomial

[write] Source 14 of batch 82: *Research report 24*, dated 3 October 2026; archive `Native_Grill_Exact_Degree_Laws_Package.zip`; pin `2d887f0fa`. It cites Report 23 revision 1, which is Part I. Its sections, statements and equations are numbered 24.*k* here.

Abstract of the source. For the pinned weak-cone native Grill template, a fixed nonempty run table of length m with g distinct exponents determines $N = 2m + g + 5$. We prove that its native unit has exact degree $63N - 4$, its largest retained comparison residual has exact degree $12N + 10$, and its complete native polynomial has exact degree $87N + 16$. The proof covers one-phase and allzero tables, identifies the unrestricted norm cancellation, and gives the full nonzero highest homogeneous part.

Adding arbitrary integer residuals of exact maximum degree D inside the same sum-of-squares finalizer gives degree $63N - 4 + 2 \max(12N + 10, D)$, including ties. The generic k -wide recoder has residual maximum $\max(k + 1, 20)$; the specified two-recoder loader, with canonical width 32, has maximum $\max(34, k + 1)$. A fresh polynomial input of degree $\delta \geq 1$ changes the native degree to $(\delta + 2)(29N - 8) + 40$. Exact witness-count corollaries concern the supplied layouts only.

These are source-specific algebraic theorems. They do not establish a new universality theorem, the semantic correctness of arbitrary loader attachments, or minimal degree among equivalent representations. A portable data-only verification packet supplies small binary fixtures, independent full coefficient expansions, loader checks, and optional regression against the previously studied large circuit.

[write] Theorem 24.1.1 at $N = 797,011$ is Part I's Theorem 23.13.1; both are printed. The programme's review `review_incoming_matrix_grill.md` recomputes the law at that N . Source 14's files are shipped with the prefix `14-degree-laws-`; fourteen of its members are byte copies of source 11's files and are shipped once, under source 11's prefix.

24.1 The theorem and its scope

This report generalizes a degree calculation for one large emitted Grill polynomial to every nonempty fixed run table in the same native source template. The dependence on table length and number of distinct exponents is exact. The values of the exponents change fixed integer coefficients; they do not change the highest-degree lanes.

Throughout, degree means total degree of a polynomial over $\mathbb{Z}$, before any equations on its coordinates are imposed. Every supplied native coordinate and the supplied input X has degree one unless a different substitution is explicitly stated. A fixed numeral, including an integer given by an enormous fixed recipe, has degree zero. Write $\text{in}(A)$ for the highest nonzero homogeneous part of a nonzero polynomial A .

Let $(n_0, \dots, n_{m-1})$ be a fixed table, $m \geq 1$, and let g be its number of distinct exponent values. Set

$$s = 2m, \quad N = s + g + 5, \quad \alpha = 3N. \quad (24.1)$$

In particular, $g \geq 1$, $N \geq 8$, and $\alpha \geq 24$. Let U and $R_1, \dots, R_{10}$ be the unit and comparison differences of the exact source specified in Sections 24.2–24.4. Define

$$F = U \left(1 + \sum_{j=1}^{10} R_j^2 \right) - 1. \quad (24.2)$$

Theorem 24.1.1 (Exact native degree). *For independent degree-one X and native coordinates,*

$$\deg U = 63N - 4, \quad \max_j \deg R_j = 12N + 10, \quad \boxed{\deg F = 87N + 16}.$$

The largest native residual is unique. The statement holds for every table accepted by the pinned emitter: each n_i is an exact integer with $0 \leq n_i \leq 2^{32} - 1$. The same displayed polynomial formulas satisfy the theorem for arbitrary fixed nonnegative integer exponents.

The last sentence is a mathematical extension of the formulas, not an extension of the unchanged implementation’s accepted input type. In particular, it must not be read as permission to pass larger exponents to that emitter. Fixed exponent recipes are not variable-exponent operations.

No positivity, Boolean-selector, history, halting, or universality assumption enters this degree theorem. Such hypotheses matter for the interpretation of zeros; they do not define the ambient polynomial ring. Restricting to a solution set, identifying coordinates, or eliminating a witness can change degree and is a different question.

Theorem 24.1.2 (Arbitrary residual gluing). *Keep the native coordinates independent and of degree one. Let $C_1, \dots, C_r$ be any finite list of integer polynomials in those coordinates and any additional independent coordinates. They may share coordinates with the native part. Let D be the exact maximum of zero and the degrees of the nonzero C_ℓ . Then*

$$\deg \left[U \left(1 + \sum_j R_j^2 + \sum_\ell C_\ell^2 \right) - 1 \right] = 63N - 4 + 2 \max(12N + 10, D). \quad (24.3)$$

No unique largest residual is required in the enlarged family.

The two theorems separate two issues that are easily conflated. A particular loader can be attached algebraically and its degree computed exactly. Showing that its zeros encode the intended input for a particular Grill table requires a separate compiler proof. Section 24.7 gives the exact degree for the supplied loader layout without inferring such a proof for arbitrary pairings.

24.2 The precise source family

The source identity is fixed by `native_history.py`, its accompanying proof note, and `native_unit_kernel.json`. The latter contains 67 binary arithmetic rows, four ports, sixteen native auxiliary coordinates, six comparisons, and the four factors of U . Section 24.11 gives the SHA-256 pins. The source is included as evidence and is never executed by the verification packet.

24.2.1 Coordinates and fixed coefficients

Write $V, h_U, h_V, \phi, \sigma, G_{\text{bd}}, Z_0$ for the seven outer native coordinates, respectively named `Vfinal`, `H_U`, `H_V`, `phase_initial`, `height_slack`, `global_bound`, and `Z0`. Let $S_0, \dots, S_{s-1}$ be the independent supplied

selector hats and let $\widehat{z}_0, \dots, \widehat{z}_{g-1}$ be the independent group hats. They are coordinates, not Boolean quantities in this calculation.

Order the distinct exponent values $e_0, \dots, e_{g-1}$ by first occurrence. Define the nonempty groups

$$I_j = \{2i + 1 : n_i = e_j\}, \quad G_j = \sum_{a \in I_j} (S_a - 1), \quad z_j = \widehat{z}_j - 1.$$

The fixed slope and offset coefficients are

$$b(n) = 2^{2n+1}, \quad d(n) = \frac{2^{2n+1} - 2}{3} = \frac{2(4^n - 1)}{3}.$$

The quotient is an integer. In particular $b(0) - 1 = 1$ and $d(0) = 0$. Set

$$K = 2^{\max\{3, \lceil \log_2(s+4) \rceil, 2 \max_i n_i + 2\}}. \quad (24.4)$$

This is a fixed positive integer, including for an allzero table.

The base and height registers are

$$\begin{aligned} L &= X + Z_0, & U_f &= LV + X, & D_0 &= U_f + \phi + \sigma, \\ B &= KD_0, & J &= \sum_{a=0}^{s-1} (S_a - 1), & P &= (B - 1)J + 1. \end{aligned} \quad (24.5)$$

We use D_0 for the source height register to avoid confusion with the residual-degree parameter D of Theorem 24.1.2. Define the affine transport expressions

$$A_U = 2h_U + \sum_j G_j, \quad A_V = h_V + \sum_j (b(e_j) - 1)z_j + \sum_j d(e_j)G_j.$$

The first three residuals, left minus right, are

$$\begin{aligned} R_1 &= h_U + h_V + \sum_j \widehat{z}_j + G_{\text{bd}} - P, \\ R_2 &= BA_U + 1 - h_U - PU_f, \\ R_3 &= BA_V + 1 - h_V - PV. \end{aligned} \quad (24.6)$$

For $m > 1$ put

$$T_\phi = \sum_{i=1}^{m-1} (m - i)(S_{2i} + S_{2i+1} - 2).$$

The phase residual is exactly

$$R_4 = \begin{cases} \phi - 1, & m = 1, \\ m(S_0 + S_1) - (B - 1)T_\phi + \phi - (J + 3m), & m > 1. \end{cases} \quad (24.7)$$

The one-phase reduction is an unrestricted polynomial identity in this template. Using an unreduced syntactic phase bound would obscure this genuine exceptional degree.

24.2.2 The joined history lanes

For a fixed positive integer r , use polynomial notation

$$\text{rep}(r) = \sum_{a=0}^{r-1} P^a, \quad \text{pack}(v_0, \dots, v_{r-1}) = \sum_{a=0}^{r-1} v_a P^a.$$

These are polynomial sums, including at $P = 0$ and $P = 1$; no division is used. Put

$$\begin{aligned} S &= \text{pack}(S_0 - 1, \dots, S_{s-1} - 1), & M_c &= J \text{rep}(s), \\ T &= h_U + P h_V, & G_p &= \text{pack}(G_0, \dots, G_{g-1}), \\ H_b &= h_V \text{rep}(g), & M_b &= (B - 1)G_p, \\ Z_b &= \text{pack}(z_0, \dots, z_{g-1}), & R_M &= (D_0 - 1)J(1 + P). \end{aligned}$$

The indexed G_j are group selectors; G_{bd} is the separate global-bound coordinate. No supplied coordinate is identified with a selector.

The exact kernel ports are

$$\begin{aligned} H &= H_b + P^g S + P^{g+s} T + P^{g+s+2} B + P^{g+s+4} L, \\ M &= M_b + P^g M_c + P^{g+s} R_M + P^{g+s+2} (B - 1) + P^{g+s+4} (L - 1), \\ Z &= Z_b + P^g S + P^{g+s} T, \quad \text{scale} = P^N. \end{aligned} \tag{24.8}$$

The source uses shared powers, repunits, and a Horner schedule. These evaluation choices represent exactly the displayed polynomials and do not affect their degrees.

24.3 The leading history lanes

Let

$$J_1 = \sum_{a=0}^{s-1} S_a, \quad p = K L V J_1. \tag{24.9}$$

Every factor is nonzero in the free polynomial ring, and $\deg p = 3$. Equation (24.5) gives exact degrees

$$\deg L = 1, \quad \deg U_f = \deg D_0 = \deg B = 2, \quad \deg J = 1, \quad \deg P = 3, \quad \text{in}(P) = p.$$

The nonzero coefficient K may be enormous, but its size has no role in total degree.

Lemma 24.3.1 (Pack degrees). *For $r \geq 1$, $\deg \text{rep}(r) = 3(r - 1)$. A pack of r independent supplied hats minus constants has exact degree $3(r - 1) + 1$. The same degree assertion holds for a pack of r nonempty, disjoint selector-group sums minus constants.*

Proof. The last power of P in the repunit is uniquely largest. In either pack, the last coefficient has a nonzero degree-one homogeneous part. Its contribution has degree $3(r - 1) + 1$, whereas every earlier contribution has degree at most $3(r - 2) + 1$. Thus the last term cannot cancel. Disjointness is more than is needed here; nonvanishing of the last group sum suffices. $\square$

In particular,

$$\text{in}(S) = p^{s-1} S_{s-1}, \quad \deg S = 3s - 2, \quad \text{in}(T) = p h_V, \quad \deg T = 4.$$

The three terms of Z in (24.8) have degrees

$$3g - 2, \quad 3(g + s) - 2, \quad 3(g + s) + 4.$$

The last is strictly largest even when $m = g = 1$. Therefore, with $Z_* = \text{in}(Z)$,

$$Z_* = p^{N-4}h_V, \quad \deg Z = 3N - 11 = \alpha - 11. \quad (24.10)$$

The width lane is strictly largest in both H and M :

$$\text{in}(H) = \text{in}(M) = p^{N-1}L, \quad \deg H = \deg M = \alpha - 2. \quad (24.11)$$

For clarity, the common T lane has degree $\alpha - 11$, the radix lane has degree $\alpha - 7$, and the R_M lane of M has degree $\alpha - 9$. The lower group lanes cannot exceed these width lanes. Thus this step requires neither a positive exponent nor a nonzero appendant.

Finally put

$$q = 16P^N, \quad Q = \text{in}(q) = 16p^N, \quad \deg q = \alpha. \quad (24.12)$$

This notation distinguishes the source register q from its homogeneous part Q .

24.4 The unit and its unrestricted cancellation

The sixteen independent auxiliary coordinates of the kernel are

$$F_0, F_1, F_2, f, h, i, j, o, w, \tau, \eta, \zeta, \gamma, y, \beta, \rho.$$

They correspond, in order, to the source suffixes `F0`, `F1`, `F2`, `f`, `h`, `i`, `j`, `o`, `w`, `tau_gap`, `eta`, `zeta`, `ga`, `y_aux`, `odd_half`, and `bound_beta`, all in the native `AND` namespace.

The following equations are a compact regrouping of the pinned 67 rows, not a new kernel. Define

$$\begin{aligned} b &= 2\beta, & \kappa &= \eta + \zeta, & v &= (b + 1)q, & u &= wq, \\ c &= \kappa v + \eta, & a &= uv + v, & \Delta &= 4a + 3, & A &= a^2 + \Delta. \end{aligned} \quad (24.13)$$

Its first two norm expressions and packed register are

$$\begin{aligned} R_{15} &= (u + ac + \gamma\Delta)^2 - Ac^2, \\ R_{16} &= A(f^2 - 1), \\ F_3 &= 16Z + 8, & B_p &= F_0 + q(F_1 + q(F_2 + qF_3)). \end{aligned} \quad (24.14)$$

The remaining factors are

$$\begin{aligned} H_{17} &= jc - (2B_p + 1), \\ P_{17} &= R_{16}(H_{17}^2 - y^2) + y^2, \\ U_1 &= \tau^2 + 4(uv)(\kappa v)(\tau - \kappa), \\ q_b &= q - (F_0 + F_1 + F_2 + F_3), \\ U &= R_{15}P_{17}U_1q_b. \end{aligned} \quad (24.15)$$

The six residuals after the four outer residuals are

$$\begin{aligned} R_5 &= B_p + \rho - u, & R_6 &= \kappa - (B_p + 1 + h uv), \\ R_7 &= (ic^2)^2 - R_{16}, & R_8 &= H_{17} - (of - c), \\ R_9 &= F_1 + F_3 - (16H + 12), & R_{10} &= F_2 + F_3 - (16M + 10). \end{aligned} \quad (24.16)$$

The R_{15} , R_{16} , P_{17} names are historical kernel-register names; they are not indices in the ten-residual list.

24.4.1 The cancellation that a syntactic pass misses

From (24.13),

$$\text{in}(u) = wQ, \quad \deg u = \alpha + 1; \quad \text{in}(c) = \kappa bQ, \quad \deg c = \alpha + 2; \quad \text{in}(a) = wbQ^2, \quad \deg a = 2\alpha + 2.$$

The two operands defining R_{15} have the same leading square a^2c^2 . It cancels identically over the entire polynomial ring:

$$R_{15} = u^2 + 2uac + 2u\gamma\Delta + 2ac\gamma\Delta + \gamma^2\Delta^2 - \Delta c^2. \quad (24.17)$$

No equation on witnesses has been used. The six terms in order have degree bounds

$$2\alpha + 2, \quad 4\alpha + 5, \quad 3\alpha + 4, \quad 5\alpha + 7, \quad 4\alpha + 6, \quad 4\alpha + 6.$$

For $\alpha \geq 24$, the fourth is uniquely largest. Since $\text{in}(\Delta) = 4wbQ^2$,

$$\text{in}(R_{15}) = 8\gamma w^2 b^3 \kappa Q^5, \quad \deg R_{15} = 5\alpha + 7. \quad (24.18)$$

A gate-wise rule that replaces the degree of every subtraction by the maximum of operand degrees would overestimate this factor. Its upper bound becomes sharp only after the verified identity (24.17), or after a full expansion that discovers the same cancellation.

[write] *Attribution.* Identity (24.17) is identity (4) of the research tree's `gpcp_shared_selectors774.md` (commit `ee507599f`) and Part I's (23.10); see the note there. This Part does not cite it. The parametric exact law built on it is new for the Grill template; the tree has an exact law of the same shape, by the same method, for its PCP affine slope-class history: factor degrees $5d_0 + 7$, $12d_0 - 4$, $3d_0 + 5$, d_0 with $d_0 = 2N$ and exact degree $58N + 28$ (`pcp_affine_slope_class_history.md`, shipped inside archive 11).

24.4.2 The other three unit factors

Equations (24.10)–(24.15) imply

$$\begin{aligned} \text{in}(R_{16}) &= w^2 b^2 f^2 Q^4, & \deg R_{16} &= 4\alpha + 6, \\ \text{in}(B_p) &= 16Q^3 Z_*, & \deg B_p &= 4\alpha - 11, \\ \text{in}(H_{17}) &= -32Q^3 Z_*, & \deg H_{17} &= 4\alpha - 11, \\ \text{in}(P_{17}) &= 1024w^2 b^2 f^2 Q^{10} Z_*^2, & \deg P_{17} &= 12\alpha - 16, \\ \text{in}(U_1) &= 4wb^2 \kappa(\tau - \kappa)Q^3, & \deg U_1 &= 3\alpha + 5, \\ \text{in}(q_b) &= Q, & \deg q_b &= \alpha. \end{aligned} \quad (24.19)$$

For example, jc has degree $\alpha + 3$, below $4\alpha - 11$, and y^2 cannot compete with H_{17}^2 . The form $\tau - \eta - \zeta$ is nonzero because the supplied coordinates are independent. All four unit factors are therefore nonzero.

Degrees add under multiplication in an integer polynomial ring. Adding the four factor degrees gives

$$\deg U = (5\alpha + 7) + (12\alpha - 16) + (3\alpha + 5) + \alpha = 21\alpha - 4 = 63N - 4. \quad (24.20)$$

The argument explains why $d(0) = 0$ causes no difficulty: it removes only a lower affine term in A_V , not one of these leading factors.

24.5 Residuals and the full leading homogeneous part

The exact residual degrees are as follows. All signs refer to the left-minus-right conventions above.

Residual	Source role	Exact degree
R_1	Global bound	3
R_2	U transport	5
R_3	V transport	4
R_4	Phase	1 if $m = 1$; 3 if $m > 1$
R_5	Packed bound	$4\alpha - 11$
R_6	Packed transport	$4\alpha - 11$
R_7	Strong norm coefficient	$4\alpha + 10$
R_8	Auxiliary root comparison	$4\alpha - 11$
R_9	Padded H	$\alpha - 2$
R_{10}	Padded M	$\alpha - 2$

For the first three, the uniquely largest terms are $-P$, $-PU_f$, and $-PV$. For $m > 1$, the phase residual has degree-three homogeneous part

$$-KLV \sum_{i=1}^{m-1} (m-i)(S_{2i} + S_{2i+1}),$$

which is nonzero. For $m = 1$, equation (24.7) gives exact degree one directly. The top terms of R_5, R_6, R_8 are inherited from B_p or H_{17} ; their alternative operands are strictly smaller. The last two residuals inherit $-16 \ln(H)$ and $-16 \ln(M)$.

The decisive residual is R_7 . Its first term has leading form

$$\ln((ic^2)^2) = i^2 \kappa^4 b^4 Q^4 \quad \text{and degree } 4\alpha + 10,$$

whereas R_{16} has degree $4\alpha + 6$. Thus R_7 is uniquely largest among the ten. Consequently

$$\deg \left(1 + \sum_{j=1}^{10} R_j^2 \right) = 8\alpha + 20.$$

Combining this with (24.20), and observing that subtraction of one cannot remove a positive-degree leading part, proves

$$\deg F = 21\alpha - 4 + 8\alpha + 20 = 29\alpha + 16 = 87N + 16.$$

Proposition 24.5.1 (Exact highest form). *In the notation above,*

$$\boxed{\ln(F) = 2^{138} p^{29N-8} \gamma w^5 \beta^{15} \kappa^{10} f^2 i^4 (\tau - \kappa) h_V^2.} \quad (24.21)$$

In particular this is a nonzero polynomial of total degree $87N + 16$.

Proof. Before substituting $b = 2\beta$, $Q = 16p^N$ and $Z_* = p^{N-4}h_V$, multiplication of the four unit forms and $\ln(R_7)^2$ gives

$$2^{15} \gamma w^5 b^{15} \kappa^{10} f^2 i^4 (\tau - \kappa) Q^{27} Z_*^2.$$

The scalar power becomes $2^{15+15+108} = 2^{138}$, and the power of p becomes $27N + 2(N-4) = 29N - 8$. All factors are nonzero. The factors outside that power of p have total degree 40, so the total is $3(29N - 8) + 40 = 87N + 16$. $\square$

24.5.1 A deterministic ray certificate

Set every supplied coordinate, including X , equal to one indeterminate t . Then $L = 2t$ at highest degree, $J_1 = st$, $p = 2Kst^3$, $\kappa = 2t$, and $\tau - \kappa = -t$. Formula (24.21) gives the exact leading coefficient

$$[t^{87N+16}]F(t, \dots, t) = -2^{148}(2Ks)^{29N-8} \neq 0. \quad (24.22)$$

This single ray witnesses exact degree for every table in the degree-one theorem. A zero residue modulo one chosen prime does not contradict (24.22): the prime may divide its coefficient. For example, a table of length 17 has $s = 34$, so the coefficient vanishes modulo 17. The symbolic integer formula remains nonzero.

24.6 Residual gluing and integer zero sets

The argument needed for arbitrary added residuals is elementary, but the hypothesis on signs matters.

Lemma 24.6.1 (Positive sums of squares). *Let $T_1, \dots, T_v \in \mathbb{Z}[\mathbf{z}]$, and let $e > 0$ be their exact maximum degree, ignoring zero polynomials. Then*

$$\deg \left(1 + \sum_{a=1}^v T_a^2 \right) = 2e.$$

Proof. Let t_a be the degree- e homogeneous part of T_a , possibly zero. The degree- $2e$ part of the sum is $\sum_a t_a^2$, with at least one nonzero t_a . If that sum were identically zero, evaluating at every point of $\mathbb{R}^{|\mathbf{z}|}$ would force each real square, and hence each t_a , to vanish at every point. A real polynomial that vanishes everywhere is the zero polynomial, a contradiction. Adding the constant one does not alter degree $2e$. $\square$

Apply the lemma to the native and added residuals together. Their exact maximum is $\max(4\alpha + 10, D) > 0$. Multiplying by the nonzero U adds $21\alpha - 4$ to the degree, proving Theorem 24.1.2. Tied degrees, shared coordinates, and opposite signs of leading residuals cause no cancellation: the top forms are squared with positive coefficients.

This is a familiar sum-of-squares fact, not a claim of novelty about sums of squares or Diophantine representation generally. It is useful here because it turns an exact residual maximum into an exact degree for this unit-based finalizer.

24.6.1 Why the location of added squares matters

At an integer tuple, let $S = 1 + \sum_j R_j^2 + \sum_\ell C_\ell^2$. Then S is a positive integer. Thus

$$US = 1 \iff U = 1, R_j = 0 \text{ for all } j, C_\ell = 0 \text{ for all } \ell.$$

The forward direction uses both integrality and positivity: $U = 1/S$ is an integer only when $S = 1$. No assumption that U is positive off the zero set is needed. The correct extension is consequently

$$F + U \sum_\ell C_\ell^2,$$

not, in general, $F + \sum_\ell C_\ell^2$. This zero-set observation still does not provide the semantic proof of a loader.

The degree lemma cannot be transferred unchanged to arbitrary coefficient fields or signed differences of squares. Over the complex numbers, $T^2 + (iT)^2 = 0$; over the integers, $T^2 - T^2 = 0$. Nor does a quotient by residual equations retain the ambient polynomial degree. The pinned checks compute the polynomial before any such quotient or substitution.

24.7 Generic recoders and the specified loader

Consider the complete 34-comparison recoder in the pinned 130-row receipt. The generic width replacement is

$$Q_r = q_r^k, \quad B_r = 2^{k-1}Q_r, \quad k \geq 4,$$

with k a fixed integer. The remaining 127 source rows are unchanged. Recoder auxiliaries are independent degree-one coordinates. For the uses considered here, input and output ports are either degree-one affine supplied inputs or the fixed integer one.

Proposition 24.7.1 (Exact recoder residual maximum). *The exact maximum degree of the retained recoder residuals is*

$$D_{\text{recoder}}(k) = \max(k + 1, 20).$$

Proof. Propagation through the complete source gives the following upper bounds for the 34 comparisons in source order:

$$(k + 1, k + 1, 1, k + 1, k, 14, 3, 1, 5, 4, 2, 4, 6, 10, 2, 1, 2, k, \\ 7, 2, 1, 3, 20, 4, 1, 7, 6, 3, 4, 6, 10, 2, 2, 1).$$

When both ports are one, the penultimate bound becomes one; this does not change the maximum. Every bound is at most $\max(k + 1, 20)$.

Two exact source motifs attain the possible maxima. The first residual is

$$(2^{k-1}q_r^k - 1)J_r + 1 - P_r,$$

with leading form $2^{k-1}q_r^k J_r$ and degree $k + 1$. Here J_r, P_r are supplied recoder coordinates, not the computed native registers J, P . The AND norm residual has leading form

$$2^{24}w_r^2\kappa_r^2s_r^4q_r^6P_r^6$$

and exact degree 20. Here w_r, κ_r, s_r are the independent recoder coordinates named `and_w`, `and_k`, and `and_s`; κ_r is not the native sum $\eta + \zeta$. Its other operand has degree two. These two nonzero forms attain the upper envelope, including $k = 19$, where they tie. $\square$

The fixed Pell/AND baseline is essential. For example, at $k = 4$ the maximum is 20, not 5. The supplied data-only checker represents each upper bound as $ak + b$ valid throughout $k \geq 4$, rejects a comparison of crossing degree lines, and authenticates the exact source motifs used for attainment. A list of tested widths alone would not prove the all-width upper bound.

24.7.1 The two-recoder layout

The supplied loader uses a canonical recoder at width 32 and an unrestricted recoder at width k . The canonical input is $x + 1$ and its output is an independent spread coordinate. Both ports of the

unrestricted recoder are one. All six external coordinates and all supplied loader witnesses are assigned degree one.

The canonical recoder maximum is 33. The ordinary-frame residual has the unique degree-34 term

$$-(2^{32} - 1)p_e s_e q_c^{32}, \quad (24.23)$$

where q_c is the canonical recoder's supplied q coordinate. The other terms in that residual have degrees at most 33, 2, and 1. The nonzero fixed coefficient and both independent external factors in (24.23) must be retained when taking degree.

The seven outer loader comparisons, excluding the two groups of 34, have upper bounds in their own source order

$$1, \quad 34, \quad k+1, \quad k, \quad 1, \quad k, \quad 2. \quad (24.24)$$

Their roles are the canonical length guard, ordinary frame, exponent quotient, exponent gap, exponent-to-tape equality, unary scale, and exact unary queue. The exponent-to-tape equation uses a tape coordinate N_{tape} , unrelated to the native lane count N .

For completeness, writing $B_e = 2^{k-1}q_e^k$, their last five source shapes are

$$(B_e - 1)v_e + \ell - J_e, \quad \ell + g_e - (B_e - 1), \quad \ell - (N_{\text{tape}} + 1), \quad K_e T - q_e^k, \\ (K_e - 1)X - ((K_e - 1)C_e + L_e V_e(T - 1)).$$

Here K_e, V_e are fixed integer recipes, $V_e \neq 0$, and the other displayed loader letters are supplied coordinates; K_e is consistently the fixed radix for width k . These formulas explain (24.24) directly. The native-width residual is $X + Z_0 - L_e T$, of exact degree two.

Theorem 24.7.2 (Exact loader maximum and composition). *For this precise two-recoder layout, with the fixed canonical width 32 and consistent fixed width- k radix constants,*

$$D_{\text{loader}}(k) = \max(34, k+1), \quad k \geq 4.$$

Attaching all its residuals and the native-width equality inside the native finalizer gives exact degree

$$63N - 4 + 2 \max(12N + 10, 34, k+1). \quad (24.25)$$

Proof. The first unrestricted recoder residual attains $k+1$, and the canonical frame residual attains 34. Proposition 24.7.1 and (24.24) bound everything else by their maximum. The width equality cannot dominate. Apply Theorem 24.1.2. $\square$

24.7.2 Matching widths and deliberately mismatched tests

If a separately validated compiler has $k = m \geq 4$, then

$$12N + 10 = 24m + 12g + 70 > \max(34, m+1).$$

The complete composition therefore retains exact degree $87N + 16$. The inequality is purely algebraic; the hypothesis that the compiler matches remains separate.

For the frozen width $k = 397488$, the loader maximum is 397489. Its algebraic attachment to a native table is loader-dominated for $N \leq 33123$ and native-dominated for $N \geq 33124$. These integer cutoffs follow from comparing $12N + 10$ with 397489. Arbitrary mismatched attachments are useful degree tests and are not asserted to encode the intended ordinary input.

The degree convention also matters for program slices. The six external coordinates are counted as degree one here even if five will later be fixed to represent a particular language. Specializing those five parameters to integer values is a different polynomial and can lower the loader's degree. The free-coordinate theorem is the stated claim.

24.8 Fresh polynomial inputs and coordinate identifications

Replacing a supplied coordinate by a computed polynomial is not the same operation as retaining that coordinate and adding equations that constrain it. The latter fits Theorem 24.1.2; the former requires a new degree analysis.

Theorem 24.8.1 (A fresh polynomial input). *Substitute $X = \chi$ in the native polynomial, where χ is a nonzero polynomial of exact degree $\delta \geq 1$ in fresh coordinates independent of all native witnesses. Give those fresh coordinates and the native witnesses degree one. Put*

$$d = \delta + 2, \quad \alpha = dN.$$

Then

$$\begin{aligned} \deg U &= 21\alpha + 20 - 8d, \\ \max_j \deg R_j &= 4\alpha + 10, \\ \boxed{\deg F|_{X=\chi} &= d(29N - 8) + 40.} \end{aligned}$$

The highest-form formula (24.21) remains valid with $p = \text{in}(P)$ for the substituted source.

Proof. Let $L_* = \text{in}(\chi + Z_0)$. For $\delta > 1$, $L_* = \text{in}(\chi)$; for $\delta = 1$, $L_* = \text{in}(\chi) + Z_0$. In both cases $L_* \neq 0$ and $\deg L_* = \delta$, because the coordinates of χ are fresh. Thus

$$\deg P = d, \quad p = KL_*VJ_1 \neq 0, \quad Q = 16p^N.$$

The pack argument now uses step size d in place of three. It yields

$$\text{in}(Z) = p^{N-4}h_V, \quad \deg Z = \alpha - 4d + 1, \quad \text{in}(H) = \text{in}(M) = p^{N-1}L_*.$$

The last two forms have degree $d(N-1) + \delta = \alpha - 2$. Their lower radix lanes have degree $\alpha - 2d - 1$, and the R_M lane has degree $\alpha - 3d$, so neither competes with the width lane.

The kernel formulas for u, c, a and the cancellation (24.17) are unchanged, with the new α . Therefore $\deg R_{15} = 5\alpha + 7$. The changed Z degree gives

$$\deg H_{17} = 4\alpha - 4d + 1, \quad \deg P_{17} = 12\alpha + 8 - 8d.$$

The other factors still have degrees $3\alpha + 5$ and α . Adding proves the asserted unit degree. Since $d \geq 3$ and $N \geq 8$, the strong norm residual remains uniquely maximal with degree $4\alpha + 10$. Hence the final degree is

$$(21\alpha + 20 - 8d) + 2(4\alpha + 10) = 29dN - 8d + 40.$$

The same leading-factor multiplication proves (24.21). Every factor is nonzero in the larger polynomial ring, establishing exactness without relying on a particular ray. $\square$

24.8.1 Sharp examples and failed ray certificates

For the one-phase table [0], $N = 8$ and the ordinary degree is 712. Substitution of $X = \xi_1^2$ changes it to

$$4(29 \cdot 8 - 8) + 40 = 936.$$

Thus a claim that arbitrary computed loaders preserve $87N + 16$ would already fail in the smallest case.

For $\chi = \xi_1^3 + \xi_1 \xi_2 + 7$, the degree is 1160. For $\chi = \xi_1^2 - \xi_2^2$, it is 936, witnessed on the ray $\xi_1 = 2t, \xi_2 = t$ with all native coordinates equal to t . On the all-ones ray $\xi_1 = \xi_2 = t$, however, χ vanishes identically and the specialized polynomial has degree 712. The symbolic highest form survives, while that particular test ray does not see it. The portable checks expand the entire specialized polynomial in these examples, with no special norm-cancellation rule.

Freshness is a sufficient hypothesis designed to prevent accidental identities with native factors. It is not a claim to characterize every degree-preserving substitution. For example, setting $\tau = \eta + \zeta$ annihilates the highest form of U_1 , and therefore the displayed highest form of F . Such an identification lies outside both the free-coordinate theorem and the fresh-input theorem.

24.9 Witness counts and concrete instances

The native builder allocates seven named outer coordinates, s selector hats, g group hats, and sixteen kernel auxiliaries. Excluding its supplied X port, its positive-witness ledger is

$$W_{\text{native}} = 7 + s + g + 16 = s + g + 23 = N + 18.$$

Consequently Theorem 24.1.1 can be rewritten as

$$\deg F = 87W_{\text{native}} - 1550. \quad (24.26)$$

The count describes the supplied source variables. Positivity is their semantic type; it is not imposed when computing polynomial degree.

The two recoders contribute $49 + 49$ coordinates. The spread coordinate, canonical guard, tape coordinate, ℓ, v_e, g_e, T, X contribute eight more, for 106 loader witnesses. The loader's X is reused as the supplied native input and is not counted twice. There are six external coordinates, excluded from the existential count. Thus

$$W_{\text{total}} = W_{\text{native}} + 106 = N + 124.$$

Whenever the native residual dominates, including the separately justified matching case $k = m \geq 4$,

$$\deg F_{\text{composed}} = 87W_{\text{total}} - 10772. \quad (24.27)$$

These are exact affine rewritings of this ledger, not lower bounds relating degree to the minimum possible number of variables. Alternative encodings, elimination, or auxiliary introduction can change both quantities.

24.9.1 Small fixtures and the large regression

Run table	m	g	N	Exact native degree
[0]	1	1	8	712
[1]	1	1	8	712
[0, 0]	2	1	10	886
[0, 0, 0]	3	1	12	1060
[0, 1, 1]	3	2	13	1147
[2, 0, 1]	3	3	14	1234
[0, 0, 0, 0, 0]	5	1	16	1408
$[2^{32} - 1]$	1	1	8	712
$[0, 2^{32} - 1]$	2	2	11	973

The four fixtures [0], [1], [0, 1, 1], and [2, 0, 1] are checked against complete frozen binary DAGs and corresponding receipt data. The additional rows are clean-room formula cases, not additional emitted binary fixtures. This distinction is preserved in the certificates.

For [0], the four unit factors have degrees 127, 272, 77, 24, so $\deg U = 500$. The maximum residual degree is 106, and $500 + 2 \cdot 106 = 712$. This explicitly verifies that the phase collapse and the vanishing zero-exponent offset do not invalidate the theorem.

The previously studied full program has

$$m = 397488, \quad g = 2030, \quad N = 797011,$$

with $W_{\text{native}} = 797029$ and $W_{\text{total}} = 797135$. Its exact degree is

$$87 \cdot 797011 + 16 = 69339973.$$

A direct regression reads its 3,600,546-gate frozen DAG, all 86 residual squares, and the actual finalizer. It reproduces highest ray coefficients 3 modulo 17 and 53942795 modulo 1000000007. This is a useful consistency check with Report 23 revision 1; the general theorem is proved above, not inferred from that one large example.

[write] *Consistency with the research tree's recorded bounds.* Every syntactic upper bound that the tree records for a native Grill unit polynomial equals $90N + 17 = (87N + 16) + (3N + 1)$, the exact degree plus the cancellation gap $\alpha + 1$:

Instance	m	g	N	$87N + 16$	$90N + 17$ (recorded)
programs 0, 1 (composed205)	1	1	8	712	737
program 011	3	2	13	1,147	1,187
program 201	3	3	14	1,234	1,277
exact-width loader	1,568	6	3,147	273,805	283,247
Part I, literal program	397,488	2,030	797,011	69,339,973	71,731,007

The loader row is an inference: it assumes that the loader's 3,165 native witnesses follow the ledger $N + 18$ and that its loader residuals stay below $12N + 10$; with that assumption the recorded 283,247 is reproduced exactly. The other rows use the tree's recorded m and g . The programme's intake review recomputes $87N + 16 = 69,339,973$ at Part I's N .

24.10 Verification boundaries and reproducibility

The accompanying packet is self-contained for its default core checks. It includes the four small DAGs and metadata, their receipt fixtures, the unit kernel, recoder receipt, and the inert native and loader source needed for source authentication. Only newly written local verification code is executed. No upstream Python is imported, compiled, or dynamically loaded.

24.10.1 What the separate checks establish

The native leading-term checker structurally matches every one of the 67 kernel rows, the ports, the constants, and the actual finalizer in each frozen fixture. It propagates valid upper bounds and corresponding specialized coefficients modulo two primes. Its only special cancellation is the unrestricted identity (24.17), itself checked by exact sparse expansion. A nonzero coefficient at the upper degree certifies a matching lower degree; a zero modular coefficient is treated as inconclusive.

The direct expansion checker evaluates every gate over $\mathbb{F}_{17}[t]$, retaining all nonzero coefficients. It uses no norm rewrite and no assumed degree truncation. For each small fixture, every coefficient agrees with a separately evaluated receipt. It also checks allzero and maximum-uint32 formula examples and the degree-two input substitution. The fresh-input checker compares complete coefficient dictionaries for the cubic and difference-of-squares inputs discussed in Section 24.8.

The loader checker proves the parametric source-row upper bounds as degree pairs $ak + b$ for all $k \geq 4$, verifies the attaining recoder motifs, and tests representative widths including the transition cases. The seven outer loader formulas are transcribed from authenticated source and proved directly in Section 24.7. The checker does not claim to execute or parse the producer’s Python into a general theorem about that source.

These are finite implementation regressions supporting the source transcription. The symbolic arguments in Sections 24.3–24.8 prove the parameterized statements. Neither an all-gate numerical pass nor a set of modular examples alone is an all-table proof.

24.10.2 Core replay and optional full regression

The archive’s top-level README gives the exact replay command and expected results. Normal Python and Python with optimization enabled are both tested; correctness checks use explicit runtime failures rather than removable assertions. Outputs are written outside the sealed release tree. An exact inventory verifier rejects missing, changed, or unexpected files.

The optional full regression requires an explicit source directory containing the unchanged full-DAG files supplied in Report 23 revision 1. Those large files are not duplicated here. The optional command authenticates the files it consumes and has no automatic workspace search or fallback to a different source directory. A missing optional source does not invalidate the self-contained core; requesting the full regression with missing or mismatched data is an error, never a silent skip.

[write] In this collection Part I’s DAG is not shipped either; the README gives its rebuild, after which the optional regression can be run in an extraction of the two arrival archives.

The packet records original research-file hashes separately from portable-file hashes. Changes needed for relative paths, output discipline, and source selection are documented rather than presented as byte-identical originals. Mutation tests exercise corrupted and missing source files, altered expected results, inventory changes, and invalid optional-full requests. Final fresh-extraction runs verify the packaged bytes, not only an unarchived working directory.

24.11 Source provenance and further questions

The native kernel derives from the composed205 receipt at immutable ProveIt commit

2d887f0fa768fd67f3e545d83f8998b5780e530d.

Its receipt Git blob is 929d89c9f1ac1b727e207d8b518cfab54c8c4125. The source and kernel relationship is described by the included native proof note. The degree claims here use the explicit polynomial identities, not a new import of a semantic universality theorem.

Pinned source	SHA-256
<code>native_history.py</code>	<code>d6f88f09f5ffd48747b60bd53b98814431bcc5515cd16431438e2768a7f67340</code>
<code>native_history_proof.md</code>	<code>714274c3f7a9009f0526bcd4f854396670d6e8f1e2f58fd15cc737779145747</code>
<code>native_unit_kernel.json</code>	<code>2d88343037c08fe8b073f67dca531ee73309c0735c1d98a23c20b9ddb1eb08ae</code>
<code>input_loaders.py</code>	<code>597a028e3a64e8ed97502f0b5e66411f293cdda0c9c1d7ebc470f9c38c897c14</code>
<code>input_recoder130_receipt.json</code>	<code>175c498990a8e63de7a91d1e3d321ef90a19f129f305074f0c6de10967d0a182</code>

The exact source files named in this table are part of the core packet; their inclusion is not a claim that the entire historical proof dependency graph is independently formalized here. The original general theorem text has SHA-256 `4f4281ae58ba6df422f3e56301faa094001e522502362a17aaa302f568de51af`. The release’s provenance record separately pins the frozen research inventory and every adapted artifact.

24.11.1 Questions left open

The degree formulas suggest several concrete next problems.

1. *More general substitutions.* Characterize when coordinate identifications preserve (24.21), and determine exact degrees on exceptional loci such as $\tau = \eta + \zeta$. Nonvanishing of a substituted top form is sufficient, but a vanishing top form requires the next homogeneous layers.
2. *Weighted and multigraded costs.* Assign different weights to input, loader, and native coordinates, derive the resulting piecewise-linear degree function, and locate all leading-lane transitions. The fresh-input theorem is one controlled weighted direction.
3. *Certified source transcription.* Replace manually reviewed outer-formula transcription with a small formally specified arithmetic-source parser or proof-producing checker, while retaining independence from producer execution.
4. *Semantic compiler families.* Determine which width- k loaders and table families are genuinely compatible, then combine those semantic proofs with the exact algebraic gluing law. An arbitrary table-loader pairing is not sufficient.
5. *Alternative representations.* Investigate the cost of replacing high-degree registers by auxiliary variables and quadratic equations. Any tradeoff must recount witnesses, preserve the integer-unit finalizer’s semantics, and distinguish a new representation from a degree computation on this one.

None of these questions is settled by the current degree or arity identities. In particular, this report gives no lower bound on the minimum degree of a Diophantine representation and no improvement to the general MRDP theorem.

References of Report 24

- [1] Native weak-cone source and correctness-transfer note. `native_history.py`, `native_history_proof.md`, and `native_unit_kernel.json`. Exact frozen data included in the accompanying packet. [write] Shipped as `code/11-grill-poly-arithmetic-native_history.py.txt` (inert original) with its executable portable version, `11-grill-poly-arithmetic-native_history_proof.md` and `data/11-grill-poly-arithmetic-native_unit_kernel.json` (source 14’s copies are byte-identical to source 11’s and are shipped once).
- [2] V. Reshetnikov, ProveIt repository, composed205 native receipt, immutable commit `2d887f0fa768fd67f3e545d83f8998b5780e530d`. [Pinned composed205 receipt](#).

- [3] Paid ordinary-input loader and complete recoder receipt. `input_loaders.py` and `input_recoder130_receipt.json`. Exact frozen data included in the accompanying packet. [write] The recoder receipt is the research tree's `native_binary_input_dilation130.json`, not shipped here.
- [4] *A fixed universal Grill polynomial*. Research report 23, revision 1, 3 October 2026. The optional full regression uses its unchanged arithmetic archive. [write] Printed as Part I of this report.

Part III

Entire native fibers: Pell parametrization and height laws

[write] Source 07 of batch 82: *Research report 25*, dated 3 October 2026; archive `Entire_Native_Witness_Fiber_Package.zip`; pin `ad634b2d1`. It is the sequel of Report 22. Its sections, statements and equations are numbered 25.*k* here.

Abstract of the source. For fixed valid padded AND ports and prescribed dyadic scale, we classify every positive witness of the retained twenty-two-coordinate native arithmetic component. Seventeen coordinates are forced. The remaining five are in bijection with two integer indices: $m = M\ell$, $\ell \geq 1$, and $n = p$ or $n = 4mk \pm p$, $k \geq 1$, where $M = pc/\gcd(c, A^2 - 1)$ and $c = \psi_A(p)$. The maximum supplied-coordinate height is exactly the last Pell coordinate, with no native fixed-coordinate plateau. This yields an exact finite floor formula and the unsmoothed two-term law

$$N(H) = \kappa \log H + \frac{\zeta(1/2)}{M\sqrt{\log \alpha}} \sqrt{\log H} + O((\log H)^{1/3}).$$

The coefficient κ includes an exceptional baseline family and an exact convergent series. We prove a corresponding logarithmic ranked-height expansion, allowing ties. Native soundness and positive completeness are explicitly inherited from pinned source proofs; the Pell rank, congruence and analytic arguments are included with their attribution. A companion theorem counts the sum of all coordinate bit lengths. A portable standard-library verification packet accompanies the article.

[write] The classification is conditional on the pinned native soundness and positive-completeness theorems, as the abstract and Section 25.2 say. The programme's review `review_incoming_negative_index_fibers.md` (commit `0055e1c4d`) finds the classification consistent and does not certify the second-term or total-bit-length theorems. In this Part $\alpha = A + \sqrt{\Delta}$ and $\lambda = \log \alpha$; Report 34 in Part IV uses the two letters the other way round (Section 0.3). The source's files are shipped with the prefix `07-native-fiber-`.

25.1 The object being counted

The native component is a fixed integer arithmetic circuit, not an unspecified Diophantine representation. Fix relation arguments

$$P = 2^b \ (b \in \mathbb{Z}_{\geq 0}), \quad h_0 = \widehat{H} - 1, \quad v_0 = \widehat{M} - 1, \quad z_0 = \widehat{Z} - 1, \quad 0 \leq h_0, v_0 < P, \quad z_0 = h_0 \text{ AND } v_0. \quad (25.1)$$

All four relation arguments are positive integers. Its computed padded ports are

$$q = 16P, \quad A_{\text{port}} = 16h_0 + 12, \quad B_{\text{port}} = 16v_0 + 10, \quad F_3 = 16z_0 + 8. \quad (25.2)$$

The twenty-two *supplied* native coordinates, all required to lie in $\mathbb{Z}_{>0}$, are

$$\begin{aligned} &F_0, F_1, F_2, a, c, d, f, h, i, j, k, o, r, s, w, \\ &\tau, \eta, \zeta, ga, y, odd_half, bound_beta. \end{aligned} \quad (25.3)$$

Here y denotes the literal coordinate `y_aux`. The symbol ζ in this coordinate list is unrelated to the Riemann zeta function. Computed quantities, including q, F_3, X, Y , are not additional supplied

coordinates. Positivity is imposed on the supplied coordinates; intermediate subtraction registers follow the literal circuit.

We write $\mathcal{F}$ for the full positive fiber at the fixed arguments (25.1). The height and its ordinary counting function are

$$\text{ht}(w) = \max\{\text{the twenty-two coordinates of } w\}, \quad N(H) = \#\{w \in \mathcal{F} : \text{ht}(w) \leq H\}.$$

Different tuples count separately, even if their heights coincide. All logarithms are natural, and all asymptotic constants may depend on the fixed ports and scale.

For an integer $B \geq 2$, define the Pell sequences by

$$\chi_B(t) + \psi_B(t)\sqrt{B^2 - 1} = (B + \sqrt{B^2 - 1})^t \quad (t \in \mathbb{Z}). \quad (25.4)$$

The polynomial and addition identities used below follow either by multiplying these expressions or by the recurrence $u_{t+2} = 2Bu_{t+1} - u_t$. In particular $\chi_B(-t) = \chi_B(t)$ and $\psi_B(-t) = -\psi_B(t)$.

25.1.1 Precisely stated inherited mathematics

All upstream native sources are pinned to the same repository commit

ad634b2d10ad666260f9fdff04ec94b75169ee4b.

The selector soundness theorem [1], its prescribed-scale interface [2], the relaxed-norm proof [3] and fixed-minus proof [4] are supplied as inert text in the accompanying packet. We use these specific conclusions:

1. Every positive native witness has the forced main indices and exponent/rounding values displayed in Section 25.3. These are forward soundness conclusions about *every* witness, established before any preferred converse witness is selected.
2. Every valid tuple (25.1) admits a strictly positive native extension at its prescribed scale. Thus the uniquely forced quotients and differences below are positive integers.
3. The native bootstrap, at the base $A = a + 2$, establishes $c > A(A^2 - 1)^2$ before invoking the relaxed norm. The source obtains the main index $p = 2r + 1$, and $p \equiv 3 \pmod{4}$.

The new classification is conditional on these proved native representation results. It does not purport to rebuild the entire representation theorem from elementary arithmetic. We give the relevant relaxed-norm rank proof and signed congruence argument in full because their exact hypotheses are essential to counting the whole fiber.

The predecessor Report 22 established uniqueness of the canonical outer history and a bijection from its complete positive witness fiber to this native fiber, as well as native infinitude. Section 25.12 records the resulting transport of the present classification. The original Report 22 and its frozen circuits are unchanged.

[write] Report 22 is archive `Canonical_Histories_and_Infinite_Fibers_Package.zip` of batch 82 (cluster M2), placed by commit 7d2b1b245 as source 23 of `signal-machine-collision-certificates`, in its Part VIII. Its theorem “Fixed-scale native infinitude” (three native coordinates vary while nineteen stay fixed) is a corollary of Theorem 25.2.1, which fixes exactly seventeen and parametrizes the other five; Report 22’s elementary argument is a second route and is printed with Report 22.

25.2 The complete result

Using the forced values in Section 25.3, set

$$\begin{aligned} A = a + 2, \quad \Delta = A^2 - 1, \quad p = 2r + 1, \quad c = \psi_A(p), \quad \alpha = A + \sqrt{\Delta}, \quad \lambda = \log \alpha, \\ g = \gcd(c, \Delta) = \gcd(p, \Delta), \quad M = \frac{pc}{g}. \end{aligned} \quad (25.5)$$

The native hypotheses give $A \geq 3$, $p \geq 3$, $p \equiv 3 \pmod{4}$, and $M \geq c > 2p$.

Theorem 25.2.1 (Entire positive native fiber). *Exactly seventeen coordinates of (25.3) are fixed throughout $\mathcal{F}$. The other five, f, i, j, o, y , are parametrized without repetition by*

$$m = M\ell \quad (\ell \geq 1), \quad n = p \quad \text{or} \quad n = 4mk - p, \quad 4mk + p \quad (k \geq 1). \quad (25.6)$$

For each such pair define

$$\begin{aligned} f_m &= \chi_A(m), & R_m &= \Delta\psi_A(m), & i_m &= R_m/c^2, \\ U_{m,n} &= \chi_{R_m}(n)/R_m, & j_{m,n} &= (U_{m,n} + p)/c, \\ o_{m,n} &= (U_{m,n} + c)/f_m, & y_{m,n} &= \psi_{R_m}(n). \end{aligned} \quad (25.7)$$

All five reconstructed coordinates are positive integers. Adjoining the seventeen forced coordinates gives every member of $\mathcal{F}$, and

$$\text{ht}(m, n) = y_{m,n} = \psi_{R_m}(n). \quad (25.8)$$

In fact all seventeen fixed supplied coordinates are strictly smaller than $c^2 \leq R_m < y_{m,n}$.

Thus even the baseline $n = p$ is present for every $m = M\ell$, including $m = M$; the canonical converse choice $m = 2cp$ is not a restriction on the answer.

Put

$$D_m = R_m^2 - 1, \quad \varepsilon_m = R_m + \sqrt{D_m}, \quad \beta_m = \log \varepsilon_m, \quad t_m(H) = \frac{\text{arsinh}(H\sqrt{D_m})}{\beta_m}.$$

For $H \geq 1$, let L_H be the largest $\ell \geq 1$ with $\psi_{R_{M\ell}}(p) \leq H$, or zero if no such ℓ exists.

Theorem 25.2.2 (Finite count and two height laws). *The integer L_H is finite and the exact count is*

$$N(H) = \sum_{\ell=1}^{L_H} \left(\left\lfloor \frac{t_m(H) + 4m - p}{4m} \right\rfloor + \left\lfloor \frac{t_m(H) + p}{4m} \right\rfloor \right), \quad m = M\ell. \quad (25.9)$$

The same formula holds with $t_m(H)$ replaced by $\lfloor t_m(H) \rfloor$. One may also replace the upper summation bound by

$$K_0(H) = \left\lfloor \frac{\log H}{M(p-1)\lambda} \right\rfloor; \quad (25.10)$$

any added summands are zero. For $H < 1$, $N(H) = 0$.

Define the positive constant

$$\kappa = \frac{1}{M(p-1)\lambda} + \sum_{\ell=1}^{\infty} \frac{1}{2M\ell\beta_{M\ell}}, \quad d_*^{\text{count}} = \frac{\zeta(1/2)}{M\sqrt{\lambda}} < 0. \quad (25.11)$$

The series converges absolutely. Pointwise, including exact jump thresholds,

$$N(H) = \kappa \log H + d_*^{\text{count}} \sqrt{\log H} + O((\log H)^{1/3}). \quad (25.12)$$

If H_v is the height of the v -th tuple in nondecreasing height order, counting ties with multiplicity, then

$$\log H_v = \frac{v}{\kappa} - \frac{\zeta(1/2)}{M\sqrt{\lambda}\kappa^{3/2}} \sqrt{v} + O(v^{1/3}). \quad (25.13)$$

This is a logarithmic asymptotic, not a multiplicative equivalent for H_v .

25.3 Seventeen forced coordinates and three varying comparisons

25.3.1 Explicit formulas and universal forcing

The padded ports, checksum and packing comparison first force

$$\begin{aligned} F_1 &= 16(h_0 - z_0) + 4, & F_2 &= 16(v_0 - z_0) + 2, \\ F_0 &= 16(P - h_0 - v_0 + z_0) - 15, & r &= F_0 + qF_1 + q^2F_2 + q^3F_3. \end{aligned} \quad (25.14)$$

Since $h_0 + v_0 - z_0 = h_0$ OR $v_0 < P$, $F_0 \geq 1$; the other fields are positive as well. The checksum gives $F_0 + F_1 + F_2 + F_3 + 1 = q$. Because F_0 is odd and q is even, r is odd, hence $p = 2r + 1 \equiv 3 \pmod{4}$.

The forward selector theorem, Sections 2–3 of [1], gives

$$\begin{aligned} X &= 2^p, & Y &= \left\lfloor \frac{(X+1)^{2r}}{X^r} \right\rfloor = \binom{2r}{r} + \sum_{v=1}^r \binom{2r}{r+v} X^v, \\ w &= X/q, & s &= Y/q, \\ a &= Y(X+1), & c &= \psi_A(p), \quad d = \chi_A(p), \\ E &= XY, & B &= 2XY^2 + 1, \quad k = \psi_B(r+1). \end{aligned} \quad (25.15)$$

The notation B renames the first Pell base to distinguish it from the external scale P . The remaining forced coordinates are

$$\begin{aligned} h &= \frac{k - r - 1}{E}, & \eta &= c - Yk, & \zeta &= (Y+1)k - c, \\ ga &= \frac{d - X - ac}{4a + 3}, & \tau &= \frac{\chi_B(r+1) - 1}{2}, \\ odd_half &= \frac{s - 1}{2}, & bound_beta &= X - r. \end{aligned} \quad (25.16)$$

In particular, the seventeen fixed names are

$$F_0, F_1, F_2, r, w, s, a, c, d, k, h, \eta, \zeta, ga, \tau, odd_half, bound_beta.$$

This enumeration includes no computed register.

Here universal soundness is doing real work: merely writing one canonical witness would not prove these values unique. The source first recovers the indices $p = 2r + 1$ and $r + 1$, and then the exponent X and rounded value Y , for every positive witness. Subsequent divisions have positive denominators. The first norm determines the positive square root

$$(2\tau + 1)^2 - (B^2 - 1)k^2 = 1, \quad 2\tau + 1 = \chi_B(r + 1),$$

so it forces τ as well. Finally the source's positive converse at the same valid ports proves that the unique values in (25.14)–(25.16) are indeed positive integers. This invocation does not impose its auxiliary-index choice on other witnesses.

25.3.2 The complete literal comparison list

Let $U = jc - p$ and $R = ic^2$. The following are all sixteen comparisons of the native block, written as equalities of their actual polynomials. The right column gives incidence in the potentially varying set $\{f, i, j, o, y\}$.

No.	Comparison	Incidence
1	$r = F_0 + qF_1 + q^2F_2 + q^3F_3$	none
2	$F_0 + F_1 + F_2 + F_3 + 1 = q$	none
3	$s = 2\text{odd_half} + 1$	none
4	$r + \text{bound_beta} = X$	none
5	$(E^2 + X)(Yk)^2 = \tau(\tau + 1)$	none
6	$c = Yk + \eta$	none
7	$k = \eta + \zeta$	none
8	$k = r + 1 + hE$	none
9	$a = Y(X + 1)$	none
10	$d = X + ac + ga(4a + 3)$	none
11	$d^2 = 1 + \Delta c^2$	none
12	$(ic^2)^2 = \Delta(f^2 - 1)$	f, i
13	$(ic^2)^2(U^2 - y^2) = 1 - y^2$	i, j, y
14	$U = of - c$	f, j, o
15	$F_1 + F_3 = 16h_0 + 12$	none
16	$F_2 + F_3 = 16v_0 + 10$	none

The parentheses in comparison 12 are essential: its right side is $\Delta(f^2 - 1)$, not $\Delta f^2 - 1$. It is the relaxed main auxiliary norm; comparison 11 is the fixed main Pell norm.

The literal-DAG audit independently expands all sixteen residuals, propagates transitive dependencies, checks the twenty-two supplied names, and checks the acyclic sixty-four-gate source on four retained fixtures. Consequently exactly thirteen comparisons are unaffected when the five coordinates change. All recomputed registers are determined by the unchanged acyclic DAG. Once the forced values are installed, the entire positive fiber is therefore exactly the set of positive solutions of

$$R^2 = \Delta(f^2 - 1), \quad (RU)^2 - (R^2 - 1)y^2 = 1, \quad U = jc - p = of - c, \quad R = ic^2. \quad (25.17)$$

The general valid-port theorem comes from [2]; it is not inferred by extrapolating finite fixture tests.

25.4 The exact main auxiliary progression

The relaxed norm alone can have unwanted rational Pell behavior. The large native bound $c > A\Delta^2$ is what excludes it. The following proof is the rank argument of [3], restated at $A = a + 2$; after its bootstrap the argument only needs the hypotheses displayed here.

Lemma 25.4.1 (Inherited relaxed-norm rank lemma). *Suppose $A \geq 2$, $\Delta = A^2 - 1$, $c = \psi_A(p) > A\Delta^2$, and positive integers R, f satisfy $c^2 \mid R$ and $R^2 = \Delta(f^2 - 1)$. Then for a unique $m \in \mathbb{Z}_{>0}$,*

$$p \mid m, \quad f = \chi_A(m), \quad R = \Delta\psi_A(m).$$

Proof. Write $\Delta = d_0 s_0^2$ with $d_0 > 1$ squarefree. Choose the least positive norm-one unit with integer coefficients in $\mathbb{Z}[\sqrt{d_0}]$, say $F + y_0\sqrt{d_0}$. It exists because $A + s_0\sqrt{d_0}$ is one such unit. Every positive integer-coefficient norm-one unit is its power: division by the appropriate power leaves an integer-coefficient unit in $[1, F + y_0\sqrt{d_0})$, contradicting minimality unless it equals one. Hence for some $e \geq 1$,

$$A = \chi_F(e), \quad S = \psi_F(e), \quad s_0 = y_0 S, \quad \Delta = (F^2 - 1)S^2, \quad \psi_F(ep) = Sc.$$

The relaxed norm gives $s_0 \mid R$. Then $(R/s_0)^2 = d_0(f^2 - 1)$ and squarefreeness gives $d_0 \mid R/s_0$. Thus $f + R/(d_0 s_0)\sqrt{d_0}$ is an integer-coefficient norm-one unit, so for some $v > 0$,

$$f = \chi_F(v), \quad R = T_0 \psi_F(v), \quad T_0 = (F^2 - 1)S = \Delta/S.$$

The sequence ψ_F is strongly divisible:

$$\gcd(\psi_F(u), \psi_F(v)) = \psi_F(\gcd(u, v)).$$

Indeed the norm gives $\gcd(\chi_F(t), \psi_F(t)) = 1$, and the addition identity reduces the gcd by the Euclidean algorithm. Put $b_0 = \gcd(v, ep)$ and $h_1 = \psi_F(b_0)$. Since $c^2 \mid T_0\psi_F(v)$,

$$c \leq \gcd(c^2, T_0\psi_F(ep)) = c \gcd(c, \Delta) \leq T_0 h_1, \quad h_1 \geq c/\Delta.$$

If $b_0 < ep$, then $2b_0 \leq ep$, and doubling yields

$$2h_1^2 < \psi_F(2b_0) \leq \psi_F(ep) = Sc < Ac.$$

Here $S < A$, as follows from $\Delta = (F^2 - 1)S^2 < A^2$ and $F \geq 2$. But $h_1^2 \geq c^2/\Delta^2 > Ac$, a contradiction. Therefore $ep \mid v$. Write $v = em$; Pell composition gives the stated formulas and $p \mid m$. Strict increase of χ_A proves uniqueness. $\square$

Proposition 25.4.2. *For the native fixed parameters, the possible main auxiliary indices are exactly $m = M\ell$, $\ell \geq 1$.*

Proof. The lemma gives $m = pk$ and $R = \Delta\psi_A(pk)$. Expanding $(d + c\sqrt{\Delta})^k$ and dividing its square-root coefficient by c gives

$$\frac{\psi_A(pk)}{c} \equiv kd^{k-1} \pmod{c}.$$

All terms after the linear term contain c^2 . Since $d^2 - \Delta c^2 = 1$, $\gcd(d, c) = 1$. Therefore

$$\begin{aligned} c^2 \mid \Delta\psi_A(pk) &\iff c \mid \Delta kd^{k-1} \iff c \mid \Delta k \\ &\iff (c/g) \mid k \iff M \mid m, \quad g = \gcd(c, \Delta). \end{aligned}$$

This is an equivalence, not only a necessary divisibility condition. Also the binomial expansion modulo Δ gives

$$c \equiv pA^{p-1} \pmod{\Delta}, \quad \gcd(A, \Delta) = 1,$$

so $g = \gcd(p, \Delta)$ and $g \mid p$. Hence M is a multiple of c and $m \geq M \geq c > 2p$. Conversely every $m = M\ell$ has $p \mid m$ and $c^2 \mid R_m$, so $i_m = R_m/c^2$ is a positive integer and the first equation of (25.17) holds. It remains to solve the other two equations; this is done next. $\square$

No evenness condition on m or oddness condition on f_m has been imposed. Those features of some canonical constructions are not requirements of the whole fiber.

[write] *A domain distinction* (from the programme's review `review_incoming_negative_index_fibers.md`, commit 0055e1c4d, which finds the classification consistent). The progression $m = M\ell$, $M = pc/g$, is for the ordinary source, where the coefficient Δ multiplies $\psi_A(m)$. It must not be copied to the tree's normalized strong equation, where $c^2 \mid \psi_A(m)$ forces m to be a multiple of pc . The review's auxiliary-only example $A = 2$, $p = 3$, $c = \psi_2(3) = 15$ has $g = 3$, $M = 15$, against 45 in the normalized case.

25.5 The two normalized Pell progressions

Fix any $m = M\ell$. Then $c \mid m$, $R = R_m$ is a positive multiple of c^2 , and $f = f_m > 2c$. The latter follows from $m > 2p$ and $\chi_A(2p) = 1 + 2\Delta c^2 > 2c$.

25.5.1 Generation and the odd-index congruences

For $B \geq 2$, the least positive integer-coefficient norm-one unit in $\mathbb{Z}[\sqrt{B^2 - 1}]$ is $B + \sqrt{B^2 - 1}$: a positive integer second coordinate is at least one, making the first coordinate at least B . The minimal-unit argument from Lemma 25.4.1 therefore proves that every positive solution is $(\chi_B(n), \psi_B(n))$ for a unique $n \geq 1$. This concerns the integer-coefficient group, not the full ring of integers of the quadratic field; no unjustified fundamental-unit claim is needed.

For a native witness, $U = jc - p \geq c - p > 0$, so the second equation in (25.17) gives

$$RU = \chi_R(n), \quad y = \psi_R(n), \quad n \geq 1. \quad (25.18)$$

Modulo R , the recurrence gives $\chi_R(2h) \equiv (-1)^h$ and $\chi_R(2h + 1) \equiv 0$. Since $R > 1$ and $R \mid \chi_R(n)$, n must be odd.

For $n = 2h + 1$, write $\chi_X(2h + 1) = XQ_h(X^2)$, where

$$Q_0(t) = 1, \quad Q_1(t) = 4t - 3, \quad Q_{h+2}(t) = (4t - 2)Q_{h+1}(t) - Q_h(t).$$

Induction gives

$$Q_h(0) = (-1)^h(2h + 1), \quad Q_h(1 - A^2) = (-1)^h\psi_A(2h + 1).$$

Since $c \mid R$ and $R^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f}$, these identities imply

$$U \equiv (-1)^h n \pmod{c}, \quad U \equiv (-1)^h \psi_A(n) \pmod{f}. \quad (25.19)$$

The reconstruction of j and o requires exactly

$$U \equiv -p \pmod{c}, \quad U \equiv -c \pmod{f}. \quad (25.20)$$

These polynomial identities and the sign analysis below are the fixed-minus argument in [4], with its normalized index renamed n .

25.5.2 Step down and necessity

Lemma 25.5.1 (Plus-sign step down). *If $B \geq 2$, $0 < k_0 \leq m$ and $\chi_B(v) \equiv \chi_B(k_0) \pmod{\chi_B(m)}$, then $v \equiv \pm k_0 \pmod{4m}$.*

Proof. The identities give $\chi_B(v + 2m) \equiv -\chi_B(v)$ modulo $\chi_B(m)$ and $\chi_B(-v) = \chi_B(v)$. Thus reduce to a representative $\pm \chi_B(l)$, $0 \leq l \leq m$. For $k_0 < m$ and $l < m$,

$$\chi_B(l) + \chi_B(k_0) \leq 2\chi_B(m - 1) < \chi_B(m).$$

Strict monotonicity therefore excludes the negative representative and forces $l = k_0$ for the positive one. The case $l = m$ is zero modulo the modulus and cannot occur when $k_0 < m$. When $k_0 = m$, only $l = m$ is zero. Translating these representatives gives exactly the two residues asserted. $\square$

Squaring the second congruence in (25.19)–(25.20) and using $\chi_A(2t) = 1 + 2\Delta\psi_A(t)^2$ yields

$$\chi_A(2n) \equiv \chi_A(2p) \pmod{f}.$$

Since $0 < 2p < m$, Lemma 25.5.1 gives

$$n = ep + 2mt, \quad e \in \{1, -1\}, \quad t \in \mathbb{Z}. \quad (25.21)$$

This division by two is a division of an integer equality after obtaining the residue modulo $4m$; it is not cancellation of a possibly nonunit in a residue ring.

The addition identities give

$$\psi_A(n) \equiv e(-1)^t c \pmod{f}, \quad n \equiv ep \pmod{c}.$$

With $p = 2r + 1$ and $h = (n - 1)/2$, both signs satisfy $(-1)^h e = (-1)^{r+mt}$. The two original, unsquared minus congruences become

$$(-1)^{r+mt} p \equiv -p \pmod{c}, \quad (-1)^{r+(m+1)t} c \equiv -c \pmod{f}.$$

Because $c > 2p$ and $f > 2c$, the respective exponents must both be odd. Their difference is t , which is therefore even. Consequently

$$n \equiv p \text{ or } -p \pmod{4m}. \quad (25.22)$$

The same argument recovers the necessary oddness of r , consistent with the padded fields.

25.5.3 Converse and full-fiber bijection

Conversely let $n > 0$ have either residue in (25.22). It is odd, so $U = \chi_R(n)/R$ is a positive integer. Write $n = ep + 4mk$. The same identities give

$$\psi_A(n) \equiv ec \pmod{f}, \quad n \equiv ep \pmod{c}, \quad (-1)^{(n-1)/2} e = (-1)^r = -1.$$

Thus both congruences (25.20) hold. Hence $j = (U + p)/c$ and $o = (U + c)/f$ are positive integers, and (25.18) supplies the norm. All three varying comparisons hold, while the other thirteen keep their zero values. This proves positive completeness for every allowed pair in (25.6).

The representatives p and $4m - p$ are distinct and lie in $(0, 4m)$ because $0 < p < 2m$. Their increasing merge is

$$p, 4m - p, 4m + p, 8m - p, 8m + p, \dots$$

The zero-index Pell solution is excluded by $y > 0$; negative indices have negative y , and a negative first coordinate contradicts $U > 0$. Strict increase of $\chi_A(m)$ recovers m from f , and strict increase of $\psi_{R_m}(n)$ then recovers n from y . There is no double counting. Moreover f, i vary as m increases, and j, o, y vary within each fixed- m progression. This proves the exact seventeen/five partition and the bijection of Theorem 25.2.1.

25.6 The native maximum is always the last Pell coordinate

First control the fixed coordinates. The native bounds give

$$q < r < X < a < A < c.$$

The formulas for the fields, $w, s, a, \text{odd_half}, \text{bound_beta}$ therefore make each less than c . The relations give $k < c/Y < c$, $h < k$, $\eta < c$, and $\zeta < k$. Also

$$\frac{d}{c} = \sqrt{\Delta + c^{-2}} < A, \quad d < Ac < c^2, \quad ga < \frac{d}{4a + 3} < c,$$

because $4a + 3 > A$. The coordinate c itself is below c^2 .

For the remaining coordinate τ , note $E = XY < a$ and $X < 2E + 1$. Thus

$$\tau^2 < \tau(\tau + 1) = (E^2 + X)(Yk)^2 < (E + 1)^2(Yk)^2,$$

whence $\tau < (E + 1)Yk < (E + 1)c < Ac < c^2$. This accounts for all seventeen fixed supplied coordinates.

For the five varying ones, the relaxed norm and $\Delta \geq 3, f \geq 2$ give

$$R^2 = \Delta(f^2 - 1) > f^2, \quad i = R/c^2 \leq R.$$

Since $n \geq p \geq 3$,

$$y = \psi_R(n) \geq \psi_R(3) = 4R^2 - 1 > R \geq c^2.$$

The normalized norm yields

$$0 < U \leq y, \quad U^2 = y^2 - \frac{y^2 - 1}{R^2}.$$

Using $p < c$ and $c < f$,

$$U + p \leq y + c - 1 \leq cy, \quad U + c \leq y + f - 1 \leq fy.$$

Therefore $j, o \leq y$. This proves (25.8) for the actual full native tuple.

An expression $\max(C, y)$ remains correct if one fixes other coordinates and calls their maximum C . Within the native twenty-two-coordinate fiber, however, the inequalities just proved always give $C < y$: there is no native initial plateau. A genuinely independent external fixed-coordinate maximum can create such a cutoff; Section 25.12 keeps that case separate.

The smallest native height is attained uniquely at $(m, n) = (M, p)$. This follows from strict increase of R_m in m and of $\psi_R(n)$ in either argument in the present range. No assertion about componentwise minimality of the entire five-coordinate tuple is needed for this height conclusion.

25.7 Exact finite counts and the fixed-slice phase

For fixed m ,

$$\psi_{R_m}(n) = \frac{\sinh(n\beta_m)}{\sqrt{D_m}},$$

so $y_{m,n} \leq H$ if and only if $n \leq t_m(H)$. Counting the two distinct residues yields the summand in (25.9). Since the smallest admissible index is p , only $\ell \leq L_H$ contribute. The baseline height $\psi_{R_{M\ell}}(p)$ increases strictly to infinity, proving finiteness.

For $n \geq 1$ the geometric-sum identity is

$$\psi_{R_m}(n) = \varepsilon_m^{n-1} \frac{1 - \varepsilon_m^{-2n}}{1 - \varepsilon_m^{-2}} \geq \varepsilon_m^{n-1}. \quad (25.23)$$

Also $R_m > f_m = \cosh(m\lambda)$, so $\beta_m > m\lambda$. Thus a baseline point under height H satisfies $\log H \geq (p - 1)m\lambda$, proving (25.10). The outer floors are unchanged on replacing t_m by its integer part because their shifts and denominators are integers.

25.7.1 The exact staircase inside one main-index slice

For fixed m , put

$$\mathcal{P}_m(x) = 1 - \left\{ \frac{x-p}{4m} \right\} - \left\{ \frac{x+p}{4m} \right\},$$

where braces denote fractional part. The function is bounded and $4m$ -periodic. For every $H \geq 1$, the fixed- m count is exactly

$$N_m(H) = \frac{t_m(H)}{2m} + \mathcal{P}_m(t_m(H)). \quad (25.24)$$

In particular $N_m(H) = \log H / (2m\beta_m) + O(1)$. The admissible density is $1/(2m)$ among all positive indices, or $1/m$ among odd indices.

The smooth inverse phase has the convergent expansion, for $H\sqrt{D_m} > 1$,

$$t_m(H) = \frac{1}{\beta_m} \left[\log(2\sqrt{D_m}H) + \sum_{k \geq 1} \frac{(-1)^{k-1} \binom{2k}{k}}{2^k 4^k D_m^k H^{2k}} \right]. \quad (25.25)$$

To verify it, expand the derivative of $\operatorname{arsinh} x - \log(2x)$ by the convergent binomial series for $x > 1$, and integrate toward infinity. The first terms are $1/(4x^2) - 3/(32x^4) + 5/(96x^6)$. This smooth expansion must not be substituted approximately into a discontinuous floor with a claim of uniform correctness at jumps. The exact phase or exact integer comparisons are needed there.

The ranks within this slice have explicit indices

$$n_{2k+1} = 4mk + p \ (k \geq 0), \quad n_{2k} = 4mk - p \ (k \geq 1),$$

and exact heights $\psi_{R_m}(n_v)$. There are no initial repeated heights caused by fixed native coordinates. The least slice triple (j, o, y) occurs at $n = p$; selecting it mathematically is not a new arithmetic gate or a finite-fold implementation.

25.8 The leading coefficient and its exceptional baseline

Before obtaining the second term, it is useful to prove the leading law with a simpler uniform estimate. Write $T = \log H$ and separate

$$N(e^T) = B(T) + Q(T),$$

where B counts $n = p$ over varying m , and Q counts $n = 4mk \pm p$, $k \geq 1$.

Since

$$R_m = \frac{\sqrt{\Delta}}{2} \alpha^m (1 - \alpha^{-2m}),$$

we have $\log R_m = m\lambda + O(1)$. For fixed p , $\psi_R(p)$ is a polynomial of degree $p-1$ with positive leading coefficient 2^{p-1} ; moreover $R^{p-1} \leq \psi_R(p) \leq (2R)^{p-1}$ for $R \geq 2$. Hence

$$\log \psi_{R_m}(p) = (p-1)m\lambda + O(1), \quad B(T) = \frac{T}{M(p-1)\lambda} + O(1). \quad (25.26)$$

The $p-1$ is essential: the varying Pell denominator must not be discarded.

The smallest nonbaseline index is $4m-p$. Since $m > 2p$ and $p \geq 3$, $4m-p-1 \geq 3m$. Equation (25.23) gives

$$\log \psi_{R_m}(4m-p) \geq 3\lambda m^2.$$

Thus nonbaseline slices are supported only at $m \leq J(T) = \sqrt{T/(3\lambda)}$, whereas the baseline occupies $O(T)$ slices.

Uniformly in m and $H \geq 1$,

$$0 \leq t_m(H) - \frac{\log H}{\beta_m} \leq 1. \quad (25.27)$$

The lower bound follows from $\operatorname{arsinh}(H\sqrt{D_m}) \geq \log H$. For the upper bound, $\operatorname{arsinh}(H\sqrt{D_m}) - \log H$ decreases with $H \geq 1$ and equals $\operatorname{arsinh}\sqrt{D_m} = \beta_m$ at $H = 1$. The two floors now give $N_m(e^T) = T/(2m\beta_m) + O(1)$ with an absolute uniform error. Subtracting the baseline indicator preserves this bound. Summing only over $m \leq J(T)$ therefore gives

$$Q(T) = T \sum_{M\ell \leq J(T)} \frac{1}{2M\ell\beta_{M\ell}} + O(\sqrt{T}).$$

Because $\beta_{M\ell} > M\ell\lambda$, the series tail after $K \geq 1$ terms satisfies

$$0 < \sum_{\ell > K} \frac{1}{2M\ell\beta_{M\ell}} \leq \frac{1}{2\lambda M^2 K}. \quad (25.28)$$

Extending the sum costs $O(\sqrt{T})$ and proves the weaker law $N(e^T) = \kappa T + O(\sqrt{T})$. Also

$$0 < \kappa - \frac{1}{M(p-1)\lambda} \leq \frac{\pi^2}{12\lambda M^2}.$$

One cannot obtain the correct coefficient by simply summing fixed-slice leading terms: that misses the exceptional baseline. Nor can one sum a fixed-slice $O(1)$ error over the $O(T)$ baseline range.

25.9 A pointwise second term without smoothing

We now strengthen the remainder and identify its square-root term. The method is the classical hyperbola decomposition and Euler summation, applied to exact Pell heights. The exact sign-specific boundary below avoids any averaging of floor phases.

25.9.1 Exact energy and uniform row and column estimates

For this section set

$$q_0 = M\lambda, \quad b_0 = \frac{1}{2} \log \Delta, \quad a_0 = 4\lambda M^2, \quad \beta_\ell = \beta_{M\ell}, \quad g_\ell = 4M\ell\beta_\ell.$$

The elementary bounds are

$$q_0\ell \leq \beta_\ell \leq q_0\ell + b_0. \quad (25.29)$$

The lower bound was proved above. For the upper bound use $R_{M\ell} = \sqrt{\Delta} \sinh(q_0\ell)$ and $\cosh(q_0\ell + b_0) > \sqrt{\Delta} \sinh(q_0\ell)$.

For $\sigma \in \{-1, 1\}$ define the nonbaseline energy

$$E_\sigma(\ell, k) = \log \psi_{R_{M\ell}}(4M\ell k + \sigma p), \quad \ell, k \geq 1.$$

Taking logarithms in (25.23) gives the exact identity

$$E_\sigma(\ell, k) = g_\ell k + (\sigma p - 1)\beta_\ell + \theta_{\ell, k, \sigma}, \quad 0 \leq \theta_{\ell, k, \sigma} \leq d_0, \quad (25.30)$$

where $d_0 = -\log(1 - e^{-2q_0})$. The bound holds because every normalized index here is at least p . In particular

$$|E_\sigma(\ell, k) - a_0 k \ell^2| \leq C k \ell, \quad C = 4M b_0 + (p+1)(q_0 + b_0) + d_0. \quad (25.31)$$

Also $g_\ell = a_0 \ell^2 + O(\ell)$, so

$$S = \sum_{\ell \geq 1} \frac{1}{g_\ell} < \infty, \quad \sum_{\ell > L} \frac{1}{g_\ell} = \frac{1}{a_0 L} + O(L^{-2}). \quad (25.32)$$

The exact series is retained here. Replacing every β_ℓ by $q_0 \ell$ in its definition would generally change the linear coefficient.

Let $q_\sigma(T, \ell)$ be the number of $k \geq 1$ with $E_\sigma(\ell, k) \leq T$. Exact inversion gives

$$q_\sigma(T, \ell) = \max \left(0, \left\lfloor \frac{t_{M\ell}(e^T) - \sigma p}{4M\ell} \right\rfloor \right).$$

By (25.27), uniformly for $T \geq 0$ and $\ell \geq 1$,

$$q_\sigma(T, \ell) = T/g_\ell + O(1). \quad (25.33)$$

The positive-part clipping and floor each cost a bounded amount, since $T/g_\ell \geq 0$ and $p/(4M\ell)$ is uniformly bounded.

Let $r_\sigma(T, k)$ be the number of $\ell \geq 1$ with $E_\sigma(\ell, k) \leq T$, and put $x = \sqrt{T/(a_0 k)}$ and $D_0 = C/a_0$. From (25.31),

$$\ell^2 - D_0 \ell \leq E_\sigma(\ell, k)/(a_0 k) \leq \ell^2 + D_0 \ell.$$

Every integer $\ell \leq x - D_0$ is admitted, and every admitted ℓ satisfies $\ell \leq x + D_0$. Including endpoint rounding and the possible empty range proves the uniform estimate

$$r_\sigma(T, k) = \sqrt{T/(a_0 k)} + O(1). \quad (25.34)$$

This estimates the count directly. It never assumes that a floor remains unchanged after replacing its argument by an asymptotic approximation.

The energy is strictly increasing in each positive integer variable. For fixed k , both $R_{M\ell}$ and $4M\ell k + \sigma p > 1$ increase with ℓ ; $\psi_R(n)$ increases with R and n in this range. Finiteness follows, for example, from

$$E_\sigma(\ell, k) \geq (4M\ell k - p - 1)M\ell \lambda \geq 3\lambda M^2 k \ell^2.$$

25.9.2 An exact boundary rectangle

For sufficiently large T set

$$L = \lfloor T^{1/3} \rfloor, \quad K_\sigma = q_\sigma(T, L).$$

Equations (25.33) and (25.29) give

$$K_\sigma = \frac{T}{a_0 L^2} + O(T/L^3 + 1) = \frac{T}{a_0 L^2} + O(1) \asymp T^{1/3}. \quad (25.35)$$

Let $Q_\sigma(T)$ count all points of sign σ . Monotonicity gives the exact hyperbola identity

$$Q_\sigma(T) = \sum_{\ell \leq L} q_\sigma(T, \ell) + \sum_{k \leq K_\sigma} r_\sigma(T, k) - L K_\sigma. \quad (25.36)$$

Every point in $1 \leq \ell \leq L$, $1 \leq k \leq K_\sigma$ is admitted because $E_\sigma(L, K_\sigma) \leq T$. No point with both $\ell > L$ and $k > K_\sigma$ is admitted, since already $E_\sigma(L, K_\sigma + 1) > T$. The two strips therefore cover the region and overlap in exactly that rectangle. This argument remains true when T is exactly an energy, and uses a separate exact cutoff for each sign.

Put $u = \sqrt{T/a_0}$. Substituting (25.32), (25.33), and (25.34) into (25.36) gives

$$Q_\sigma(T) = ST - \frac{T}{a_0 L} + u \sum_{k \leq K_\sigma} k^{-1/2} - LK_\sigma + O(L + T/L^2 + K_\sigma). \quad (25.37)$$

The classical Euler-summation formula is

$$\sum_{k=1}^K k^{-1/2} = 2\sqrt{K} + \zeta(1/2) + O(K^{-1/2}). \quad (25.38)$$

For completeness, the identity recorded in DLMF 25.2.8 [6] at $s = 1/2$ is

$$\zeta(1/2) = \sum_{k=1}^K k^{-1/2} - 2\sqrt{K} - \frac{1}{2} \int_K^\infty \{x\} x^{-3/2} dx.$$

The integral contribution has absolute value at most $K^{-1/2}$, proving (25.38). The alternating-series identity DLMF 25.2.3 gives $\zeta(1/2) < 0$ because its numerator is positive and its factor $1 - \sqrt{2}$ is negative.

The apparent boundary mismatch cancels quadratically:

$$-\frac{T}{a_0 L} + 2u\sqrt{K_\sigma} - LK_\sigma = -L \left(\sqrt{K_\sigma} - u/L \right)^2 = O(1). \quad (25.39)$$

Indeed (25.35) gives $K_\sigma - u^2/L^2 = O(1)$, and hence the squared difference is $O(T^{-1/3})$. The Euler-sum error is $O(u/\sqrt{K_\sigma}) = O(T^{1/3})$, as are all other errors in (25.37). Consequently

$$Q_\sigma(T) = ST + \frac{\zeta(1/2)}{2M\sqrt{\lambda}} \sqrt{T} + O(T^{1/3}). \quad (25.40)$$

The signs are disjoint, and disjoint from the exceptional baseline. Adding both copies of (25.40) to (25.26) proves (25.12), since $2S$ is the series term in (25.11).

This proof is unsmoothed and pointwise. The fixed-slice staircases remain present, but only $O(T^{1/3})$ short rows and columns contribute independent bounded discrepancies. No generic-position assumption, equidistribution claim, or unproved cancellation of floor phases is used.

25.10 Ranked heights and binary length

Let $F(T) = N(e^T)$, and write $d = d_*^{\text{count}}$. The fiber has finitely many tuples below any finite height and is infinite, so the nondecreasing ranked heights H_v exist and tend to infinity. Let $T_v = \log H_v$. Irrespective of any ties,

$$F(T_v - 1) < v \leq F(T_v). \quad (25.41)$$

The lower count is strictly below the height of the v -th tuple; no bound on jump multiplicity is needed. Applying (25.12) at both arguments gives

$$v = \kappa T_v + d\sqrt{T_v} + O(T_v^{1/3}).$$

First $T_v \sim v/\kappa$, then $T_v - v/\kappa = O(\sqrt{v})$, and therefore $\sqrt{T_v} = \sqrt{v/\kappa} + O(1)$. Substituting gives (25.13).

Equivalently, for a suitable fixed constant C and all sufficiently large v ,

$$\exp\left(\frac{v}{\kappa} - \frac{d}{\kappa^{3/2}}\sqrt{v} - Cv^{1/3}\right) \leq H_v \leq \exp\left(\frac{v}{\kappa} - \frac{d}{\kappa^{3/2}}\sqrt{v} + Cv^{1/3}\right).$$

The square-root correction is positive because $d < 0$. The unbounded $O(v^{1/3})$ uncertainty in the exponent is why a relative $1 + o(1)$ height equivalent has not been proved.

For maximum binary coordinate length at most the integer $b \geq 1$, substitute $H = 2^b - 1$ to obtain

$$N_{\text{bits}}(b) = \kappa(\log 2)b + \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{\lambda}}\sqrt{b} + O(b^{1/3}).$$

This is maximum bit length, not the sum of coordinate lengths; those are different size conventions.

25.11 The sum of all coordinate bit lengths

There is a different two-term law when the budget is the sum of all twenty-two coordinate bit lengths. For $x \in \mathbb{Z}_{>0}$ let $b(x) = \lfloor \log_2 x \rfloor + 1$, and set

$$S(w) = \sum_{x \text{ a supplied coordinate of } w} b(x), \quad N_{\text{sum}}(B) = \#\{w \in \mathcal{F} : S(w) \leq B\}.$$

This is a raw unsigned coordinate-bit budget: it includes all twenty-two supplied coordinates and excludes delimiters, signs, metadata and computed registers.

Theorem 25.11.1 (Total coordinate-bit budget). *For fixed valid native ports and real $B \rightarrow \infty$,*

$$N_{\text{sum}}(B) = \kappa_s(\log 2)B + \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{3\lambda}}\sqrt{B} + O(B^{1/3}), \quad (25.42)$$

where

$$\kappa_s = \frac{1}{M(3p-2)\lambda} + \sum_{\ell \geq 1} \frac{1}{6M\ell\beta_{M\ell}} = \frac{\kappa}{3} - \frac{1}{M\lambda(3p-2)(3p-3)}. \quad (25.43)$$

All exact integer cutoffs and tied tuples are included.

Proof. The reconstruction formulas have the exact product cancellation

$$fijoy = \frac{R_m y (U + p)(U + c)}{c^3}. \quad (25.44)$$

Put $\beta = \beta_m$ and $q_0 = M\lambda$. The hyperbolic expressions give

$$\log R_m = \beta - \log 2 + r_m, \quad \log y = (n-1)\beta + \theta_{m,n}, \quad \log U = (n-1)\beta + u_{m,n},$$

with, uniformly on every admissible pair,

$$0 \leq r_m = \log(1 + e^{-2\beta}) \leq r_* := \log(1 + e^{-2q_0}), \\ 0 \leq \theta_{m,n} \leq d_0 := -\log(1 - e^{-2q_0}), \quad -r_* \leq u_{m,n} = \log(1 + e^{-2n\beta}) - r_m \leq 0.$$

Also $U \geq 1$. Taking logarithms in (25.44) therefore proves

$$\log(fijoy) = (3n-2)\beta_m + O(1) \quad (25.45)$$

uniformly over the entire pair family. For example the absolute error is bounded by $\log 2 + 3 \log c + 3r_* + d_0 + \log(1+p) + \log(1+c)$.

Define the comparison energy $E_0(m, n) = (3n - 2)\beta_m$, and let $G(T)$ count admissible pairs with $E_0 \leq T$. For $n = p$,

$$E_0(M\ell, p) = (3p - 2)M\lambda\ell + O(1),$$

so its count is $T/[M(3p - 2)\lambda] + O(1)$. For each sign, the exact nonbaseline energy is

$$E_{0,\sigma}(\ell, k) = 12M\ell\beta_{M\ell}k + (3\sigma p - 2)\beta_{M\ell}.$$

Use the proof of Section 25.9 with g_ℓ replaced by $12M\ell\beta_{M\ell}$ and a_0 by $12\lambda M^2$. The row estimate is still $T/g_\ell + O(1)$ because the offset divided by g_ℓ is $(3\sigma p - 2)/(12M\ell)$. The reciprocal tail is still $1/(a_0L) + O(L^{-2})$, and $|E_{0,\sigma} - a_0k\ell^2| \leq Ck\ell$ follows from (25.29). The energy is strictly increasing in both variables, so the exact sign-specific rectangle (25.36) applies unchanged. The same Euler sum and quadratic cancellation yield

$$G(T) = \kappa_s T + \frac{\zeta(1/2)}{M\sqrt{3\lambda}}\sqrt{T} + O(T^{1/3}).$$

In particular, the exceptional baseline has coefficient $3p - 2$, not $3(p - 1)$, which proves the correction in (25.43).

Let C_{bits} be the fixed sum of the seventeen forced bit lengths. For the five varying coordinates,

$$\log_2(fijoy) < b(f) + b(i) + b(j) + b(o) + b(y) \leq \log_2(fijoy) + 5.$$

Together with (25.45), this gives a fixed constant C for which

$$|(\log 2)S(w) - E_0(m, n)| \leq C.$$

Hence, for every sufficiently large real B ,

$$G((\log 2)B - C) \leq N_{\text{sum}}(B) \leq G((\log 2)B + C).$$

Apply the asymptotic for G to both sides. A fixed shift changes its linear term by $O(1)$ and its square-root term by $O(B^{-1/2})$, proving (25.42). This global sandwich includes powers of two and every rounding boundary; individual floors need not be stable. $\square$

If B_v is the v -th total bit length, in its own nondecreasing order with multiplicity, put

$$a_s = \kappa_s \log 2, \quad e_s = \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{3\lambda}}.$$

The inequalities $N_{\text{sum}}(B_v - 1) < v \leq N_{\text{sum}}(B_v)$ and the same inversion as in Section 25.10 give

$$B_v = \frac{v}{a_s} - \frac{e_s}{a_s^{3/2}}\sqrt{v} + O(v^{1/3}).$$

The ordering by total bits need not agree with the ordering by maximum height. Both formulas are for their stated encoding convention.

25.12 Transport to the canonical outer history

This section is a consequence of the original Report 22’s entire-fiber projection theorem [7], rather than a new audit of its compiler or physical-clock construction.

[write] Report 22’s entire-fiber projection theorem is printed with Report 22 (source 23, Part VIII of `signal-machine-collision-certificates`). The corollary below is conditional on it and was not re-audited in this write.

Fix an accepted nonempty input and one of its retained clock models. Determinism and first-halt semantics fix the whole trace, final payload and physical clock. The selected outer height is the unique power of two in $(S, 2S]$, where S is the sum of the two encoded endpoint values and the physical clock. The physical duration bounds already dominate every true quotient, so the packed trace fits at this exact height. The retained no-wrap argument recovers the exact clock from its congruence. All outer supplied coordinates and all native boundary ports are thereby fixed.

Write w_{out} for this unique outer tuple. Report 22 checked that the five nonnative comparisons have no native witness dependency and that the full native block is retained literally. Projection to the twenty-two native coordinates is therefore a bijection: two complete witnesses with the same native projection have the same fixed outer tuple, and adjoining w_{out} to any native witness makes all native and outer comparisons vanish.

Corollary 25.12.1 (Conditional transport through the established outer bijection). *For each such accepted nonempty input, all complete witnesses are*

$$(w_{\text{out}}, w_{m,n}), \quad (m, n) \text{ as in (25.6).}$$

If C_{out} is the maximum supplied outer coordinate, their complete height is

$$\max(C_{\text{out}}, \psi_{R_m}(n)).$$

Their count is zero below C_{out} and equals $N(H)$ for $H \geq C_{\text{out}}$. Thus the two-term asymptotic and logarithmic ranked-height law have the same coefficients. At finite ranks the complete height is $\max(C_{\text{out}}, H_v)$.

This is the only plateau possibility discussed here: it comes from additional fixed outer coordinates. It is distinct from the native height law (25.8). The predecessor’s separate no-witness initial-halt circuits are outside this nonempty-fiber corollary.

25.13 Reproduction and scope

The packet distinguishes a mathematical proof, finite auxiliary evidence, and a sealed reproducibility artifact. Its README gives the precise commands and inventory. The core checker independently audits every literal native comparison with sparse integer-polynomial arithmetic, verifies the required pinned inputs, and uses explicit failures that remain active under optimized Python. It imports no upstream author program. Exact integer tests check the main-index divisibility equivalence, positive reconstruction, congruence residues, height domination and count values immediately below, at and above Pell-height jumps.

The second-term verification supplements the analytic proof with exact boundary tests and numerical auxiliary models. Small test values are not examples of a complete padded native tuple unless the native hypotheses have separately been checked; the astronomical native tuple has not been materialized.

Floating-point agreement is never used to prove the classification, floors at a boundary, or the asymptotic theorem.

The release includes source provenance, independently reviewed mathematical statements, portable checker adaptations, deterministic receipts and a strict manifest. Replays run the substantive checks in both normal and optimized Python, test rejection of altered or missing required material, and compare the entire release inventory and bytes before and after. A fresh extracted ZIP is the final replay target. The PDF build requires a local LaTeX installation; the verification code itself uses only the Python standard library. Source hashes and any portable text/code adaptation hashes are kept distinct in the provenance record.

25.13.1 Attribution and limits

Pell generation, strong divisibility, step-down congruences, fixed-base logarithmic counting and Euler–Maclaurin summation are classical. The native repository already contains the crucial relaxed-norm integrality recovery, fixed-minus necessity and parity analysis, and a positive converse. The present compiler-specific deduction is the exact seventeen/five split, the full two-index parameter set with its converse, native maximum-coordinate domination, the baseline-corrected whole-fiber coefficient, and the pointwise two-term count for that exact fiber.

These statements do not change the native circuit, add an uncharged operation, prove circuit minimality, establish a general finite-fold MRDP result, or produce a full numerical Pell witness. Nor is a literature-wide open-problem or novelty claim made. The asymptotic parameters are fixed: neither the error constants nor its onset are asserted uniform across different external inputs.

Binary powering gives effective evaluation and exact comparison in the two quadratic rings. One can locate integer cutoffs by monotone searches rather than floating-point inversion. This is computability, not a small-witness claim: a coordinate at normalized index n requires $\Theta(n \log \varepsilon_m)$ output bits, and the fixed native coordinates can already be enormous. The least-height pair and exact counts are mathematical descriptions; they do not implement a canonical choice by the retained polynomial.

25.13.2 Further mathematical questions

Several bounded questions remain. Can the $O((\log H)^{1/3})$ error be improved, or can further oscillatory terms be identified without smoothing? Can the onset and constants be controlled uniformly while the external ports vary? What is the cost of enforcing a preferred auxiliary-index choice in the retained arithmetic model, with every operation charged? How do other explicit encodings change the coefficient and secondary term? None of these improvements is claimed here.

References of Report 25

- [1] V. Reshetnikov, *Native controller binary selector 56*, pinned repository proof, Sections 2–3 and 5. [Pinned source](#). Local snapshot: `native_controller_binary_selector56.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
- [2] V. Reshetnikov, *Native binary masked selection 63*, prescribed-scale AND interface, Section 2. [Pinned source](#). Local snapshot: `native_binary_masked_selection63.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
- [3] V. Reshetnikov, *Recovering integrality after relaxing an auxiliary Pell norm*. [Pinned source](#). Local snapshot: `PELL_RELAXED_AUXILIARY_PROOF.md`. [write] Shipped as

- `07-native-fiber-source-PELL_RELAXED_AUXILIARY_PROOF.md`; it differs from the research-tree file only by one added final newline.
- [4] V. Reshetnikov, *Fixed-minus index parity*, Sections 2–3. [Pinned source](#). Local snapshot: `EXPLORATION_FIXED_MINUS_INDEX_PARITY.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
 - [5] V. Reshetnikov, *Odd-index Pell signs*. [Pinned source](#). Local snapshot: `EXPLORATION_ODD_INDEX_PELL_SIGNS.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
 - [6] NIST Digital Library of Mathematical Functions, Section 25.2, [equation 25.2.8](#) and [equation 25.2.3](#). Accessed 3 October 2026.
 - [7] *Canonical outer histories and infinite native fibers*, Research report 22, original release, 3 October 2026. Entire-fiber projection theorem and its dependency audit. This prior result is used only for Section 25.12; its original release remains unchanged. [write] Batch 82, cluster M2; source 23 of `signal-machine-collision-certificates`, in its Part VIII (placed by commit 7d2b1b245).

Part IV

Failure of positive index restoration, and the height and counting of one counterexample family

[write] This Part prints two manuscripts. Report 33 (Sections 33.*k*) proves that the signed19 child of the complete74 parent has positive zeros with negative restored index at every positive input; Report 34 (Sections 34.*k*) counts one canonical family of those zeros. Report 34 is a direct sequel of Report 33 and shipped Report 33’s source, byte-identical to the text printed here, as context.

Report 33. Failure of Positive Index Restoration: the complete74 signed19 eliminated child

[write] Source 10 of batch 82: *Research Report 33*, author line “Mathematical research report”, dated 3 October 2026; archive `Failure_of_Positive_Index_Restoration_Package.zip`; pin 2dde7850f, context 93ad43375.

Abstract of the source. Deleting a positive Diophantine witness by a polynomial graph substitution need not preserve the positive existential projection. We prove that the complete74 signed19 index-eliminated child has infinitely many strictly positive supplied-witness zeros at every strictly positive ordinary input, for every actual fixed compiler export in the pinned source. At every constructed zero, the uniquely restored index is negative. The proof incorporates the full transport, shifted packing, first and main Pell norms, strong auxiliary block, and signed input block. It uses one prime in a primitive progression, compatible linear congruences, an irrational rotation, and exact Pell identities. All nineteen supplied child coordinates are positive; a negative computed intermediate is permitted. The established parent theorem and the previously audited graph identity remain intact. No improved universal operation bound, machine-specific false acceptance, or fully materialized full child witness is claimed. A portable independently authored finite-check packet accompanies the proof and is explicitly separated from the unbounded existence argument.

[write] The programme’s review `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) finds no gap in the construction of Sections 33.4–33.9 and also passes the generalized supplement of Section 33.A. Thirty-eight members of the archive are byte copies of research-tree files (README). In this file Report 33’s macro for the native mask is `\MFnat` and for the paid mask `\MFsrc` (Section 0.2). The source’s files are shipped with the prefix `10-index-restore-`.

33.1 Result and exact scope

The relevant source is pinned to the ProveIt repository at commit

2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff.

The nonlinear-index scout removes one positive supplied coordinate from each of three complete74 parent presentations. It correctly proves an all-value polynomial graph identity, but leaves positivity of the restored coordinate open. This report resolves that open obligation negatively for its signed19 child. The names *signed20* and *signed19* concern allowed signed intermediate registers. Their supplied existential coordinates are still required to be strictly positive integers.

[write] The research tree now records this: its review `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) finds no gap in the construction and states that Report 33 “settles the signed19 case”; the programme’s README calls the signed19 candidate refuted, and raw29 and positive21 open. The tree’s later refinement `complete74_negative_index_refinement.md` (commit 813c1cff4) had excluded the signed child from its own scope (“Domain limitation”).

Theorem 33.1.1 (Full positive child zeros with negative restoration). *Fix any actual half-binomial compiler export used by the complete74 signed parent. In particular let $B = 2^d \geq 16$, with d a power of five, let b be positive and odd, and let $K_0 = D_C + BD_R \geq 0$. Retain the actual fixed masks M_C, M_F^{native} unchanged and put $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$. For every $x \in \mathbb{Z}_{>0}$, the complete signed19 child has infinitely many zeros in its nineteen strictly positive supplied coordinates such that*

$$R = k - hXY - 1 < 0.$$

The computed input root μ is positive at these zeros.

The source additionally specifies $b < B$, $0 < M_C, M_F^{\text{native}} < B - 1$, and the appropriate fixed mask congruences. The construction below uses no alteration of these numerals and works for every such export. In fact it only needs the displayed structural conditions on B, d, b, K_0 , with the remaining mask numerals fixed integers. The stronger compiler facts are retained when interpreting the result as a theorem about that particular source.

Corollary 33.1.2. *The positive existential projection of this signed19 child onto its positive ordinary input is all of $\mathbb{Z}_{>0}$. The proposed restoration map from all positive child zeros to positive parent zeros fails.*

This is a failure of the proposed positive-domain equivalence. The original source audit did not assert that equivalence. The positive parent retains $R > 0$ as a hypothesis; no constructed assignment here belongs to that parent domain. The report makes no claim about the acceptance status of a particular input for a chosen machine. It also makes no corresponding existence or nonexistence claim for the separate raw29 and positive21 children.

33.1.1 Operation counts are context only

The pinned scout and its source review give the following costs for the emitted children, using the original parent labels. These are source accounting statements, not universal bounds newly established here.

Parent label	Child witnesses	Equations	Full SOS cost	Degree
raw30	29	18	127	52
positive22	21	10	103	84
signed20	19	8	97	84

Each comparison schedule is $40M + 34A = 74$, and the full sum-of-squares cost is $74 + 3e - 1$ for e comparisons. Thus the signed candidate has $48M + 49A = 97$. These figures do not certify a universal representation on the intended positive domain: Theorem 33.1.1 invalidates this signed candidate’s proposed restoration route. The established universal74 and universal86 parent bounds are not changed by this report.

[write] At the pin the programme’s established bounds were 74 operations for a complete certificate and 86 for one universal polynomial. They are now 74 and 85 (commit 109aaca15, exact degree 175), with an 86-operation alternative of exact degree 131 (commit 55d248dc5); Report 33 changes none of them.

33.2 The exact polynomial interface

We state the entire child interface so that neither its scope nor a hidden sign convention can be imported from a different presentation. The ordinary input x and precisely the coordinates

$$J, F, \alpha, z, f, h, i, j, o, s, w, \tau, \eta, \zeta, y, Z, \delta, \rho, \sigma \quad \text{belong to } \mathbb{Z}_{>0}. \quad (33.1)$$

All other quantities below are computed expressions; they are not additional existential coordinates. Define

$$q = (B - 1)J + 1, \quad X = wq^3, \quad Y = sq^3, \quad E = XY. \quad (33.2)$$

$$k = \eta + \zeta, \quad a = Y(X + 1), \quad c = kY + \eta, \quad (33.3)$$

$$\begin{aligned} A &= a + 2, & H &= 4a + 3, & \Delta &= a^2 + H = A^2 - 1, \\ \gamma &= \rho + \sigma, & D &= X + ac + \gamma H, & R &= k - hE - 1, \end{aligned} \quad (33.4)$$

$$u = 2dx + b, \quad \kappa = u + \delta\Delta, \quad C = q - \alpha - 2dx, \quad (33.5)$$

$$W = C - Z, \quad \mu = W + a\kappa + \rho H.$$

For clarity, the source's register named **A** is our Δ , whereas our $A = a + 2$ is a mathematical abbreviation. The source's **Jrep**, **zquot**, and **y_aux** are J, z, y ; its **R10b**, **R10a**, **R12**, and **R14** are k, c, a, D . Its paid field-mask port is M_F^{src} , not the native mask M_F^{native} .

The eight retained equations are exactly

$$(K_0 + X)C = F + z(q - 1), \quad (33.6)$$

$$R = (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F^{\text{src}})J, \quad (33.7)$$

$$\tau^2 - XY^2(XY^2 + 1)k^2 = 1, \quad (33.8)$$

$$D^2 - \Delta c^2 = 1, \quad (33.9)$$

$$(ic^2)^2 = \Delta(f^2 - 1), \quad (33.10)$$

$$(ic^2)^2((jc - R)^2 - y^2) = 1 - y^2, \quad (33.11)$$

$$jc - R = of - c, \quad (33.12)$$

$$\mu^2 - \Delta\kappa^2 = 1. \quad (33.13)$$

The child polynomial F_{19} is the sum of the squares of their residuals. Over the integers, it vanishes if and only if all eight equations hold. The first factored paid comparison is

$$(EkY)(EkY + k) = \tau^2 - 1.$$

Because $E = XY$, its left side equals $k^2XY^2(XY^2 + 1)$, exactly as in (33.8). No weaker auxiliary square has been substituted in (33.10) or (33.11).

33.2.1 Why the graph identity does not supply positivity

In the signed20 parent, R is itself a positive supplied coordinate, and there is a ninth equation $k = R + 1 + hE$. Its index-eliminated descendant substitutes $R = k - hE - 1$ in both packing and $jc - R$, then removes only that now-identical ninth residual. Consequently

$$F_{19}(v) = F_{20}(v, R = k(v) - hE(v) - 1)$$

as a polynomial identity over every commutative ring. On integer zero sets it gives inverse deletion and restoration maps without sign restrictions. On positive zero sets, the inverse is valid exactly on the slice $k > hE + 1$. Theorem 33.1.1 supplies positive child zeros outside this slice.

The source proof for the parent uses positive R before deriving a packing bound. More explicitly, its shifted-mask rewrite is

$$\begin{aligned} S' &= Z + qF - 1, \\ T' &= M_C J + 1 + q(M_F^{\text{native}} J - 1), \\ R &= (q^2 - S')(q^2 - 1) + T', \quad 0 < T' < q^2 - 1. \end{aligned}$$

Positive R rules out $S' > q^2$. Without that hypothesis the same formula permits negative representatives. Importing the parent's consequent bounds into the eliminated child would be circular.

33.3 Elementary Pell identities used in the proof

For an integer $A \geq 2$ and an integer $r \geq 0$, define $\chi_A(r), \psi_A(r)$ by

$$\chi_A(r) + \psi_A(r)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^r. \quad (33.14)$$

These are integer polynomials in A . Their initial values are $(\chi_A(0), \psi_A(0)) = (1, 0)$ and $(\chi_A(1), \psi_A(1)) = (A, 1)$; both obey $v_{r+1} = 2Av_r - v_{r-1}$. The positive coefficient sequence $\psi_A(r)$ is strictly increasing for $r \geq 1$.

Lemma 33.3.1 (Norms and congruences). *Write $\Delta = A^2 - 1$. For nonnegative indices,*

$$\chi_A(r)^2 - \Delta\psi_A(r)^2 = 1, \quad (33.15)$$

$$\psi_A(r+s) = \psi_A(r)\chi_A(s) + \chi_A(r)\psi_A(s), \quad (33.16)$$

$$\psi_A(r) \equiv rA^{r-1} \pmod{\Delta} \quad (r \geq 1). \quad (33.17)$$

If A is even, $\psi_A(r) \equiv r \pmod{2}$. If $A \equiv 1 \pmod{m}$, then $\psi_A(r) \equiv r \pmod{m}$. If $r \geq 3$ is odd, then $\psi_A(r) > r$ and $\psi_A(r) \equiv r \pmod{\Delta}$.

Proof. Multiplication of (33.14) proves the addition law; conjugation proves the norm. In the binomial expansion the only term surviving in the square-root coefficient modulo Δ is rA^{r-1} . For odd r , $A^{r-1} \equiv 1 \pmod{\Delta}$. The stated parity and modulus- m formulas follow from the recurrence and initial values. Finally $\psi_A(2) = 2A > 2$, and induction in the recurrence gives $\psi_A(r+1) > \psi_A(r) + 1$ for $r \geq 2$. $\square$

Lemma 33.3.2 (Projection congruence and growth). *Let $a \geq 1$, $A = a + 2$, $H = 4a + 3$, and set*

$$e_A(r) = \chi_A(r) - a\psi_A(r).$$

Then $e_A(0) = 1$, $e_A(1) = 2$, and

$$e_A(r) \equiv 2^r \pmod{H}, \quad e_A(r) = 2\psi_A(r) - \psi_A(r-1) > \psi_A(r) \quad (r \geq 1). \quad (33.18)$$

In particular $e_A(r)$ grows exponentially with r for fixed A .

Proof. The sequence e_A has recurrence coefficient $2A$. The candidate 2^r satisfies that recurrence modulo H because $2A \cdot 2 - 1 = 4a + 7 \equiv 4 \pmod{H}$, and its initial values agree. The standard recurrence identity $\chi_A(r) = A\psi_A(r) - \psi_A(r-1)$ gives the second formula; strict growth of ψ_A gives the inequality. Formula (33.14), or its conjugate, gives exponential growth. $\square$

Lemma 33.3.3 (Odd quotient polynomial). *There is an integer polynomial $\mathcal{Q}_b(z)$ for each $b \geq 0$ such that*

$$\chi_Q(2b+1)/Q = \mathcal{Q}_b(Q^2), \quad (33.19)$$

$$\mathcal{Q}_b(0) = (-1)^b(2b+1), \quad (33.20)$$

$$\mathcal{Q}_b(1-A^2) = (-1)^b\psi_A(2b+1). \quad (33.21)$$

The first identity is a polynomial identity, with the quotient interpreted polynomially even at $Q = 0$.

Proof. The odd polynomial $\chi_Q(2b+1)$ is divisible by Q . Its quotient is a polynomial in Q^2 . More explicitly,

$$\mathcal{Q}_0(z) = 1, \quad \mathcal{Q}_1(z) = 4z - 3, \quad \mathcal{Q}_{b+1}(z) = (4z - 2)\mathcal{Q}_b(z) - \mathcal{Q}_{b-1}(z).$$

This follows from the two-step Pell recurrence. At $z = 0$ the initial values and recurrence give (33.20). At $z = 1 - A^2$, its recurrence coefficient is $2 - 4A^2$. The sequence $(-1)^b\psi_A(2b+1)$ has this coefficient and the same initial values $1, 1 - 4A^2$, proving (33.21). $\square$

Lemma 33.3.4 (A divisibility lift). *If $c = \psi_A(p)$ and $m = cp$ with $p, c \geq 1$, then $c^2 \mid \psi_A(m)$.*

Proof. Write $D = \chi_A(p)$. The square-root coefficient in $(D + c\sqrt{\Delta})^c$ is $\psi_A(cp)$. Its term with one square root is $c^2 D^{c-1}$. Every higher odd term contains at least c^3 . Thus each term is divisible by c^2 . $\square$

33.4 Fixing the input and a compatible scale

Fix an arbitrary $x \in \mathbb{Z}_{>0}$ and an actual fixed compiler export. Choose a power of five N large enough that

$$q = B^N > 2dx + 1.$$

Put

$$\begin{aligned} t &= dN, & J &= \frac{q-1}{B-1}, & p_0 &= 12t + 7, \\ X &= 2^{p_0}, & w &= 2^{p_0-3t}, & Q &= q^2 - 1. \end{aligned} \quad (33.22)$$

Here t is a power of five, $q = 2^t$, $p_0 \equiv 3 \pmod{4}$, and $p_0 \equiv 1 \pmod{6}$. Thus J, w are strictly positive integers and $X = wq^3$. Set

$$C = 1, \quad \alpha = q - 1 - 2dx, \quad z = 1, \quad F = K_0 + X - q + 1, \quad M = (M_C + qM_F^{\text{src}})J. \quad (33.23)$$

The coordinates α, z, F are positive, since $X \geq q^3 > q$ and $K_0 \geq 0$. The definition of C and the transport equation (33.6) are now exact. The fixed input index

$$u = 2dx + b$$

is odd and at least three.

33.4.1 A prime progression removes the remaining congruence obstruction

Define

$$D_0 = \frac{4q^3(X+1)}{3}. \quad (33.24)$$

This is an integer because p_0 is odd.

Lemma 33.4.1. *One has $\gcd(D_0, Q) = 1$.*

Proof. Since t is a power of five, $\gcd(p_0, 2t) = \gcd(12t + 7, 2t) = 1$. We show $\gcd(2^{p_0} + 1, 2^{2t} - 1) = 3$. Any common divisor g is odd. Choose integers r, s with $rp_0 + 2st = 1$. The integer r is odd, so, using inverses modulo g if necessary, $2 = (2^{p_0})^r (2^{2t})^s \equiv (-1)^r = -1 \pmod{g}$. Thus $g \mid 3$, and both expressions are divisible by three. Moreover $p_0 \equiv 1 \pmod{6}$ gives $X + 1 \equiv 3 \pmod{9}$. Removing its sole factor of three makes $(X + 1)/3$ coprime to Q . Since Q is odd and coprime to q , (33.24) proves the assertion. $\square$

For each prime divisor r of the odd integer Q , avoid the two distinct residues 0 and $-D_0^{-1}$ modulo r . This is possible because $r \geq 3$. The Chinese remainder theorem, followed by any lift to the prime powers if needed, provides s_0 with

$$\gcd(s_0, Q) = \gcd(1 + D_0 s_0, Q) = 1. \quad (33.25)$$

The progression

$$\ell \equiv 1 + D_0 s_0 \pmod{D_0 Q} \quad (33.26)$$

is primitive: its residue is one modulo D_0 and a unit modulo Q . Dirichlet's theorem therefore supplies arbitrarily large primes in it [1]. Fix one with $\ell > 3$ and put

$$\begin{aligned} s &= (\ell - 1)/D_0, & Y &= sq^3, & E &= XY, \\ a &= Y(X + 1), & A &= a + 2, & H &= 4a + 3 = 3\ell, \\ \Delta &= A^2 - 1, & P &= 2XY^2 + 1. \end{aligned} \quad (33.27)$$

The coordinate s is a positive integer congruent to s_0 modulo Q . In particular it is coprime to Q and not divisible by three. The latter follows because $3 \mid Q$. As D_0 is also a unit modulo three and ℓ is a unit modulo three, $\ell - 1 = D_0 s$ is nonzero modulo three; consequently $\ell \equiv 2 \pmod{3}$.

Let $O = \text{ord}_H(2)$, which exists because H is odd. Since $H = 3\ell$,

$$O \mid \text{lcm}(2, \ell - 1) = \ell - 1 = D_0 s.$$

The number $D_0 s$ is coprime to Q , to three, and to ℓ , so $\gcd(QH, O) = 1$. Separately $\gcd(QH, 2E) = 1$: the factors of Q do not divide s or the powers of two X, q ; neither three nor ℓ divides s , and both are odd. Therefore, with

$$T = \text{lcm}(4, 2E, O), \quad L = E/2, \quad (33.28)$$

we have the crucial compatibility

$$\gcd(QH, T) = 1. \quad (33.29)$$

The scale, including this one chosen prime, is now held fixed. Dirichlet's theorem is used only here; the infinitely many full child zeros below come from varying the main Pell index.

33.5 The entire outer and input block in one free progression

Before choosing the main index, set

$$\kappa = \psi_A(u), \quad \mu_0 = \chi_A(u), \quad e_u = e_A(u), \quad \delta = \frac{\kappa - u}{\Delta}. \quad (33.30)$$

Lemma 33.3.1 makes δ a strictly positive integer. Impose the two congruences

$$p \equiv p_0 \pmod{T}, \quad (33.31)$$

$$p \equiv -M - Q(q^2 - qF + e_u - 1) \pmod{QH}. \quad (33.32)$$

By (33.29), their solutions form one nonempty arithmetic progression

$$p = p_* + Vr, \quad V = TQH, \quad r \in \mathbb{Z}. \quad (33.33)$$

In particular p remains a freely varying linear index. It has not been restricted to values of a Pell sequence or to another nonlinear sparse set. Every solution has

$$p \equiv 3 \pmod{4}, \quad 2^p \equiv X \pmod{H}, \quad p \equiv p_0 \pmod{2E}. \quad (33.34)$$

Choose an integer n_0 representing $(1 - p_0)/2$ modulo $L = E/2$. For every

$$n = n_0 + Lv \quad (33.35)$$

and every p satisfying (33.31),

$$2n \equiv 1 - p \pmod{E}. \quad (33.36)$$

For sufficiently large p in (33.33), define

$$Z = q^2 - qF + \frac{p + M}{Q}, \quad \rho = \frac{Z + e_u - 1}{H}, \quad W = 1 - Z. \quad (33.37)$$

Reduction of (33.32) modulo Q makes Z integral; division by Q in the same congruence makes ρ integral. Both are eventually strictly positive: as functions of p , their slopes are respectively $1/Q$ and $1/(QH)$. The computed root is

$$W + a\kappa + \rho H = e_u + a\kappa = \chi_A(u) = \mu_0 > 0. \quad (33.38)$$

Hence the complete input definition, the definition $\kappa = u + \delta\Delta$, and the input norm (33.13) all hold. Also

$$(q^2 - Z - qF)Q + M = -p. \quad (33.39)$$

Thus the packing equation is exact with $R = -p$. Notice that $W = 1 - Z$ is eventually negative. It is a computed intermediate, not one of the nineteen supplied coordinates in (33.1). No positivity condition on W may be added to this child.

33.6 An irrational rotation supplies the strict Pell ratio

The only missing coupling between the main and first Pell indices is the strict interval

$$Y < \frac{\psi_A(p)}{2\psi_P(n)} < Y + 1. \quad (33.40)$$

It gives the two positive decomposition witnesses η, ζ . We must obtain it on the already imposed progressions, without discarding any outer equation.

Lemma 33.6.1 (Independent quadratic growth rates). *For the fixed parameters above, put*

$$\lambda = A + \sqrt{\Delta}, \quad \nu = P + \sqrt{P^2 - 1}, \quad \alpha_0 = \log \lambda, \quad \beta_0 = \log \nu.$$

Then α_0/β_0 is irrational.

Proof. The integer A is even, hence Δ is odd. On the other hand

$$P^2 - 1 = 4XY^2(XY^2 + 1), \quad v_2(P^2 - 1) = 2 + p_0 + 2v_2(Y)$$

has odd 2-adic valuation. Thus the squarefree part of Δ is odd and that of $P^2 - 1$ is even. Both numbers are nonsquares: for any integer $z \geq 2$, $z^2 - 1$ lies strictly between consecutive squares. Their real quadratic fields are therefore distinct and have intersection $\mathbb{Q}$.

If α_0/β_0 were rational, there would be positive integers r, s with $\lambda^r = \nu^s$. This common positive number would belong to the intersection $\mathbb{Q}$. Its norm in either quadratic field is one; the norm of a rational number q is q^2 , so positivity would force it to be one. This contradicts $\lambda, \nu > 1$. $\square$

Lemma 33.6.2 (Density on the imposed progressions). *There are infinitely many pairs of positive integers (p, n) satisfying (33.33), (33.35), and (33.40), with both p and n tending to infinity.*

Proof. For a positive irrational θ , every forward tail of the rotation $r\theta$ modulo one is dense. One elementary proof is to use the pigeonhole principle to obtain nonzero integer multiples arbitrarily close to zero modulo one; successive steps by such a multiple approximate any point. The closure of the integer orbit is thus the entire circle. The closure of a forward tail is the same, because positive multiples approach zero and hence approximate the negative multiples as well. In particular every nonempty open arc is visited infinitely often.

Apply this fact to

$$\theta = \frac{V\alpha_0}{L\beta_0},$$

which is irrational by Lemma 33.6.1. Write

$$c_0 = \log \frac{\sqrt{P^2 - 1}}{2\sqrt{\Delta}}.$$

Choose a nondegenerate closed interval $I = [a_0, b_0]$ strictly inside $(\log Y, \log(Y + 1))$, small enough that its length is less than $L\beta_0$. The affine expression

$$(p_* + Vr)\alpha_0 - n_0\beta_0 + c_0$$

is dense modulo $L\beta_0$ as positive integers r vary. Infinitely many such r admit an integer v for which

$$(p_* + Vr)\alpha_0 - (n_0 + Lv)\beta_0 + c_0 \in I. \quad (33.41)$$

To see that the closed interval causes no endpoint issue, apply density to its nonempty interior. Since the left side is bounded in (33.41),

$$v = \frac{V\alpha_0}{L\beta_0}r + O(1),$$

so v , and then $n = n_0 + Lv$, tend to positive infinity along the selected hits. So does $p = p_* + Vr$.

The exact Binet formula gives

$$\begin{aligned} \log \frac{\psi_A(p)}{2\psi_P(n)} &= p\alpha_0 - n\beta_0 + c_0 + \varepsilon_{p,n}, \\ \varepsilon_{p,n} &= \log(1 - e^{-2p\alpha_0}) - \log(1 - e^{-2n\beta_0}). \end{aligned}$$

The error tends to zero. The positive distance between I and the complement of $(\log Y, \log(Y+1))$ absorbs this error at every sufficiently large selected hit. Exponentiating proves (33.40). $\square$

No density assertion about prime indices or Pell-valued indices is being used. The prime fixes the scale once; the simultaneous congruences leave a full ordinary arithmetic progression for p .

33.7 All main and first witnesses are positive

For each sufficiently large selected pair from Lemma 33.6.2, define

$$c = \psi_A(p), \quad D = \chi_A(p), \quad k = 2\psi_P(n), \quad \tau = \chi_P(n), \quad (33.42)$$

$$\eta = c - kY, \quad \zeta = k - \eta, \quad \gamma = \frac{D - ac - X}{H}, \quad h = \frac{k + p - 1}{E}, \quad (33.43)$$

$$\sigma = \gamma - \rho, \quad R = -p.$$

The strict ratio (33.40) says exactly $0 < c - kY < k$. Thus η, ζ are positive integers, $k = \eta + \zeta$, and $c = kY + \eta$. The two Pell norms imply

$$D^2 - \Delta c^2 = 1, \quad \tau^2 - 1 = (P^2 - 1)\psi_P(n)^2 = XY^2(XY^2 + 1)k^2.$$

This establishes the two norms (33.8) and (33.9).

By Lemma 33.3.2 and (33.34),

$$D - ac = e_A(p) \equiv 2^p \equiv X \pmod{H}.$$

Hence γ is integral. Moreover $e_A(p) > c$ grows exponentially with p , so γ is eventually positive. The already fixed expression ρ is affine in p . Therefore $\sigma = \gamma - \rho$ is also a strictly positive integer for all sufficiently large selected pairs. In particular the actual stronger source definition $\gamma = \rho + \sigma$, with both summands positive, holds exactly.

Because $P \equiv 1 \pmod{E}$, Lemma 33.3.1 gives $\psi_P(n) \equiv n \pmod{E}$. Consequently

$$k \equiv 2n \equiv 1 - p \pmod{E},$$

so h is a positive integer and

$$k - hE - 1 = -p = R. \quad (33.44)$$

Thus R is exactly the eliminated source's restored expression, not an independently imposed variable. Equation (33.39) is therefore its retained packing comparison.

33.8 The complete strong auxiliary block

We now choose the four remaining auxiliary supplied coordinates f, i, j, o and the Pell coordinate y , verifying their integrality, positivity, and all three retained auxiliary equations.

Since A is even and $p \equiv 3 \pmod{4}$, Lemma 33.3.1 makes $c = \psi_A(p)$ odd. Define

$$m = cp, \quad f = \chi_A(m), \quad Q_{\text{aux}} = \Delta\psi_A(m), \quad i = Q_{\text{aux}}/c^2. \quad (33.45)$$

By Lemma 33.3.4, i is a strictly positive integer. The norm at index m gives

$$Q_{\text{aux}} = ic^2, \quad Q_{\text{aux}}^2 = \Delta(f^2 - 1). \quad (33.46)$$

This is precisely the actual strong square required by (33.10).

Let

$$s_a = p + 2m, \quad b_a = (s_a - 1)/2, \quad U = \frac{\chi_{Q_{\text{aux}}}(s_a)}{Q_{\text{aux}}}, \quad y = \psi_{Q_{\text{aux}}}(s_a). \quad (33.47)$$

Both m and p are odd, and $p \equiv 3 \pmod{4}$, so $s_a \equiv 1 \pmod{4}$ and b_a is even. Lemma 33.3.3 proves that U is a positive integer. We need two exact congruences.

First, $c^2 \mid Q_{\text{aux}}$, so (33.20) gives

$$U = Q_{b_a}(Q_{\text{aux}}^2) \equiv s_a \equiv p \pmod{c}. \quad (33.48)$$

Second, (33.46) gives $Q_{\text{aux}}^2 \equiv -\Delta = 1 - A^2 \pmod{f}$. Therefore (33.21), with b_a even, gives

$$U \equiv \psi_A(p + 2m) \pmod{f}.$$

The doubling formulas are

$$\chi_A(2m) = 2f^2 - 1, \quad \psi_A(2m) = 2f\psi_A(m).$$

Using the addition law (33.16), we obtain

$$U \equiv -\psi_A(p) = -c \pmod{f}. \quad (33.49)$$

Define

$$j = \frac{U - p}{c}, \quad o = \frac{U + c}{f}. \quad (33.50)$$

Equations (33.48)–(33.49) prove their integrality. For all sufficiently large selected p , one has $c > p$, $Q_{\text{aux}} \geq c^2$, and $s_a \geq 3$. Monotonicity of the positive Pell root then gives

$$U \geq \frac{\chi_{Q_{\text{aux}}}(3)}{Q_{\text{aux}}} = 4Q_{\text{aux}}^2 - 3 > c > p.$$

Thus j, o are strictly positive. Since the restored index is $R = -p$,

$$U = jc + p = jc - R = of - c,$$

which proves (33.12). Finally the norm at parameter Q_{aux} is

$$(Q_{\text{aux}}U)^2 - (Q_{\text{aux}}^2 - 1)y^2 = 1.$$

Rearranging and using $Q_{\text{aux}} = ic^2$ yields

$$(ic^2)^2(U^2 - y^2) = 1 - y^2,$$

exactly (33.11). This completes the strong auxiliary block, including the sign that is essential for a negative restored index.

33.9 Completion and domain audit

Every definition in Section 33.2 has now been checked. To collect the strict supplied-coordinate signs without suppressing a domain condition:

- J, w are positive integer powers or repunit expressions; s is positive by the prime choice

- α, z, F are positive by (33.23) and $q > 2dx + 1$
- $\delta > 0$ by the odd input Pell congruence and strict growth
- $Z, \rho > 0$ eventually by their positive affine slopes in p
- $\tau > 0$ and $\eta, \zeta > 0$ by the Pell norm and strict ratio
- $h > 0$ by its positive integral quotient in (33.43)
- $\sigma > 0$ eventually because exponential γ dominates affine ρ
- $f, i, j, o, y > 0$ by the full auxiliary construction

These are precisely the nineteen coordinates in (33.1). Finitely many eventual lower bounds on p can be met simultaneously because Lemma 33.6.2 gives unbounded selected pairs. Equations (33.6)–(33.13) all hold, and therefore $F_{19} = 0$. Yet (33.44) gives $R = -p < 0$. Different selected values of p yield different Z , whose affine slope is $1/Q > 0$. Thus there are infinitely many distinct witness tuples at the same arbitrary fixed x . This proves Theorem 33.1.1 and Corollary 33.1.2.

The computed $\mu = \chi_A(u)$ is positive, while $W = 1 - Z$ is eventually negative. All supplied variables remain positive. Neither a signed input root nor a nonpositive supplied auxiliary coordinate is used to evade a hypothesis.

33.9.1 What remains outside this result

The positive21 child retains a positive supplied W and a stronger positive input-gap structure; raw29 has its own positive supplied interface. The signed19 construction cannot be transferred to those domains because its W is negative. The general kernel family alone also does not settle their additional outer conditions. The signed restoration failure established here should therefore not be relabelled as a failure of every complete74 presentation.

The proof does not derive $X = 2^R$, an accepting typed word, or a no-wrap exponent estimate. Those are conclusions of a positive-index theorem whose sign hypothesis is unavailable here. It also does not produce a new universal Diophantine operation bound. The invalidated signed shortcut and the established positive parent occupy different mathematical domains.

33.10 Proof provenance and reproducibility

The mathematical theorem is established by Sections 33.2–33.9. Finite arithmetic tests do not prove Dirichlet’s theorem, the density step, eventual positivity, or the quantified conclusion. In particular they are not a numerical construction of one full giant witness tuple.

33.10.1 Pinned source evidence

The source definitions are taken from the signed-projection note [2]; the exact substitution, literal domains, and candidate costs are taken from the nonlinear-index scout and its independent source review [3, 4]. The complete74 factored parent receipt supplies the literal arithmetic interface [5]. The half-binomial compiler note explicitly preserves powers-of-five b, L, d and $B = 2^d$ [6]. The unchanged nonnegative sparse definitions of D_C, D_R also support $K_0 \geq 0$; the packet includes the relevant compiler source files as inert text.

A later repository contribution at commit 93ad43375a4a5610de0cfb25d64f84ec018d02f7 records the sign-free bootstrap $p \geq 13$, positivity of the auxiliary argument, raw/positive exact-sign reductions,

and a positive auxiliary realization of $R = -p$ at arbitrarily large indices [7]. We inspected that note and credit the overlapping auxiliary construction. It expressly leaves the full child/compiler system unresolved. Theorem 33.1.1 goes beyond that auxiliary-only scope by satisfying the transport, literal packing, first norm, and complete signed input block simultaneously. This later contextual note does not change the earlier equation pin; its separate cached bytes and manifest are included under `context/`.

All sixteen cached upstream files have their repository paths, exact pinned URLs, SHA256 hashes, and Git blob SHA1 identities recorded in the packet's source manifest. The Git blob hash is computed as SHA1 of the canonical bytes `blob <length>` followed by a zero byte and the file bytes. Reading these files as text or JSON is distinct from executing the upstream programs. No upstream compiler, historical checker, or emitted arithmetic schedule was executed for this report, and no upstream repository file was modified.

33.10.2 Independent proof review

The full all-input proof was independently audited against the exact source interface, including all eight literal residuals, all nineteen signs, the shifted native mask, the CRT coprimality, the unrestricted linear main-index progression, and the strong auxiliary sign. The reviewed proof and audit are byte-bound in the accompanying evidence:

Proof SHA256

b109e2fd1142cdad84a5b519acc5dc5055160f1d8e7617d648a561538fd956cd

Independent audit SHA256

cd62529a6efcdcfbab43f1426c6b3fe5930575fda2cc865f12c40a5ec6c3bda6

The audit verdict is PASS for the full signed19 theorem stated here. The article supplies the supporting identities directly, rather than requiring the reader to combine chronological exploratory reports.

33.10.3 Finite checks and limits

The portable replay uses newly authored Python, the standard library, explicit checks that remain active with Python optimization, and exact-type receipt comparison. Upstream source bytes are only authenticated and read as data. Its identity gate rejects a changed packet before running the authored check program. Normal and optimized replay run from an unrelated working directory with isolated imports and temporary outputs.

The finite evidence comprises an exact recursive Lucas primality certificate for a 98-bit prime, scale and CRT coprimality checks, and 32 complete outer/input progression fixtures at toy fixed numerals. These toy masks are not represented as actual compiler exports. Additional bounded exact identities corroborate the elementary Pell and auxiliary congruence algebra: 451 main Pell/projection cases, 160 first-norm factoring cases, 25 coefficientwise odd-quotient identities and 200 evaluations, and six strong-auxiliary congruence cases. One small auxiliary-only block is materialized; it is not a full child zero. The replay does not numerically locate a density hit or materialize the full main and auxiliary coordinates. The unbounded proof applies to the actual arbitrary fixed exports; the toy fixtures test algebraic components only.

The packet README provides the exact replay commands and scope of each receipt. A failed identity or receipt check is an error rather than a silently regenerated expected result. The article's LaTeX source and a local build script are included; PDF compilation is separate from arithmetic replay and requires a standard LaTeX installation. The archive contains no third-party PDF cache and requires no network access to replay its arithmetic evidence.

33.A A broader algebraic parameter theorem

The powers-of-five convention is sufficient for the actual-compiler result, but is not necessary for failure of restoration in the same algebraic interface. The following extension is separately independently reviewed. It does not assert that arbitrary numeral ports implement a compiler.

Theorem 33.A.1. *Consider exactly the definitions and eight equations in Section 33.2, but assume only $d \in \mathbb{Z}_{>0}$, $B = 2^d$, b a positive odd integer, $K_0 \in \mathbb{Z}_{\geq 0}$, and arbitrary fixed integers M_C, M_F^{src} . At every $x \in \mathbb{Z}_{>0}$, there are infinitely many strictly positive supplied-witness zeros with restored $R < 0$ and computed $\mu > 0$.*

Proof. Choose any positive integer N large enough that $q = B^N > 2dx + 1$, put $t = dN$, and retain $J = (q - 1)/(B - 1)$. Write

$$t = 2^{a_2} 3^{a_3} t_0, \quad \gcd(t_0, 6) = 1.$$

Because $\gcd(12, t_0) = 1$, CRT provides p_0 with

$$p_0 \equiv 7 \pmod{12}, \quad p_0 \equiv 1 \pmod{t_0}.$$

When $t_0 = 1$ the second condition is vacuous. Increase p_0 by a positive multiple of $12t_0$ until $p_0 > 3t$. This replaces only the specific choice $p_0 = 12t + 7$ in the main proof. Every prime factor of t either is two or three, or divides t_0 . Therefore

$$\gcd(p_0, 2t) = 1, \quad p_0 \equiv 3 \pmod{4}, \quad p_0 \equiv 1 \pmod{6}.$$

Set $X = 2^{p_0}$ and $w = 2^{p_0-3t}$ as before. The proof of Lemma 33.4.1 uses exactly these congruences, so $D_0 = 4q^3(X + 1)/3$ is integral and coprime to $Q = q^2 - 1$. The primitive prime progression construction works without change.

There is one parity detail worth making explicit. Here $D_0 \equiv (-1)^t \pmod{3}$, which need not be two. No particular nonzero residue is needed: both D_0 and s are units modulo Q , and $3 \mid Q$, so $\ell - 1 = D_0 s$ is nonzero modulo three. Since ℓ is also a unit modulo three, $\ell \equiv 2 \pmod{3}$ follows. Thus all coprimality culminating in $\gcd(QH, T) = 1$ remain valid.

The input $u = 2dx + b$ is odd and at least three, so the same $\delta > 0$ works without an upper bound on b . Also $F = K_0 + X - q + 1 > 0$. Arbitrary signs of the fixed masks affect only the constant $M = (M_C + qM_F^{\text{src}})J$. CRT still accepts any resulting residue in (33.32); the positive slopes of Z, ρ in (33.37) are unchanged.

Finally q, Y are even for every $t \geq 1$, whereas p_0 is odd. Hence the quadratic-field separation in Lemma 33.6.1 remains true even when t is even. The free progression, density interval, all Pell definitions, the exponential dominance over ρ , and the auxiliary construction use only the properties just established. Every subsequent step of Sections 33.5–33.9 therefore applies and proves the claim. $\square$

This extension changes the algebraic parameter range, not the main theorem's source attribution. For actual exports the paid mask remains $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$ and all compiler conditions remain in force. The independent supplement review is bundled separately from the originally approved main proof and audit.

33.B The unresolved raw29 and positive21 domains

This appendix records a separate audited necessary-condition reduction. It does not prove positivity of the restored index and does not construct a full negative zero in either domain. The source receipts

call the parent modes `raw30` and `positive22`; after deletion the children have 29 and 21 supplied coordinates. Unlike `signed19`, both retain a positive supplied W and a positive input-gap witness. The theorem of the main text has $W < 0$ and therefore cannot settle these cases.

33.B.1 Literal assumptions and imported bootstrap

In addition to the common Pell and scale equations, a positive zero in either domain has

$$\begin{aligned} q &= (B-1)J+1, & C+\alpha+2dx &= q, \\ C &= Z+W, & (K_0+X)C &= F+z(q-1), \\ R &= (q^2-Z-qF)(q^2-1)+(M_C+qM_F^{\text{src}})J, \\ \kappa &= u+\delta\Delta, & c &= \kappa+\phi, \\ \mu &= W+a\kappa+\rho H, & \mu^2-\Delta\kappa^2 &= 1, \end{aligned}$$

where W, Z, α, ϕ and the supplied coordinates are strictly positive. In `raw29`, the appropriate raw quantities are supplied and the equations are retained comparisons; in `positive21` the triangular quantities are computed positive expressions. Thus

$$q \geq B \geq 16, \quad 0 < Z < C < q, \quad 0 < W < C < q, \quad F < (K_0+X)C, \quad 0 < u < 2q. \quad (33.51)$$

The last bound uses the actual compiler's $b < B \leq q$.

The sign-free bootstrap is an imported, independently audited prerequisite here, as also recorded in the later source note [7]; it is not needed for the constructive main theorem. Its precise conclusions are

$$\begin{aligned} c &= \psi_A(p), & p &\geq 13 \text{ odd}, & R &\equiv \pm p \pmod{c}, & (33.52) \\ k &= 2\psi_P(n), & \frac{p+1}{2} &\leq n \leq p-1, & 2n &\equiv R+1 \pmod{E}, \\ c &> 2p, & c &> A^{12}, & P &= 2XY^2+1. \end{aligned}$$

This bootstrap derives positivity of the auxiliary argument and the rank hypotheses without $R > 0$. It invokes the ordinary strong norm, not a weakened replacement. We cite its rank/step-down argument rather than claiming to reprove it in this appendix; its reviewed snapshot and audit are included with the packet.

33.B.2 Why the actual compiler gives the required coefficient bound

The reduction needs $K_0 < B^2$ for actual compiler exports; it would not be legitimate to impose that inequality on arbitrary evaluator ports. Here is the source provenance. Write $R_{\text{rad}} = 2^b$ for the inner radix, and $B = R_{\text{rad}}^{L_{\text{comp}}}$. If S is the sum of the positive center coefficients, the `complete76` sparse D_C consists of four unit monomials, two copies of those coefficients, and at most one optional unit monomial. Every coefficient is therefore at most $2S+5$. All exponents, including the optional high monomial, are below L_{comp} . The half-binomial wrapper raises the radix to at least

$$R_{\text{rad}} \geq 4(K+2)(2S+6)+8 > 2S+5,$$

where K is the number of native positions. Hence there are no radix carries and $0 < D_C < B$. The inherited $D_R = R_{\text{rad}}^{H_{\text{comp}}}$ has $0 < H_{\text{comp}} < L_{\text{comp}}$, so $0 < D_R < B$ too.

The materializers are inherited from `complete78` through `complete77` and `complete76`. `Complete77` changes only the masks/export wrapper. The sparse construction does not access the cached properties

B, D_C, D_R before the half-binomial wrapper raises the radix; hence the export contains no stale smaller-radix cached value. Its `new_constants` function changes only M_C, M_F^{native} after materializing the other numerals. The literal alias chain identifies `Kconstant` with $D_C + BD_R$ and the paid MF with $M_F^{\text{native}} + B - 1$. These facts are checked against the pinned source files in the separate raw/positive audit. Consequently

$$0 < K_0 = D_C + BD_R \leq (B - 1) + B(B - 1) = B^2 - 1 \leq q^2 - 1. \quad (33.53)$$

33.B.3 Only two exact index representatives survive

The shifted-mask identity from Section 33.2 holds with $0 < T' < q^2 - 1$ using only the native mask ranges and the repunit equation; it does not require $q = B^N$. It implies $R \neq 0$ and $R < q^4$. If $R < 0$, dropping the positive mask contribution in the original packing formula and using (33.51) gives

$$\begin{aligned} -R &< (Z + qF)q^2 \\ &< Cq^3(X + K_0 + 1/q) \\ &< CXq^3(1 + 1/q) \\ &\leq (q - 1)Xq^3(1 + 1/q) < Xq^4. \end{aligned}$$

The third step follows from $K_0 + 1/q < q^2 \leq X/q$, by (33.53) and $X \geq q^3$. The positive branch already has $0 < R < q^4 < Xq^4$. Thus $|R| < Xq^4$ in either case. Since $A > XY \geq Xq^3$, one has $A^2 > 2Xq^4$, and (33.52) gives $c > 2Xq^4$. Both $|R|$ and p are therefore less than $c/2$. The congruence $R \equiv \pm p \pmod{c}$ can have only its zero multiple of c , so

$$R = p \quad \text{or} \quad R = -p. \quad (33.54)$$

Both alternatives are odd. Were q odd, the repunit equation with even B would make J even; both terms in the literal packing formula would then be even. Hence q is even, so X, Y, A are even and Δ is odd. This is a parity conclusion, not a proof that q is a power of two.

33.B.4 A negative zero must lie in a bounded window

If $R = -p$, the first-index congruence in (33.52) becomes

$$2n + p - 1 = t_-E$$

for a positive integer t_- . This quotient is not the supplied witness h and is unrelated to the main proof's binary exponent $t = dN$. The interval for n gives

$$2p \leq t_-E \leq 3p - 3.$$

Combining with $p < Xq^4$ yields the necessary restrictions

$$E/3 + 1 \leq p < Xq^4, \quad s < 3q, \quad 1 \leq t_- < 3q/s. \quad (33.55)$$

Packing further requires

$$p = (Z + qF - 1 - q^2)(q^2 - 1) - T', \quad F \geq q.$$

At $F = q$ it requires $Z \geq 2$. Thus $F < q$, a conclusion of the old positive-index proof, cannot be silently imposed here. The window is finite for fixed q, X, Y and compiler numerals, but is not uniformly finite as the witnesses vary.

33.B.5 The retained positive input gives an exact dichotomy

Since the input root and coefficient are positive, the Pell norm supplies an index $v \geq 1$ with $\kappa = \psi_A(v)$ and $\mu = \chi_A(v)$. For $A \geq 2$ the unit $A + \sqrt{A^2 - 1}$ is fundamental: a smaller positive solution would have square-root coefficient at least one and root at least A . Thus the standard positive Pell parametrization applies. The retained $c > \kappa$ gives $v < p$. Also

$$p = |R| < Xq^4 < \Delta,$$

because $\Delta > (XY)^2 \geq X^2q^6$. From $0 < u < 2q < A - 1$, both u and uA belong to $(0, \Delta)$. The discriminant congruence of Lemma 33.3.1 now gives

$$\psi_A(v) \equiv \begin{cases} v & v \text{ odd,} \\ vA & v \text{ even} \end{cases} \pmod{\Delta}.$$

As $\kappa \equiv u \pmod{\Delta}$, uniqueness of the representatives in $(0, \Delta)$ proves

$$v = u \quad \text{or} \quad v = uA. \tag{33.56}$$

For the even case, multiply $vA \equiv u$ by A and use $A^2 \equiv 1 \pmod{\Delta}$ before selecting representatives. Since u is odd and A even, uA is indeed even. On the negative branch this alternative forces $uA < p < Xq^4$, hence $uY < q^4$ and $us < q$.

[write] *The same dichotomy in the research tree.* The tree's `complete74_negative_index_refinement.md`, Section 2 (commit `813c1cff4`, 3 October 2026), proves for `raw30` and `positive22` that the input Pell index is exactly $e = u$ or $e = uA$, by the same discriminant-representative argument. It is later than this Report's pins and not cited by it, and this Report is not cited there: the two proofs are independent, and no priority is claimed either way. The tree's note goes further: on $r = -p$ it proves $4^d > X \geq q^3$ for the defect $d = 2n - p$ (so $d \geq 7$), and that for fixed q, X, Y and a fixed bounded representative at most one main index p satisfies the ratio interval. The programme's review of this Report notes the same coincidence.

The positive branch actually excludes $v = uA$: it has $p = R < q^4 < A$, whereas $v = uA \geq A$. On $R = -p$, no such exclusion has been proved. The remaining projections are only

$$W \equiv 2^v \pmod{H}, \quad 0 < W < q, \quad X \equiv 2^p \pmod{H}.$$

Neither $p < \Delta$ nor $v < p$ rules out exponent wrapping. The unresolved task is to exclude or realize the negative branch while simultaneously meeting (33.55), (33.56), the exact Pell ratio, and every outer and input equation. This appendix establishes neither outcome.

References of Report 33

- [1] P. G. L. Dirichlet, *There are infinitely many prime numbers in all arithmetic progressions with first term and difference coprime*, original proof 1837, English translation by R. Stephan, arXiv:0808.1408v2 (2014). <https://arxiv.org/abs/0808.1408v2>. The prime-progression theorem is used only in Section 33.4.
- [2] ProveIt repository (VladimirReshetnikov), *A 101-operation polynomial from a signed projection*, `complete75_signed_projection_elimination101.md`, pinned commit specified in Section 1, especially Sections 1–3. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_signed_projection_elimination101.md.

- [3] ProveIt repository (VladimirReshetnikov), *Complete74 nonlinear index projection scout*, `complete74_nonlinear_index_projection_scout.md` and its JSON receipt, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_nonlinear_index_projection_scout.md.
- [4] ProveIt, *Independent review of the complete74 nonlinear index scout*, `review_complete74_nonlinear_index_projection.md`, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_complete74_nonlinear_index_projection.md.
- [5] ProveIt, `complete74_factored_first_norm.json`, literal parent source receipt, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_factored_first_norm.json.
- [6] ProveIt repository (VladimirReshetnikov), *Modified compiler masks for the half-binomial75*, `complete75_half_binomial_compiler.md`, same pinned commit, Section 1. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md.
- [7] ProveIt, *Signed first-index bootstrap for the complete74 nonlinear projection*, `review_complete74_nonlinear_index_bootstrap.md`, later commit 93ad43375a4a5610de0cfb25d64f84ec018d02f7. Its theorem is auxiliary-only as to existence of negative solutions. https://github.com/VladimirReshetnikov/ProveIt/blob/93ad43375a4a5610de0cfb25d64f84ec018d02f7/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_complete74_nonlinear_index_bootstrap.md.
- [8] ProveIt, `pell_kernel_half_binomial42.md` and the fixed-minus parity note, `EXPLORATION_FIXED_MINUS_INDEX_PARITY.md`, same pinned commit. These motivate the Pell auxiliary identities; Section 33.3 proves the identities used here independently. Exact repository locations and hashes are in the bundled source manifest.

Report 34. Height and Counting of a Signed19 Family: canonical witnesses with negative restored index

[write] Source 05 of batch 82: *Research Report 34*, author line “Mathematical research report”, dated 3 October 2026; archive `Counterexample_Height_Expansions_and_Rotation_Discrepancy_Package.zip`; pins `2dde7850f` (construction) and `9fa99be93` (transfer appendix). Its sections, statements and equations are numbered 34.*k* here.

Abstract of the source. We fix the input, compiler export, repunit and one prime-scale choice in the signed19 construction of Report 33, and count only its prescribed canonical auxiliary witnesses. Each main index in one Chinese-remainder progression admits at most one first-Pell index in the required strict ratio window. The maximum of the nineteen supplied coordinates is eventually the last auxiliary Pell coordinate, whose double logarithm has an explicit exponentially corrected expansion and a Lambert W inverse. The height count satisfies $N(B) = \kappa \log \log B + O((\log \log B)^{1-\varepsilon})$ for some effective fixed-parameter $\varepsilon > 0$. A logarithmic-form bound makes the exact selection eventually identical to a fixed irrational-rotation test. The resulting exact discrepancy term is unbounded by Kesten’s theorem, ruling out a smooth two-term count with bounded remainder. The finer two-term law with $o(\log \log \log B)$ remainder is equivalent to a specific unresolved $o(\log R)$ discrepancy estimate. This is a theorem about one explicit subfamily, not the whole signed19 fiber.

[write] Report 34 has no executable mathematical checker by design and has not been reviewed by the programme. Its letters differ from Part III’s: here $\lambda = A + \sqrt{\Delta}$ and $\alpha = \log \lambda$; κ is a counting coefficient and κ_{in} is Report 33’s κ ; $\alpha_{\text{src}}, \delta_{\text{src}}$ are Report 33’s coordinates α, δ ; and its paid mask is written M_F^{src} as in Report 33. The source’s files are shipped with the prefix `05-signed19-heights-`.

34.1 The result and the object being counted

Report 33 proves that, for every positive ordinary input and every actual fixed compiler export in its pinned source, the complete74 signed19 index-eliminated child has infinitely many positive supplied-witness zeros with negative restored index [1]. We study the size and number of one canonical family within that construction. Report 33 and its source are unchanged.

The term *negative restoration* means that the omitted, algebraically computed index is $R = -p < 0$. It does not mean that any of the nineteen supplied coordinates is negative. A negative computed intermediate is allowed by the signed presentation. Nothing here identifies a machine input’s acceptance status or changes a positive-parent theorem.

Fix all construction data in Section 34.2. In particular the prime used to choose the scale is fixed once and for all. Write

$$\begin{aligned} \Delta &= A^2 - 1, & D_P &= P^2 - 1, & \lambda &= A + \sqrt{\Delta}, & \nu &= P + \sqrt{D_P}, \\ \alpha &= \log \lambda, & \beta &= \log \nu, & L &= E/2, & V &= TQH_0, \\ \delta &= \log(1 + 1/Y), & d_{\text{rot}} &= \frac{\delta}{L\beta}, & \kappa &= \frac{\delta}{2\alpha V L \beta}, & C_0 &= \frac{\alpha}{2\Delta}. \end{aligned} \tag{34.1}$$

All logarithms are natural. The growth logarithm α is distinct from the source coordinate α_{src} ; δ and κ likewise differ from δ_{src} and κ_{in} . The letter B below is a height bound, whereas B_{comp} denotes the fixed compiler radix.

For each eligible pair (p, n) , let $\mathbf{w}(p, n)$ be the full tuple in Section 34.2, and put

$$\text{ht}(\mathbf{w}) = \max\{\text{its nineteen supplied coordinates}\}.$$

We count a sufficiently late tail $\mathcal{C}$ of these tuples, retaining all pairs in that tail which satisfy the exact congruence and strict ratio test. Define

$$N(B) = \#\{\mathbf{w} \in \mathcal{C} : \text{ht}(\mathbf{w}) \leq B\}.$$

The tail begins at one fixed main-index progression representative p_b . Its onset may be enlarged to ensure positivity, height dominance and the exact rotation reduction. Finite changes to this convention affect N by $O(1)$ only.

Theorem 34.1.1 (Canonical family height and count). *For the fixed data just specified, the following statements hold.*

- (i) Each $p = p_b + Vr$, $r \in \mathbb{Z}_{\geq 0}$, has at most one positive $n \equiv n_0 \pmod{L}$ satisfying

$$Y < \frac{\psi_A(p)}{2\psi_P(n)} < Y + 1. \quad (34.2)$$

The proportion of progression terms admitting such an n is $d_{\text{rot}} > 0$.

- (ii) On $\mathcal{C}$, the height is exactly $y = \psi_{Q_{\text{aux}}}(S)$ and strictly increases with p . With $b_0 = (\log \Delta)/\alpha - 1$,

$$\log y = C_0 p^2 e^{2\alpha p} (1 + O(e^{-\alpha p})), \quad (34.3)$$

$$\log \log y = 2\alpha p + 2 \log p + \log C_0 + \sqrt{\Delta} \left(1 + \frac{b_0}{p}\right) e^{-\alpha p} + O(e^{-2\alpha p}). \quad (34.4)$$

- (iii) If B_j is the j th height in increasing order, then

$$N(B) \sim \kappa \log \log B, \quad \log \log B_j \sim j/\kappa. \quad (34.5)$$

For some effectively computable $\varepsilon > 0$ depending on the fixed data,

$$N(B) = \kappa \log \log B + O((\log \log B)^{1-\varepsilon}), \quad \log \log B_j = \frac{j}{\kappa} + O(j^{1-\varepsilon}). \quad (34.6)$$

- (iv) There is an exact fixed-rotation discrepancy $D(R)$, defined in Section 34.7, such that, for $t = \log \log B$,

$$N(B) = \kappa t - 2\kappa \log t + D(R(B)) + O(1), \quad R(B) \sim \frac{t}{2\alpha V}. \quad (34.7)$$

The smooth all-height formula $N(B) = \kappa t - 2\kappa \log t + O(1)$ is false. The weaker formula with $o(\log t)$ in place of $O(1)$ holds if and only if $D(R) = o(\log R)$; that condition is not established here.

The height inverse is given more precisely in Section 34.5. All constants and thresholds may depend on the fixed input, compiler, repunit and prime-scale choice. No parameter-uniform exponent or useful numerical onset is asserted. Equations (34.5) and (34.6) are statements on the double-logarithmic scale, not multiplicative equivalents for the heights.

34.2 The canonical family and its source interface

This section defines every tuple being counted. We inherit from the audited construction in Report 33 its integrality, positivity on a sufficiently late tail, and satisfaction of the full child equations. We recall the exact formulas and the relevant hypotheses rather than reproducing its whole existence proof.

34.2.1 Fixed compiler and scale choices

For an integer $D \geq 2$ define the Pell sequences by

$$\chi_D(v) + \psi_D(v)\sqrt{D^2 - 1} = (D + \sqrt{D^2 - 1})^v \quad (v \in \mathbb{Z}_{\geq 0}). \quad (34.8)$$

Their norm is $\chi_D(v)^2 - (D^2 - 1)\psi_D(v)^2 = 1$.

Fix $x \in \mathbb{Z}_{>0}$ and an actual compiler export from Report 33: $B_{\text{comp}} = 2^d \geq 16$, with d a power of five, b positive and odd, $K_0 \geq 0$, and the fixed masks M_C, M_F^{native} . Put $M_F^{\text{src}} = M_F^{\text{native}} + B_{\text{comp}} - 1$. All further compiler requirements remain those of the inherited source; no toy mask replaces these numerals.

Choose and fix a power of five N_0 such that $q = B_{\text{comp}}^{N_0} > 2dx + 1$. Define

$$\begin{aligned} t_0 &= dN_0, & p_0 &= 12t_0 + 7, & X &= 2^{p_0}, & Q &= q^2 - 1, \\ J &= \frac{q - 1}{B_{\text{comp}} - 1}, & w &= 2^{p_0 - 3t_0}, & \alpha_{\text{src}} &= q - 1 - 2dx, & z &= 1, \\ F &= K_0 + X - q + 1, & M &= (M_C + qM_F^{\text{src}})J, & C &= 1. \end{aligned} \quad (34.9)$$

Thus $q = 2^{t_0}$ and $X = wq^3$. Set $D_0 = 4q^3(X + 1)/3$. The construction proves $D_0 \in \mathbb{Z}_{>0}$ and $\gcd(D_0, Q) = 1$. Choose s_0 with

$$\gcd(s_0, Q) = \gcd(1 + D_0 s_0, Q) = 1,$$

and fix one prime $\ell > 3$ in the primitive progression $1 + D_0 s_0 \pmod{D_0 Q}$, large enough that $s = (\ell - 1)/D_0 > 0$. Dirichlet's theorem supplies such primes; this report neither varies nor counts them. Define

$$\begin{aligned} Y &= sq^3, & E &= XY, & a &= Y(X + 1), & A &= a + 2, & H_0 &= 4a + 3 = 3\ell, \\ \Delta &= A^2 - 1 = a^2 + H_0, & P &= 2XY^2 + 1, \\ O &= \text{ord}_{H_0}(2), & T &= \text{lcm}(4, 2E, O), & L &= E/2. \end{aligned} \quad (34.10)$$

The inherited scale argument gives

$$\gcd(QH_0, T) = 1. \quad (34.11)$$

This fact is what allows a free main-index progression after imposing all outer constraints.

34.2.2 Indices and all nineteen coordinates

The positive odd input index $u = 2dx + b \geq 3$ is fixed. Set

$$\kappa_{\text{in}} = \psi_A(u), \quad \mu_0 = \chi_A(u), \quad e_u = \mu_0 - a\kappa_{\text{in}}, \quad \delta_{\text{src}} = \frac{\kappa_{\text{in}} - u}{\Delta}. \quad (34.12)$$

Choose p in the single simultaneous progression

$$p \equiv p_0 \pmod{T}, \quad p \equiv -M - Q(q^2 - qF + e_u - 1) \pmod{QH_0}. \quad (34.13)$$

By (34.11), its step is $V = TQH_0$. In particular $p \equiv 3 \pmod{4}$. Fix $n_0 \equiv (1 - p_0)/2 \pmod{L}$. Retain the pairs (p, n) with $n \in \mathbb{Z}_{>0}$, $n \equiv n_0 \pmod{L}$, and the exact test (34.2).

For each retained pair define

$$\begin{aligned}
c &= \psi_A(p), & D &= \chi_A(p), & k &= 2\psi_P(n), & \tau &= \chi_P(n), \\
\eta &= c - kY, & \zeta &= k - \eta, & h &= \frac{k + p - 1}{E}, & R &= -p, \\
Z &= q^2 - qF + \frac{p + M}{Q}, & \rho &= \frac{Z + e_u - 1}{H_0}, \\
\gamma &= \frac{D - ac - X}{H_0}, & \sigma &= \gamma - \rho, & W &= 1 - Z.
\end{aligned} \tag{34.14}$$

The *canonical auxiliary choice* in this report is exactly

$$\begin{aligned}
m &= pc, & f &= \chi_A(m), & Q_{\text{aux}} &= \Delta\psi_A(m), & i &= \frac{Q_{\text{aux}}}{c^2}, \\
S &= p + 2m, & U &= \frac{\chi_{Q_{\text{aux}}}(S)}{Q_{\text{aux}}}, & y &= \psi_{Q_{\text{aux}}}(S), \\
j &= \frac{U - p}{c}, & o &= \frac{U + c}{f}.
\end{aligned} \tag{34.15}$$

The supplied tuple, in source order, is

$$\mathbf{w}(p, n) = (J, F, \alpha_{\text{src}}, z, f, h, i, j, o, s, w, \tau, \eta, \zeta, y, Z, \delta_{\text{src}}, \rho, \sigma). \tag{34.16}$$

All its coordinates are positive integers for sufficiently large eligible p . This is the construction theorem inherited from Report 33, Sections 4–9, and its independent audit [2, 3]. In particular, Q_{aux}/c^2 , U , j , and o are integers by the divisibility and odd-quotient Pell identities proved there. Their integrality is not inferred from the asymptotic analysis below.

[write] Report 33's Sections 4–9 are Sections 33.4–33.9 of this Part. The construction packet and audit cited here are shipped once, as source 10's `10-index-restore-repro-evidence-FULL-SIGNED-COUNT EREXAMPLE.md` and `10-index-restore-repro-evidence-INDEPENDENT-AUDIT.md`; source 05's copies, `evidence/FULL-SIGNED-COUNTEREXAMPLE.md` and `evidence/CONSTRUCTION-INDEPENDENT-AUDIT.md`, are byte-identical and not shipped.

There is no choice of a second auxiliary index, a different multiple of pc , another prime, or a new scale hidden in $N(B)$. Once (p, n) is selected, every entry of (34.16) is prescribed. Distinct p already yield distinct tuples because Z is affine in p with positive slope.

34.2.3 Link to the complete child equations

For clarity, the child consists of the sum of squares of the following eight residuals, with the computed definitions in (34.9)–(34.15) and $\mu = W + a\kappa_{\text{in}} + \rho H_0$:

$$\begin{aligned}
(K_0 + X)C &= F + z(q - 1), \\
R &= (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F^{\text{src}})J, \\
\tau^2 - XY^2(XY^2 + 1)k^2 &= 1, & D^2 - \Delta c^2 &= 1, \\
(ic^2)^2 &= \Delta(f^2 - 1), \\
(ic^2)^2((jc - R)^2 - y^2) &= 1 - y^2, \\
jc - R &= of - c, & \mu^2 - \Delta\kappa_{\text{in}}^2 &= 1.
\end{aligned} \tag{34.17}$$

The restored expression is $R = k - hE - 1 = -p$. Also $c = kY + \eta$, $\gamma = \rho + \sigma$, $D = X + ac + \gamma H_0$, $\kappa_{\text{in}} = u + \delta_{\text{src}}\Delta$, and $\mu = \mu_0 > 0$. Thus the family satisfies the whole signed19 interface, rather than just an auxiliary subsystem. The validity and positivity proof of these statements is the inherited theorem. Only the size and count of this prescribed family are newly analyzed here.

34.3 The exact first index window

The two quadratic fields are distinct. Indeed A is even, so Δ is odd, while

$$D_P = 4XY^2(XY^2 + 1)$$

has odd 2-adic valuation $2 + p_0 + 2v_2(Y)$. The squarefree part of D_P is therefore even and that of Δ is odd. Both define real quadratic fields, and their intersection is $\mathbb{Q}$. Consequently α/β is irrational: a rational relation would put equal positive powers of λ and ν in that intersection, and the norm-one property would force such a positive rational power to be 1, which is impossible.

Put $c_p = \psi_A(p)$ and

$$g = \log \frac{\sqrt{D_P}}{2\sqrt{\Delta}}.$$

The hyperbolic Pell formulas make (34.2) exactly equivalent to

$$a_p < n\beta < b_p, \quad a_p = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2(Y+1)}, \quad b_p = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2Y}. \quad (34.18)$$

The derivative of $t \mapsto \operatorname{arsinh}(e^t)$ lies strictly between 0 and 1. Integrating across a logarithmic interval of width δ gives

$$0 < b_p - a_p < \delta < L\beta. \quad (34.19)$$

Here $\delta \leq \log 2$, $L \geq 1$, and $\beta = \operatorname{arcosh} P > \log 2$. Distinct allowed n have spacing L , so at most one can satisfy (34.18). This is an exact uniqueness result, before any limiting approximation. Both endpoints tend to infinity, and any large-index solution has

$$n = (\alpha/\beta)p + O(1) > 0. \quad (34.20)$$

For $v \in \{Y, Y+1\}$ define

$$\epsilon_v(p) = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2v} - (\alpha p + g - \log v).$$

Binet's formula and $\operatorname{arsinh} z = \log(2z) + (4z^2)^{-1} + O(z^{-4})$ give

$$\epsilon_v(p) = \left(\frac{4v^2\Delta}{D_P} - 1 \right) e^{-2\alpha p} + O(e^{-4\alpha p}). \quad (34.21)$$

The -1 term is the Binet correction. It is essential to the first endpoint coefficient.

Write the main progression as $p = p_b + Vr$, $r \geq 0$, and set

$$\theta = \frac{V\alpha}{L\beta}, \quad \xi = \frac{\alpha p_b + g - \log Y - n_0\beta}{L\beta}, \quad I = (0, d_{\text{rot}}) \pmod{1}. \quad (34.22)$$

The limiting interval contains a point of $n_0\beta + L\beta\mathbb{Z}$ precisely when

$$\{\xi + r\theta\} \in I. \quad (34.23)$$

To check the phase convention, put $z = (\alpha p + g - \log Y - n_0 \beta) / (L \beta)$. The condition is that some integer lies strictly between $z - d_{\text{rot}}$ and z , equivalent to $0 < \{z\} < d_{\text{rot}}$.

Because θ is irrational, the proportion of fixed-interval hits is d_{rot} . This transfers to the exact moving window without a logarithmic-form theorem: for any $\epsilon > 0$, after finitely many indices the exact and limiting indicators can differ only within ϵ -neighborhoods of the two endpoints of I . Equidistribution bounds the upper density of these discrepancies by 4ϵ . Letting ϵ tend to zero proves assertion (i) and positive density. This argument alone does not imply finitely many discrepancies; Section 34.7 supplies that stronger fact separately.

34.4 The maximum supplied coordinate and its growth

34.4.1 Eventual exact height and no ties

The strict ratio gives $k < c/Y$, hence $\eta, \zeta = O(c)$ and $\tau = O(c)$ by the first norm. Also $h = O(c + p)$; Z and ρ are affine in p ; and $\sigma = O(c + p)$ by the formula for γ . The remaining nonauxiliary coordinates in (34.16) are fixed. All these estimates have fixed-data implied constants.

For the auxiliary coordinates, $Q_{\text{aux}} = ic^2$ with $i \geq 1$, and

$$Q_{\text{aux}}^2 = \Delta(f^2 - 1), \quad y \geq \psi_{Q_{\text{aux}}}(3) = 4Q_{\text{aux}}^2 - 1 > Q_{\text{aux}} \geq c^2.$$

Since $\Delta \geq 3$ and $f \geq 2$, we have $f < Q_{\text{aux}}$ and $i \leq Q_{\text{aux}}$. The auxiliary norm is exactly

$$U^2 = (1 - Q_{\text{aux}}^{-2})y^2 + Q_{\text{aux}}^{-2} < y^2. \quad (34.24)$$

Therefore $j = (U - p)/c < y$ and

$$o = \frac{U + c}{f} < \frac{y + c}{2} < y.$$

Finally $y \geq 4c^4 - 1$ dominates every $O(c + p + 1)$ coordinate because c grows exponentially and $p = o(c)$. This checks all nineteen entries of (34.16). Thus $\text{ht}(\mathbf{w}) = y$ with strict dominance on a sufficiently late tail.

Extend the formulas to real $p \geq 1$ by

$$\begin{aligned} c &= \frac{\sinh(\alpha p)}{\sqrt{\Delta}}, \quad m = pc, \quad Q_{\text{aux}} = \sqrt{\Delta} \sinh(\alpha m), \quad S = p + 2m, \\ b_{\text{aux}} &= \text{arcosh } Q_{\text{aux}}, \quad y(p) = \frac{\sinh(Sb_{\text{aux}})}{\sinh b_{\text{aux}}}. \end{aligned} \quad (34.25)$$

Here $c(1) = m(1) = 1$, $Q_{\text{aux}}(1) = \Delta \geq 3$, and $S(1) = 3$. The functions m , Q_{aux} , and S increase strictly. For $S > 1$ and $b > 0$, $\sinh(Sb)/\sinh b$ increases in both arguments: the derivative of its logarithm in b is $S \coth(Sb) - \coth b > 0$, because $z \coth z$ increases on $(0, \infty)$. Thus $y(p)$ is strictly increasing, and the canonical family has no height ties after the fixed tail onset.

34.4.2 An exponentially corrected expansion

Put $d_0 = \sqrt{\Delta}$ and $\ell_0 = \log d_0$. From (34.25),

$$b_{\text{aux}} = \alpha m + \ell_0 + O(e^{-2\alpha m}), \quad \log(2 \sinh b_{\text{aux}}) = \alpha m + \ell_0 + O(e^{-2\alpha m}).$$

The logarithm of the numerator in y contributes the remaining term $\log(1 - e^{-2Sb_{\text{aux}}})$, which is smaller. Consequently

$$\log y = (2m + p - 1)(\alpha m + \ell_0) + O((m + p)e^{-2\alpha m}). \quad (34.26)$$

The error is superexponentially small in p . Substituting

$$m = \frac{p}{2d_0} e^{\alpha p} (1 - e^{-2\alpha p})$$

gives the first two scales explicitly:

$$\log y = C_0 p^2 e^{2\alpha p} + \frac{p}{2d_0} [\alpha(p - 1) + \log \Delta] e^{\alpha p} + O(p^2). \quad (34.27)$$

Divide by $C_0 p^2 e^{2\alpha p}$ and take the logarithm. The resulting first correction is

$$d_0 \left(1 + \frac{(\log \Delta)/\alpha - 1}{p} \right) e^{-\alpha p},$$

and the rest is $O(e^{-2\alpha p})$. This proves (34.3)–(34.4), including the coefficient $C_0 = \alpha/(2\Delta)$.

34.5 The real height cutoff and Lambert inversion

Strict monotonicity gives, for every sufficiently large B , a unique real p_B with $y(p_B) = B$. The exact height condition on the tail is $p \leq p_B$.

Let W be the positive real branch of the Lambert function, satisfying $W(z)e^{W(z)} = z$, and set

$$t_B = \frac{1}{\alpha} W(\sqrt{2\alpha\Delta \log B}). \quad (34.28)$$

It solves the leading model exactly:

$$C_0 t_B^2 e^{2\alpha t_B} = \log B, \quad 2\alpha t_B + 2 \log t_B + \log C_0 = \log \log B.$$

Write $b_0 = (\log \Delta)/\alpha - 1$. Equation (34.4), with the positive derivative $2\alpha + 2/p$ of the leading logarithmic model, first brackets $p_B - t_B$ by $O(e^{-\alpha t_B})$. Taylor expansion of the displayed smooth correction then gives

$$p_B = t_B - \frac{\sqrt{\Delta}(1 + b_0/t_B)e^{-\alpha t_B}}{2\alpha + 2/t_B} + O(e^{-2\alpha t_B}). \quad (34.29)$$

The derivative of the first correction is $O(e^{-\alpha t_B})$ in this neighborhood; multiplying by the first bracket error produces the stated $O(e^{-2\alpha t_B})$ residual. In particular $p_B < t_B$ eventually.

A coarser expression is

$$\begin{aligned} p_B &= \frac{1}{\alpha} W(\sqrt{2\alpha\Delta \log B}) + O\left(e^{-W(\sqrt{2\alpha\Delta \log B})}\right) \\ &= \frac{\log \log B - 2 \log \log \log B + \log(8\alpha\Delta)}{2\alpha} + O\left(\frac{\log \log \log B}{\log \log B}\right). \end{aligned} \quad (34.30)$$

The latter follows either by expanding W or by substitution into the strictly increasing leading model. It is an inverse-height expansion, not a counting second term.

The difference $|p_B - t_B|$ is eventually smaller than V , so using t_B instead of p_B in a progression cutoff changes the count by at most one. It need not give equality at every jump. We keep p_B in the exact floor formula below.

The density in Section 34.3 already proves

$$N(B) \sim \frac{d_{\text{rot}}}{V} p_B \sim \kappa \log \log B.$$

If p_j is the j th selected main index, then $p_j \sim Vj/d_{\text{rot}}$. Inserting this in (34.4) proves $\log \log B_j \sim j/\kappa$. Only the double logarithm is substituted; an $o(p_j)$ index error must not be silently exponentiated into a multiplicative height equivalent.

34.6 Height and cutoff expansions to every order

The first correction above admits an all-orders refinement. It affects the continuous height and its inverse, without improving the rotation discrepancy. It is useful to separate a convergent analytic *normal form* from the smaller remainder of the exact quantity.

Put $d_0 = \sqrt{\Delta}$, $\epsilon = e^{-\alpha p}$, and define

$$A_p = d_0(1 - 1/p), \quad B_p = \frac{d_0 \log \Delta}{\alpha p}, \quad G_0(p) = 2\alpha p + 2 \log p + \log C_0. \quad (34.31)$$

The symbols A_p, B_p here are coefficient functions, distinct from the fixed Pell base A and the height bound B .

Proposition 34.6.1 (All orders at fixed parameters). *For each fixed integer $N \geq 1$,*

$$\log \log y(p) = G_0(p) + \sum_{n=1}^N a_n(p) e^{-n\alpha p} + O(e^{-(N+1)\alpha p}), \quad (34.32)$$

where $a_n(p)$ is a polynomial in $1/p$ of degree at most n . Define

$$S_0(C) = 2, \quad S_1(C) = -C, \quad S_n(C) = -CS_{n-1}(C) + S_{n-2}(C), \quad a_n(p) = -\frac{S_n(A_p) + S_n(B_p)}{n}. \quad (34.33)$$

For $t = t_B$ from (34.28), the exact cutoff has, for every fixed N ,

$$p_B = t + \sum_{n=1}^N c_n(t) e^{-n\alpha t} + O(e^{-(N+1)\alpha t}). \quad (34.34)$$

Each $c_n(t)$ is rational in t and the fixed constants, bounded as $t \rightarrow \infty$, and analytic as a function of $1/t$ near zero. A triangular recurrence is given in (34.38). Constants may depend on N and all the fixed data.

Proof. The leading expression $F(p) = (2m + p - 1)(\alpha m + \frac{1}{2} \log \Delta)$ in (34.26) has the exact factorization

$$F(p) = C_0 p^2 \epsilon^{-2} (1 + A_p \epsilon - \epsilon^2) (1 + B_p \epsilon - \epsilon^2). \quad (34.35)$$

Indeed, $m = p(\epsilon^{-1} - \epsilon)/(2d_0)$, so the two original factors respectively equal $(p/d_0)\epsilon^{-1}(1 + A_p \epsilon - \epsilon^2)$ and $(\alpha p/(2d_0))\epsilon^{-1}(1 + B_p \epsilon - \epsilon^2)$. Since $F(p) \sim 2\alpha m^2$ and $m \geq p$, dividing the error in (34.26) by $F(p)$ and taking a logarithm gives

$$\begin{aligned} \log \log y(p) &= G_0(p) + \log(1 + A_p \epsilon - \epsilon^2) + \log(1 + B_p \epsilon - \epsilon^2) + \mathcal{R}(p), \\ \mathcal{R}(p) &= O(e^{-2\alpha m}/m) = o(e^{-K\alpha p}) \quad \text{for every fixed } K > 0. \end{aligned} \quad (34.36)$$

The roots $r_{\pm}(C) = (-C \pm \sqrt{C^2 + 4})/2$ satisfy

$$1 + C\epsilon - \epsilon^2 = (1 - r_+(C)\epsilon)(1 - r_-(C)\epsilon).$$

Their power sums are $S_n(C)$ in (34.33), and hence

$$\log(1 + C\epsilon - \epsilon^2) = - \sum_{n \geq 1} \frac{S_n(C)}{n} \epsilon^n$$

whenever $|\epsilon| \max(|r_+|, |r_-|) < 1$, taking branches that vanish at zero. For $p \geq 1$, both A_p and B_p lie in a fixed bounded nonnegative interval. A constant $K'_0 \geq 1$ therefore bounds all four root moduli uniformly. For $|\epsilon| \leq 1/(2K'_0)$, the sum of the two tails after order N has absolute value at most

$$4 \sum_{n > N} \frac{(K'_0 |\epsilon|)^n}{n} \leq \frac{8(K'_0)^{N+1}}{N+1} |\epsilon|^{N+1}.$$

This proves uniform convergence of the two-logarithm normal form. The recurrence shows that S_n is a polynomial of degree at most n , proving the coefficient assertion and (34.32).

For inversion put $p = t + u$ and $w_0 = e^{-\alpha t}$. Define

$$\begin{aligned} H_t(u, w_0) &= 2\alpha u + 2 \log(1 + u/t) \\ &\quad + \log(1 + A_{t+u} w_0 e^{-\alpha u} - w_0^2 e^{-2\alpha u}) \\ &\quad + \log(1 + B_{t+u} w_0 e^{-\alpha u} - w_0^2 e^{-2\alpha u}). \end{aligned} \tag{34.37}$$

Introduce $z_0 = 1/t$, replacing $1/(t + u)$ by $z_0/(1 + z_0 u)$. The resulting function $H(u, z_0, w_0)$ is jointly analytic near $(0, 0, 0)$, using logarithms near 1. It satisfies

$$H(0, z_0, 0) = 0, \quad \partial_u H(0, z_0, 0) = 2\alpha + 2z_0,$$

which is nonzero near zero. The analytic implicit-function theorem supplies a unique analytic small root $\mathcal{U}(z_0, w_0)$ on a fixed neighborhood, with $\mathcal{U}(z_0, 0) = 0$. Its series in w_0 converges uniformly on a smaller polydisk and has uniformly bounded fixed-order remainders.

Explicitly, write $u_{<n} = \sum_{j=1}^{n-1} c_j(t) w_0^j$. Then

$$c_n(t) = - \frac{[w_0^n] H_t(u_{<n}, w_0)}{2\alpha + 2/t}. \tag{34.38}$$

Adding $c_n(t) w_0^n$ affects the coefficient of order n only through the displayed linear derivative. Expansion of the rational functions, exponentials and logarithms proves inductively that these coefficients are rational in t and the fixed constants; joint analyticity proves regularity and boundedness at $1/t = 0$.

It remains to justify inversion for the exact height, rather than just its normal form. The already proved first-order inverse gives $u_{\text{ex}} = p_B - t = O(w_0)$. Equation (34.36) gives

$$H_t(u_{\text{ex}}, w_0) = -\mathcal{R}(t + u_{\text{ex}}).$$

In this real neighborhood $\partial_u H_t \geq \alpha > 0$ for large t . The mean value theorem bounds the distance from u_{ex} to $\mathcal{U}(1/t, w_0)$ by $\alpha^{-1} |\mathcal{R}(t + u_{\text{ex}})|$. Since $m(t + u_{\text{ex}})/m(t) \rightarrow 1$, this is

$$O(e^{-\alpha m(t)}/m(t)) = o(w_0^K) \quad \text{for every fixed } K > 0.$$

The uniformly convergent normal-form series therefore gives the actual fixed-order expansion (34.34). $\square$

For reference, $a_1 = A_p + B_p$ and $a_2 = -2 - (A_p^2 + B_p^2)/2$, while

$$c_1(t) = -\frac{d_0[1 + ((\log \Delta)/\alpha - 1)/t]}{2\alpha + 2/t}.$$

These reproduce the earlier first corrections. The convergent series sum to the *normal forms*; the exact height and cutoff still differ by beyond-all-orders terms. Neither series is asserted to equal the exact quantity. Smooth expansions of arbitrarily high fixed order do not remove the counting discrepancy analyzed next.

34.7 An eventual exact rotation count

We now strengthen the density argument using a standard effective theorem on linear forms in logarithms. For fixed positive algebraic numbers $z_1, \dots, z_s$, there are effective constants $c, C > 0$, depending on those numbers, such that every nonzero real form

$$\Lambda = \sum_{i=1}^s b_i \log z_i, \quad b_i \in \mathbb{Z},$$

satisfies $|\Lambda| \geq c\mathcal{B}^{-C}$, where $\mathcal{B} = \max(3, |b_1|, \dots, |b_s|)$. The sufficient statement is Theorem 1.1, equation (1.2), and its fixed-number consequence in Bugeaud [5]; Baker–Wüstholz [6] is a classical additional reference. No multiplicative-independence assumption is needed beyond nonvanishing of the particular form.

For $v = Y, Y + 1$ and an integer n near (34.20), consider

$$\Lambda_v(p, n) = p \log \lambda - n \log \nu + \log \frac{\sqrt{D_P}}{2v\sqrt{\Delta}}. \quad (34.39)$$

This form is nonzero. If it vanished, squaring the associated positive multiplicative equality would give

$$\lambda^{2p} = \frac{4v^2\Delta}{D_P} \nu^{2n}.$$

Its left side lies in $\mathbb{Q}(\sqrt{\Delta})$ and its right side in $\mathbb{Q}(\sqrt{D_P})$, hence both are rational. The norm-one property forces the positive rational λ^{2p} to equal 1, contrary to $p > 0$.

The coefficients in (34.39) are $O(p)$ by (34.20). Thus, uniformly for either endpoint and all relevant large p ,

$$|\Lambda_v(p, n)| \geq c_1 p^{-C_1} \quad (34.40)$$

for effective fixed-data constants. This eventually exceeds the $O(e^{-2\alpha p})$ endpoint shifts in (34.21). A disagreement between an exact and limiting indicator would require an allowed lattice point within that exponentially small distance of a limiting endpoint, contradicting (34.40). Therefore only finitely many disagreements occur.

Enlarge p_b within its progression beyond these exceptions and the positivity and height thresholds. Use the phase ξ corresponding to this new p_b in (34.22). For every $r \geq 0$, the exact selection is now precisely (34.23). Define

$$D(R) = \sum_{r=0}^{R-1} \mathbf{1}_I(\{\xi + r\theta\}) - d_{\text{rot}} R \quad (R \in \mathbb{Z}_{\geq 0}), \quad (34.41)$$

with the empty sum zero. In $D(R)$ and $R(B)$, R denotes a nonnegative number of progression terms, not the restored coordinate $R = -p$ used in the construction. For all sufficiently large B ,

$$R(B) = \max \left(0, 1 + \left\lfloor \frac{p_B - p_b}{V} \right\rfloor \right). \quad (34.42)$$

The exact identity for this tail is

$$N(B) = d_{\text{rot}} R(B) + D(R(B)). \quad (34.43)$$

This retains the fixed phase, the open strict-ratio interval and the true real cutoff. No asymptotic expression has been passed through a floor as if exact.

Combining (34.30) and (34.42) gives, for $t = \log \log B$,

$$d_{\text{rot}} R(B) = \kappa t - 2\kappa \log t + O(1), \quad R(B) \sim t/(2\alpha V),$$

which proves (34.7). Changing to another finite initial convention merely adds a fixed count offset and does not remove the discrepancy.

34.8 Effective errors and the second term obstruction

34.8.1 A fixed parameter power saving

For $h \geq 1$, choose an integer k_h nearest to $h\theta$. The nonzero form

$$hV\alpha - k_h L\beta$$

has integer coefficients $O(h)$, and its nonvanishing follows from irrationality of θ . The same logarithmic-form bound therefore supplies effective $c_2 > 0$ and $\mu \geq 1$ such that

$$\|h\theta\| \geq c_2 h^{-\mu}. \quad (34.44)$$

The exponent can be weakened to ensure $\mu \geq 1$. The finite geometric sum gives

$$\left| \sum_{r=0}^{R-1} e^{2\pi i h(\xi + r\theta)} \right| \leq \min \left(R, \frac{1}{2\|h\theta\|} \right) \ll h^\mu.$$

The unnormalized Erdős–Turán inequality [9, 10], with an absolute implied constant, yields for each integer $K \geq 1$

$$\begin{aligned} |D(R)| &\ll \frac{R}{K+1} + \sum_{h=1}^K \frac{1}{h} \left| \sum_{r=0}^{R-1} e^{2\pi i h(\xi + r\theta)} \right| \\ &\ll \frac{R}{K} + K^\mu. \end{aligned} \quad (34.45)$$

Taking K of order $R^{1/(\mu+1)}$ gives

$$D(R) = O(R^{1-\varepsilon}), \quad \varepsilon = \frac{1}{\mu+1} > 0. \quad (34.46)$$

Since $R(B)$ is of order t and $\varepsilon \leq 1/2$, the analytic $\log t$ correction is absorbed by this error. Equations (34.43) and (34.7) prove the counting part of (34.6).

There are no height ties, so $N(B_j) = j$. The leading law gives $t_j = \log \log B_j \sim j/\kappa$; substituting this into $j = \kappa t_j + O(t_j^{1-\varepsilon})$ proves the ranked error in (34.6). This effective error may be very weak, and remains much larger than the scale $\log \log B$ of the putative second term.

34.8.2 Why a bounded smooth remainder is impossible

Kesten's interval bounded-remainder theorem states that an interval for an irrational rotation has bounded discrepancy if and only if its length belongs to $\mathbb{Z} + \theta\mathbb{Z}$ [7, 8]. It applies to a fixed phase: translating the interval absorbs that phase. Open versus half-open endpoint conventions change at most two orbit hits.

In our case $d_{\text{rot}} \notin \mathbb{Z} + \theta\mathbb{Z}$. Otherwise integers a_1, b_1 would give

$$\delta = a_1 L\beta + b_1 V\alpha, \quad \frac{Y+1}{Y} = \nu^{a_1 L} \lambda^{b_1 V}.$$

The numbers λ and ν and their inverses are algebraic integers, since their monic integer polynomials have constant term 1. The right side is an algebraic unit. A rational algebraic unit must be 1 or -1 , whereas $(Y+1)/Y > 1$. This contradiction proves the claim. Kesten's theorem now implies that $D(R)$ is unbounded.

As B ranges through large real values, $R(B)$ attains every sufficiently large integer: take $B = y(p_b + V(R-1))$. The bounded residual in (34.7) cannot cancel an unbounded discrepancy. Hence

$$N(B) = \kappa \log \log B - 2\kappa \log \log \log B + O(1)$$

is false as an all-height assertion. The conclusion is about the claimed smooth remainder, not a failure of the leading law or of the precise height inverse.

34.8.3 The exact unresolved criterion

For this fixed rotation, the following statements are equivalent:

$$\begin{aligned} N(B) &= \kappa t - 2\kappa \log t + o(\log t) && \text{as all real } B \rightarrow \infty, \quad t = \log \log B; \\ D(R) &= o(\log R) && \text{as integers } R \rightarrow \infty. \end{aligned} \tag{34.47}$$

Sufficiency follows from (34.7) and $R(B) \sim t/(2\alpha V)$. For necessity take $B = y(p_b + V(R-1))$. Then $R(B) = R$ exactly, and $\log t = \log R + O(1)$ by (34.4). Subtracting the smooth terms in (34.7) gives the required $o(\log R)$ estimate.

Unboundedness of $D(R)$ does not disprove this finer condition. An $O(\log R)$ bound alone would also be insufficient to identify the second coefficient, since its error has exactly that scale. Equidistribution supplies only $o(R)$, and the effective power saving (34.46) is still too large. No $o(\log R)$ estimate is claimed.

34.9 Comparison and further research

34.9.1 The scope differs from Report 25

Report 25 classifies an entire fixed positive native fiber: seventeen supplied coordinates are forced, and all remaining tuples are parametrized by two Pell indices [4]. Its height law follows from exact finite floor counts, exceptional baseline terms, hyperbola decomposition and Euler summation. In its notation the leading count is logarithmic in height, with a proved unsmoothed square-root second term and a smaller power error.

Here the input, compiler, repunit and prime-scale choice are fixed, but only the prescription $m = p\psi_A(p)$ and $S = p + 2m$ is counted. The free main index runs through one CRT progression,

selected by an irrational-rotation window. No classification of all admissible auxiliary indices or all signed19 witnesses has been proved. The count is double-logarithmic in height, and its second-term question is governed by the explicit discrepancy (34.41). Report 25's two-index floor-sum error cannot be imported into this problem.

Pell growth, Binet expansion and careful height inversion are standard shared tools. The specific issue resolved here is the persistence of the exact rotation discrepancy through exponentially small strict-ratio endpoint perturbations, and the consequent bounded-remainder obstruction. This comparison is a statement of mathematical scope, not a literature-wide novelty claim.

34.9.2 Concrete questions left open

1. **The finer discrepancy scale.** Determine whether $D(R) = o(\log R)$ for the explicit fixed θ, ξ, I in (34.22). By (34.47), this is exactly the missing estimate for the stated triple-logarithmic second coefficient. A result only on a subsequence or after averaging would not establish the all-height claim.
2. **Usable effective constants.** Specialize explicit logarithmic-form constants to one fixed compiler, input and scale, and bound the last possible endpoint disagreement. The present proof gives effectivity in principle, without a practical numerical exponent or onset. It does not materialize the enormous supplied tuples.
3. **Larger families.** Characterize other permitted auxiliary choices at a fixed main index, then count their union without duplication. The current canonical prescription offers a lower-bound subfamily inside the full fiber, not an upper bound or an exhaustive parametrization.
4. **Dependence on the scale.** Study how density and height constants vary over prime-scale choices, with genuinely uniform error bounds if possible. Summing fixed-scale asymptotics without such control is not justified by this report.

The valid positive-index parent retains its positivity hypothesis. The separate raw29 and positive21 restoration questions remain outside the present theorem. No new universal operation count, degree bound, full numerical witness or machine-specific acceptance claim follows.

34.A Transfer to asymmetric signed19 orientations

A separate pinned source comparison [11] extends the applicability of this fixed family. At ProveIt commit

[write] The pinned sources of this appendix are research-tree files shipped inside archive 05 as byte copies and not shipped here: `complete74_asymmetric_scale_transfer.md` and `.json`, `complete74_equation_orientation_census.md` and `.json` (the latter 1,508,808 bytes), and `review_complete74_equation_orientation_census.md`, all in the research tree's `native-stream-queue/` directory and equal to the current files. The transfer proof itself is shipped as `05-signed19-heights-evidence-TRANSFER-PROOF.md`. Report 34 has not been reviewed by the programme.

9fa99be93fb479294d5d5a2458e59e5fba8444b6,

the asymmetric scale replaces $X = wq^3$ by $X = wq$, retaining $Y = sq^3$. The supplied w has no other consumer. Hence the coordinate map

$$\Phi : \quad w_{\text{new}} = q^2 w_{\text{old}} \quad (34.48)$$

preserves X and every subsequent computed quantity, including R , while leaving all other supplied coordinates unchanged.

There are four relevant equation orientations at each scale. Define

$$T_2 = (ic^2)^2, \quad K = \Delta(f^2 - 1), \quad U_0 = jc - R, \quad U_1 = of - c.$$

Each orientation retains *both* protecting comparisons $T_2 = K$ and $U_0 = U_1$, and selects its auxiliary norm from

$$C_e(U_v^2 - y^2) = 1 - y^2, \quad C_0 = T_2, \quad C_1 = K, \quad (e, v) \in \{0, 1\}^2.$$

For each parent define the corresponding child by substituting $r = R = k - hXY - 1$ everywhere and deleting only the resulting identical index comparison. These are mathematically defined descendants, not a claim that every eliminated packet has been published upstream.

For each *matching* orientation, the scale map (34.48) preserves every residual, so

$$G_{(1,3)}^{(e,v)}(\Phi(\mathbf{w})) = G_{(3,3)}^{(e,v)}(\mathbf{w})$$

is an all-value polynomial identity over any commutative ring. Across *different* orientations, the retained protecting comparisons give equality of comparison-zero loci over every commutative ring, and hence equality of sum-of-squares zero loci over the reals. This is not an assertion that the differently oriented sum-of-squares polynomials agree away from their zero sets.

The Report 33 tuples therefore give positive supplied-witness zeros with $R = -p$ in all four symmetric orientations and, after Φ , all four asymmetric orientations. In the family of this report, q and w are fixed. The map changes just one fixed coordinate to another fixed value, while leaving $y(p)$ and every index-selection test unchanged. After increasing the finite height threshold if necessary, the maximum is still exactly y . All the canonical-family height and counting conclusions of Report 34 consequently transfer with the same constants and discrepancy, up to the finite-tail convention.

This uses only the forward map. It asserts neither an integral inverse on every asymmetric-child zero nor a classification of that child's full fiber. The valid twenty-coordinate parents still supply a positive r and retain their nine comparisons. No conclusion for raw29, positive21, normalized87 or coupled88 is imported. No parent degree or operation ledger is assigned to these children.

34.B Evidence and the portable release

The construction source is pinned to ProveIt commit

2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff.

The report was prepared from the frozen construction, its independent audit, and the separately audited analytic proof packet. The original reviewed analytic proof, before path normalization in the packaged copy, has SHA256 identity

ff9d116c90b058be082c2c9d6ee051cc
f537d80821c14d10b28aec9a96d5a533.

The two lines concatenate to the complete digest. The whole-packet audit reports a mathematical pass for that exact version, including the effective error and the fixed-phase bounded-remainder application. `PROVENANCE.json` maps this original identity to the packaged digest and records every path-only portability normalization of the accompanying analytic notes.

The evidence has three distinct roles:

1. The inherited construction and its audit establish full child membership, positive supplied coordinates, and the fixed arithmetic progression.
2. The analytic proof and independent height, rotation and whole-packet checks establish the asymptotic formulas and external-theorem applications.
3. The local release verifier checks file identities; the offline build script rebuilds the PDF from the supplied LaTeX. These are packaging checks, not finite tests of the unbounded mathematical assertions.

No executable mathematical checker is included or needed for this pure analytic follow-on. No upstream code, source arithmetic schedule or historical finite checker was run to establish its claims. The source component checks are analytic notes. No third-party paper PDF is redistributed in the archive; the bibliography links to the research sources. The report is a new release and does not alter Report 33.

References of Report 34

- [1] *Failure of Positive Index Restoration: The complete⁷⁴ signed¹⁹ eliminated child*, Research Report 33, 3 October 2026. Frozen delivered report; its LaTeX source is included as context, unchanged. The construction used here is in Sections 4–9. [write] Printed in Part IV of this report, as Sections 33.1–33.B; its Sections 4–9 are Sections 33.4–33.9.
- [2] *Full negative-restoration family for the signed¹⁹ child*, frozen analytic construction packet, 3 October 2026. Included as `evidence/FULL-SIGNED-COUNTEREXAMPLE.md`; SHA256 recorded in the release provenance. Based on the ProveIt source at the commit identified in Appendix 34.B. [write] Shipped once, as `10-index-restore-repro-evidence-FULL-SIGNED-COUNTEREXAMPLE.md`.
- [3] *Independent audit of the full signed¹⁹ negative-restoration family*, 3 October 2026. Included as `evidence/CONSTRUCTION-INDEPENDENT-AUDIT.md`. This audit supports the inherited construction; it is distinct from the whole-packet growth audit. [write] Shipped once, as `10-index-restore-repro-evidence-INDEPENDENT-AUDIT.md`.
- [4] *Entire native fibers: Pell parametrization and height laws*, Research Report 25, 3 October 2026. Frozen delivered report, Theorems 2.1–2.2; its LaTeX source is included unchanged as context. The comparison here is not a new audit of that entire article. [write] Printed as Part III of this report; its Theorems 2.1–2.2 are Theorems 25.2.1–25.2.2.
- [5] Y. Bugeaud, B' , arXiv:2209.00275v1 (2022), Theorem 1.1, equation (1.2), printed pp. 2–3, and the ensuing fixed-number consequence. <https://arxiv.org/abs/2209.00275v1>. The theorem collects estimates attributed there to Waldschmidt and Matveev; its fixed-number consequence is the sufficient bound used in this report.
- [6] A. Baker and G. Wüstholz, *Logarithmic forms and group varieties*, Journal für die reine und angewandte Mathematik **442** (1993), 19–62. <https://doi.org/10.1515/crll.1993.442.19>. Classical background reference; the explicit sufficient statement inspected for this report is [5].
- [7] H. Kesten, *On a conjecture of Erdős and Szűs related to uniform distribution mod 1*, Acta Arithmetica **12** (1966), 193–212. <https://doi.org/10.4064/aa-12-2-193-212>.
- [8] M. Kelly and L. Sadun, *Pattern equivariant cohomology and theorems of Kesten and Oren*, arXiv:1404.0455v2 (2014), Theorem 1. <https://arxiv.org/abs/1404.0455v2>. A new proof of Kesten’s criterion, with a fixed starting point in its setup.

- [9] P. Erdős and P. Turán, *On a problem in the theory of uniform distribution I*, Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen **51** (1948), 1146–1154, Theorem III, printed p. 1150. https://www.renyi.hu/~p_erdos/1948-02.pdf.
- [10] P. Erdős and P. Turán, *On a problem in the theory of uniform distribution II*, Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen **51** (1948), 1262–1269. https://www.renyi.hu/~p_erdos/1948-03.pdf.
- [11] *Report 33 transfer to the asymmetric signed19 orientation children*, source-pinned transfer proof, 3 October 2026. Included as `evidence/TRANSFER-PROOF.md`; source identities are recorded in `PROVENANCE.json`. The pinned asymmetric scale transfer and equation orientation census descriptions are identified in Appendix 34.A. [Pinned asymmetric transfer source](#). [write] Shipped as `05-signed19-heights-evidence-TRANSFER-PROOF.md`.